Critical Invariant Polygons and the Farey–Ito Boundary of Stochastic Spectra

Contact Return Normal Forms, Projective Holonomy, and the Karpelevič–Ito Theorem

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Abstract

Let T be an orientation-preserving real-linear contraction of an oriented plane with nonreal eigenvalues, and let its polygonal complexity be the least number of vertices of a nondegenerate convex polygon P satisfying $TP \subseteq P$. We first develop an intrinsic contact theory for radially critical maps. Hereditary saturation, one-sided contact selection, exact contact mutations, a finite cyclic reduction, and the unimodular lattice sail lead to a contact-return normal form and a projective no-skipping principle. In an adapted complex coordinate, the resulting monodromy yields a heterogeneous Ito product and an exact lifted phase identity carried by Farey neighbors. We then apply this structure to determine, for every n, the union Θ_n of the spectra of all real row-stochastic $n \times n$ matrices. A strictly convex log-sine potential equalizes the monodromy factors and gives the sharp radial equation. Sparse stochastic matrices attain every equality point, while scalar Farey refinement proves nesting from order $n-1$ to order n. Together with radial filling and the unit-circle analysis, this establishes the full Farey–Ito description of the Karpelevič region from critical invariant polygons.

Keywords. critical invariant polygons; elliptic contractions; contact mutation; projective holonomy; Farey sequences; stochastic matrices; eigenvalue regions; Karpelevič–Ito theorem.

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Part I: Critical Invariant Polygons and Farey Return Normal Forms

1 The intrinsic theorem

Let V be a real plane. For every real-linear map $A : V \to V$, define its polygonal complexity by

$$\nu_{\text{poly}}(A) = \min\Big\{|\operatorname{Ext} P| : P \subset V \text{ is a nondegenerate compact convex polygon and } AP \subseteq P\Big\}, \tag{1.1}$$

with the value ∞ when the class is empty. An invertible map $T : V \to V$ is *elliptic* if

$$(\operatorname{tr} T)^2 < 4 \det T,$$

and it is an *elliptic contraction* when, in addition, its spectral radius belongs to $(0,1)$. Defining (1.1) for all linear maps is essential: the radial comparison maps tT in definition 1.1 need not remain contractions.

Definition 1.1 (Radial polygonal criticality)**.** For $N \geq 3$, an elliptic contraction T is *N-critical* if

$$\nu_{\text{poly}}(T) = N, \qquad \nu_{\text{poly}}(tT) > N \quad \text{for every } t > 1. \tag{1.2}$$

The second inequality allows the value ∞.

Both conditions are invariant under real-linear conjugacy. They are therefore properties of the planar dynamics, not of a chosen Euclidean metric, complex coordinate, polygon, or vertex numbering.

We write $\operatorname{Ext} K$ for the set of extreme points of a compact convex set K, ∂K for its boundary, $\operatorname{relint} F$ for relative interior, and $\operatorname{aff} S$ for affine hull.

Definition 1.2 (Strict polygon)**.** A planar convex polygon is called *strict* when it has nonempty interior and its displayed cyclic vertex list is exactly its set of extreme points. Thus adjacent sides are not collinear, every side is a maximal boundary segment, and the interior of the normal cone at every vertex is nonempty. A supporting line through a vertex is *strict* when its contact face is that vertex alone.

For a strict polygon P with an oriented boundary, let $\mathcal{E}(P)$ be its cyclic set of sides. For $e \in \mathcal{E}(P)$ write $t(e)$ and $h(e)$ for its tail and head, let $s(e)$ be the next side, and put

$$e^{\triangleright} = (t(e), h(e)].$$

A *one-sided contact representation* of $TP \subseteq P$ is a cyclic-order-preserving bijection

$$\chi : \operatorname{Ext} P \longrightarrow \mathcal{E}(P)$$

such that $Tv \in \chi(v)^{\triangleright}$ for every vertex v. Its strict side set and contact rotation are

$$I = \left\{ e \in \mathcal{E}(P) : T\chi^{-1}(e) \in \operatorname{relint} e \right\}, \qquad \sigma(e) = \chi(h(e)). \tag{1.3}$$

Both maps $e \mapsto h(e)$ and χ preserve cyclic order. Their composition σ is therefore an automorphism of the finite cyclic order; after identifying $\mathcal{E}(P)$ with $\mathbb{Z}/N\mathbb{Z}$, it is a translation. A legal contact mutation at e is available when $e \in I$ and $s(e) \notin I$; geometrically it replaces the vertex $h(e)$ by the strict contact point $T\chi^{-1}(e)$.

Theorem 1.3 (Critical-polygon contact-return normal form)**.** *Let $N \geq 4$ and let T be an N-critical elliptic contraction. One of the two orientations of V admits a strict invariant N-gon P and a one-sided contact representation with the following properties.*

(i) Hereditary saturation. *Every strict polygon R with at most N vertices and $TR \subseteq R$ has exactly N vertices; every side of R meets TR, and every vertex of TR lies on ∂R.*

(ii) Exact mutation law. *Every legal mutation produces another strict invariant N-gon with the same contact rotation σ, and its strict set is exactly*

$$I' = (I \setminus \{e\}) \cup \{\sigma(e)\}. \tag{1.4}$$

(iii) Interval reduction. *Among the finitely many strict-contact sets realized by finite mutation sequences, there is a representative for which I is a nonempty cyclic interval. If δ is the number of σ-orbits and $\varphi = |I|$, then $\varphi \geq \delta$.*

(iv) Return dichotomy. *If σ is the identity, every contact is strict. Assume σ is not the identity. If $\varphi = \delta$, then I is a complete transversal of the σ-orbits and every base has return time N/δ. If $\varphi > \delta$, the first-return map of σ to I is the cyclic successor on I. Its tower partition has two heights q and $q + h$ and exhausts all N vertices.*

All assertions are invariant under invertible real-linear conjugacy. The orientation asserted here is only the contact orientation: it selects the right-half-open ownership convention and an intermediate eigenvalue for the contact-return analysis. The later complex monodromy theorem may select the conjugate eigenvalue after reflecting the completed return strip.

The restriction $N \geq 4$ is used in the proper-corridor and Jensen-sheet steps; no assertion about the $N = 3$ case is made here.

The return dichotomy is the substantive conclusion. Before projective no-skipping is used, the first-return map is an arbitrary positive rotation of I. The theorem says that a non-transversal critical return cannot skip a base.

For a nonzero complex number z that is not positive real, write

$$\arg_+(z) \in (0, 2\pi)$$

for its positive lifted argument. The orientation used to construct the right-half-open contact system will be called the *contact orientation*. It is an intermediate choice. The complex monodromy theorem is allowed to use either that orientation or the opposite one, and it makes this choice only after the return regime is known. The exact reflection of the final return strip is proved in lemmas 8.2 and 8.3; no mutation or no-skipping argument is transported across that reflection.

Theorem 1.4 (Complex monodromy and Farey carrier)**.** *Let $N \geq 4$ and let T be an N-critical elliptic contraction. There is an adapted complex coordinate in one of the two orientations of V, hence one of the two conjugate eigenvalues μ of T, such that*

$$Tz = \mu z, \qquad 0 < |\mu| < 1.$$

Let $\vartheta = \arg_+(\mu)$ and put $y = \vartheta/(2\pi)$. There are consecutive reduced fractions

$$\frac{p}{q} < y < \frac{r}{s}$$

in the Farey sequence F_N whose ordered denominators satisfy $q \leq s$. Put

$$d = \lfloor N/q \rfloor, \qquad e = s - dq \in \mathbb{Z}.$$

Then there are parameters

$$0 \leq \beta_j < 1, \qquad \alpha_j = 1 - \beta_j > 0 \qquad (1 \leq j \leq d) \tag{1.5}$$

for which the following polynomial identity holds:

$$\mu^s \prod_{j=1}^{d} (\mu^q - \beta_j) = \mu^{dq} \prod_{j=1}^{d} \alpha_j. \tag{1.6}$$

Since $\mu \neq 0$, this is equivalent to the Laurent identity

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta_j) = \prod_{j=1}^{d} \alpha_j. \tag{1.7}$$

The signed closing exponent e is not asserted to be nonnegative. In the reflected transversal regime it is negative; in that regime (1.6), rather than the Laurent notation (1.7), is the primary identity.

Put

$$A = q\vartheta - 2\pi p, \qquad M = \arg_+(\mu^q - 1).$$

Then $0 < A < M < \pi$, and for each j there is a unique $u_j \in [A, M)$ such that

$$e^{iu_j} = \frac{\mu^q - \beta_j}{|\mu^q - \beta_j|}. \tag{1.8}$$

These arguments satisfy the phase identity

$$e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(r - dp). \tag{1.9}$$

The factors with $\beta_j = 0$ that are introduced only to complete a return strip are algebraic padding and are not claimed to be additional strict contacts.

The stochastic statement is deferred to section 9. Its proof will consist only of identifying N-criticality through the invariant-polygon criterion and applying theorem 1.4.

1.1 Relation with the stochastic eigenvalue literature

The final application belongs to the classical study of eigenvalue regions of stochastic matrices initiated by Dmitriev–Dynkin and Karpelevič; see [1, 4]. Farey descriptions and explicit matrix realizations appear in several later forms [2, 3, 5, 6]. The result proved here is organized in the opposite direction: the parent statement is intrinsic to a critical planar linear map, while stochastic matrices enter only through the invariant-polygon criterion in section 9.

1.2 Logical architecture

The proof is divided into five modules with one-way dependencies:

$$\boxed{\text{criticality}} \Longrightarrow \boxed{\text{hereditary saturation}} \Longrightarrow \boxed{\text{one-sided contacts and mutations}}$$
$$\Longrightarrow \boxed{\text{finite rotation section}} \Longrightarrow \boxed{\text{projective no-skipping}}.$$

The Farey product is then the scalar monodromy of the final return strip. The finite-rotation module is proved through consecutive lattice-sail vectors; it does not use a parity-indexed Euclidean remainder chain. The projective module contains no modular arithmetic. Detailed determinant and endpoint audits occur only inside the mutation theorem and the corridor realization where they are logically necessary.

For review purposes, each module exports one invariant-level contract:

module	input	exported conclusion
hereditary saturation	N-criticality	every invariant polygon with at most N vertices has all sides and all image vertices saturated
contact theory	saturation	a one-sided cyclic contact map and an exact, geometrically realized mutation law
finite rotation section	one reduced strict interval	a bijective two-height tower partition with unimodular return data
projective no-skipping	tower partition, exact return-edge ledger, and replacement heredity	a globally defined invariant deformation with one escaped image vertex; hence every nontransversal first return is the adjacent successor
complex monodromy	adjacent return strip	the Farey carrier, heterogeneous product, and lifted phase identity

No later module uses the internal indices of an earlier proof; it uses only the theorem displayed in the last column. This is the stability mechanism of the argument.

1.3 Foundational tools

The proof uses only finite-dimensional linear algebra, elementary planar convexity, and the index formula for a rank-two sublattice of $\mathbb{Z}^2$. For completeness, the precise forms of Perron–Frobenius, strict separation, polarity, Hausdorff continuity of polygonal area, and the lattice parallelogram count used below are proved in appendix A. Thus every nonstandard dependency of the main argument is available inside the paper. We use in their standard forms Cayley–Hamilton and Jordan normal form, elementary compactness and subsequence arguments, dominated convergence, Smith normal form for rank-two integer lattices, and elementary real projective geometry.

2 Elliptic coordinates and convex preliminaries

Proposition 2.1 (Adapted complex structures)**.** *Let T be elliptic. Put*

$$\rho = \sqrt{\det T}, \qquad \cos\theta = \frac{\operatorname{tr} T}{2\rho}, \qquad 0 < \theta < \pi,$$

and define

$$J_+ = \frac{\rho^{-1}T - \cos\theta\, I}{\sin\theta}, \qquad J_- = -J_+. \tag{2.1}$$

Then $J_+^2 = J_-^2 = -I$, and the two complex structures induce the two opposite orientations of V. There is an inner product for which both are isometries and

$$T = \rho(\cos\theta\, I + \sin\theta\, J_+) = \rho(\cos\theta\, I - \sin\theta\, J_-). \tag{2.2}$$

Thus T is multiplication by $\rho e^{i\theta}$ in the J_+ complex coordinate and by $\rho e^{-i\theta}$ in the J_- coordinate. In the adapted inner product,

$$T^* = \rho(\cos\theta\, I - \sin\theta\, J_+),$$

and T^* is conjugate to T by an orientation-reversing reflection.

Proof. The characteristic polynomial of $\rho^{-1}T$ is

$$X^2 - 2\cos\theta\, X + 1.$$

Cayley–Hamilton gives

$$(\rho^{-1}T - \cos\theta I)^2 = -\sin^2\theta I,$$

so $J_+^2 = -I$, and the same is true of $J_- = -J_+$. A complex structure on a real plane determines an orientation by declaring (v, Jv) positive; replacing J by $-J$ reverses that orientation. Hence J_+ and J_- induce precisely the two orientations of V.

Starting from any inner product g_0, set

$$g(x,y) = g_0(x,y) + g_0(J_+x, J_+y).$$

Then $g(J_+x, J_+y) = g(x,y)$, so J_+, and therefore also J_-, is orthogonal. Formula (2.2) follows directly from (2.1); its two complex interpretations follow from the rule that multiplication by $a+ib$ acts as $aI + bJ$. Since $J_+^* = -J_+$, the formula for T^* follows. In an orthonormal J_+-adapted basis, reflection in the first coordinate satisfies $SJ_+S^{-1} = -J_+$, and consequently $STS^{-1} = T^*$. □

Proposition 2.2 (Real-linear invariance of polygonal complexity). *For every invertible real-linear map A,*

$$\nu_{\mathrm{poly}}(ATA^{-1}) = \nu_{\mathrm{poly}}(T).$$

Consequently N-criticality is invariant under conjugacy, and $\nu_{\mathrm{poly}}(tT^*) = \nu_{\mathrm{poly}}(tT)$ for every $t > 0$ in the adapted inner product.

Proof. The correspondence $P \mapsto AP$ preserves convexity, nondegeneracy, the number of extreme points, and inclusion. Moreover

$$TP \subseteq P \iff (ATA^{-1})(AP) \subseteq AP.$$

The first assertion follows. The second uses the reflection conjugacy in proposition 2.1. □

Proposition 2.3 (Real-linear covariance of contact geometry). *Let $A : V \to V$ be invertible, put*

$$\widetilde{T} = ATA^{-1}, \qquad \widetilde{P} = AP,$$

and suppose that P is a strict polygon. Then the following statements hold.

(i) *A maps $\operatorname{Ext} P$ bijectively onto $\operatorname{Ext} \widetilde{P}$, maps every side of P onto a side of $\widetilde{P}$, and preserves relative interiors and proper inclusions.*

(ii) *For every face F of P, the image AF is the corresponding face of $\widetilde{P}$. In particular, side contacts, vertex contacts, strict support contacts, and the assertions "every side is touched" and "every image vertex lies on the boundary" are transported exactly by A.*

(iii) *If $TP \subseteq P$, then $\widetilde{T}\widetilde{P} \subseteq \widetilde{P}$. If a polygon P' is obtained from P by clipping with a closed half-plane, replacing a vertex by a contact point, or performing a finite sequence of such operations, then AP' is obtained from AP by the corresponding transformed operations. All vertex counts, incidence equalities, strictness conditions, and invariance statements are equivalent before and after the transformation.*

(iv) *Give $\widetilde{P}$ the boundary orientation transported by A. If $\chi : \operatorname{Ext} P \to \mathcal{E}(P)$ is a one-sided contact representation, then*

$$\widetilde{\chi}(Av) = A\chi(v)$$

is a one-sided contact representation for $\widetilde{T}\widetilde{P} \subseteq \widetilde{P}$. If $A_{\mathcal{E}} : \mathcal{E}(P) \to \mathcal{E}(\widetilde{P})$ is the induced side bijection, then

$$\widetilde{I} = A_{\mathcal{E}}(I), \qquad \widetilde{s} = A_{\mathcal{E}} s A_{\mathcal{E}}^{-1}, \qquad \widetilde{\sigma} = A_{\mathcal{E}} \sigma A_{\mathcal{E}}^{-1}. \tag{2.3}$$

Consequently legal mutations, mutation sequences, strict-contact intervals, orbit counts, first-return maps, return times, and tower partitions are carried bijectively by $A_{\mathcal{E}}$.

If A reverses the originally displayed ambient orientation, then the transported orientation on the target is the opposite displayed orientation. Re-expressing the target in the original displayed orientation reverses cyclic order and may interchange right- and left-half-open ownership; this separate handedness issue is recorded in lemma 2.4.

Proof. An invertible real-linear map is a homeomorphism that maps line segments to line segments and preserves convex combinations. A point $x \in P$ is extreme precisely when an equality $x = tu + (1-t)v$ with $u, v \in P$ and $0 < t < 1$ forces $u = v = x$; applying A and A^{-1} proves the first assertion about extreme points. Maximal boundary segments, relative interiors, and proper inclusions are therefore preserved as well.

If a nonzero functional ℓ supports P on the face $F = \{x \in P : \ell(x) = \max_P \ell\}$, then $\widetilde{\ell} = \ell \circ A^{-1}$ supports AP on

$$AF = \{y \in AP : \widetilde{\ell}(y) = \max_{AP} \widetilde{\ell}\}.$$

Thus all face and strict-support assertions are equivalent. The identity

$$\widetilde{T}(AP) = A(TP)$$

proves the invariance and image-contact claims. Finally, $A(P \cap H) = AP \cap AH$ for every half-plane H, and replacing a point x by a point c in a displayed convex hull commutes with applying A. Iteration proves the statement for finite surgery sequences.

With the transported orientation, A preserves tail, head, cyclic successor, and every right-half-open side:

$$A\big((t(e), h(e)]\big) = (At(e), Ah(e)].$$

Thus $\widetilde{\chi}$ has the required ownership property, and the strict set and contact rotation satisfy (2.3). The condition for a legal mutation is expressed only through membership in I and the successor map, so it is preserved. Conjugacy of first-return maps and equality of all return times then follow by induction on the first-return time; the tower partition is the image of the original partition. The final orientation statement follows because the sign of every oriented area is multiplied by $\det A$. □

Whenever an adapted complex coordinate is fixed, we identify V with $\mathbb{C}$ and use

$$\langle z, w\rangle = \mathrm{Re}(\overline{z}w), \qquad \det(z, w) = \mathrm{Im}(\overline{z}w). \tag{2.4}$$

The determinant is positive on positively oriented ordered bases. For a nonempty compact convex set K, its support function is

$$h_K(u) = \max_{z \in K} \langle u, z\rangle. \tag{2.5}$$

Until the one-sided selection is made, we use the J_+ coordinate and write

$$Tz = \lambda z, \qquad \lambda = \rho e^{i\theta}, \qquad 0 < \rho < 1, \qquad 0 < \theta < \pi.$$

The opposite orientation labels the same real map by $\overline{\lambda} = \rho e^{-i\theta}$, whose positive lifted argument is $2\pi - \theta$. No geometric property of the real-linear dynamics depends on this choice.

Lemma 2.4 (Coordinate reversal and handedness)**.** *Let $C : \mathbb{C} \to \mathbb{C}$ be complex conjugation. For every $\lambda \in \mathbb{C} \setminus \mathbb{R}$ and every compact convex polygon P,*

$$C(\lambda P) = \overline{\lambda}\, C(P), \qquad \lambda P \subseteq P \Longleftrightarrow \overline{\lambda}\, C(P) \subseteq C(P). \tag{2.6}$$

Conjugation preserves the number of extreme points, strictness, relative interiors, supporting incidences, side touching, image-vertex touching, and Hausdorff-continuous deformation families.

If x_i is a positively indexed boundary list for P and

$$\widetilde{x}_i = \overline{x}_{-i},$$

then $\widetilde{x}_i$ is a positively indexed boundary list for $C(P)$. However, conjugation reverses the half-open convention: the image of the right-half-open side $(x_{j-1}, x_j]$ is the left-half-open side

$$[\widetilde{x}_{-j}, \widetilde{x}_{1-j}). \tag{2.7}$$

Consequently a right-admissible contact labeling is not, in general, carried to a right-admissible labeling by conjugation when endpoint contacts are present.

Proof. The identity

$$C(\lambda z) = \overline{\lambda}\, C(z)$$

gives (2.6). Since C is an invertible real-affine map, it preserves convex combinations, extreme points, faces, relative interiors, and inclusions. A line supports P if and only if its conjugate supports $C(P)$, with the conjugate contact face; this proves all incidence and saturation assertions. Applying C pointwise to a deformation family preserves Hausdorff convergence because C is an isometry in the adapted metric.

Complex conjugation reverses the cyclic order on the boundary. Hence $\widetilde{x}_i = \overline{x}_{-i}$ restores positive order. The side of $C(P)$ from $\widetilde{x}_{-j} = \overline{x}_j$ to $\widetilde{x}_{1-j} = \overline{x}_{j-1}$ is the conjugate of the unoriented segment $[x_{j-1}, x_j]$. The old half-open set excludes $\overline{x}_{j-1} = \widetilde{x}_{1-j}$ and includes $\overline{x}_j = \widetilde{x}_{-j}$, which is exactly (2.7). Thus head and tail ownership are interchanged, proving the final assertion. □

The distinction in the last sentence is used systematically. When the least-area construction initially requires conjugation, the right-handed contact system is reconstructed by lemmas 4.11 and 4.13; it is not obtained by silently conjugating an existing right-handed labeling. After mutation reduction and no-skipping, a second orientation reversal, when needed, is applied only to the finite return strip by lemma 8.3. No saturation, mutation, or projective-holonomy assertion is invoked after that final strip reflection.

Lemma 2.5 (Interior origin for a nonreal contraction)**.** *Let P be a non-singleton compact convex polygon and let $\lambda = \rho e^{i\theta}$ with $0 < \rho < 1$ and $\theta \not\equiv 0, \pi \pmod{2\pi}$. If $\lambda P \subseteq P$, then P has nonempty interior and $0 \in \operatorname{int} P$.*

Proof. Iteration gives $0 \in P$. If P were a segment, its affine line would pass through 0 and would have to contain the nondegenerate segment λP; hence multiplication by $e^{i\theta}$ would preserve a real line, forcing $\theta \equiv 0$ or π, a contradiction. Thus P has nonempty interior.

Suppose $0 \in \partial P$. Choose a nonzero real linear functional ℓ such that $\ell \geq 0$ on P and choose $z \in P$ with $\ell(z) > 0$. Invariance would give

$$\ell(e^{ik\theta} z) \geq 0 \qquad (k \geq 0).$$

If $e^{i\theta}$ has finite order m, then $m \geq 3$ and $\sum_{k=0}^{m-1} e^{ik\theta} z = 0$; summing the displayed nonnegative quantities forces them all to vanish, contrary to $\ell(z) > 0$. If $e^{i\theta}$ has infinite order, its powers are dense in the unit circle, so some $e^{ik\theta} z$ lies in the open half-plane where $\ell < 0$, again a contradiction. Hence 0 is interior. □

Lemma 2.6 (Oriented order on a convex boundary)**.** *Let $K \subset \mathbb{R}^2$ be a compact convex set with nonempty interior, with ∂K given its positive orientation. For three distinct noncollinear points $a, b, c \in \partial K$,*

$$\det(b - a, c - a) > 0 \tag{2.8}$$

if and only if a, b, c occur in that positive cyclic order on ∂K. Consequently, if Q is a strict convex polygon all of whose vertices lie on ∂K, then the positive cyclic order of the vertices of Q is exactly the order induced from ∂K.

Proof. Choose $o \in \operatorname{int} \operatorname{conv}\{a, b, c\}$. Since the triangle $\operatorname{conv}\{a, b, c\}$ is contained in K, one has $o \in \operatorname{int} K$. Every ray from o meets ∂K in exactly one point, so the radial map

$$\pi_o : \partial K \longrightarrow \mathbb{S}^1, \qquad \pi_o(z) = \frac{z - o}{|z - o|},$$

is a homeomorphism. With the positive boundary orientation it has degree $+1$, and hence preserves cyclic order. Because o lies in the interior of the triangle abc, the three rays from o to a, b, c occur counterclockwise in the order a, b, c exactly when the oriented area of that triangle is positive, which is (2.8).

For the final assertion, apply the first part both to K and to Q. For every triple of vertices of Q, the sign of the same determinant characterizes its positive cyclic order on ∂K and on ∂Q. Thus the two ternary cyclic-order relations coincide, and therefore so do the full cyclic orders. ☐

Lemma 2.7 (Triple-sign criterion for strict convex position)**.** *Let $z_0, \ldots, z_{m-1}$ be a cyclic list of points, $m \geq 3$. The list is exactly the cyclic vertex list of a strict convex m-gon, in the displayed orientation, if and only if there is a sign $\varepsilon \in \{1, -1\}$ such that*

$$\varepsilon \det(z_j - z_i, z_k - z_i) > 0 \tag{2.9}$$

for every triple of distinct indices i, j, k occurring in that cyclic order. The displayed orientation is positive precisely when $\varepsilon = 1$.

Proof. If the list is the cyclic vertex list of a strict polygon, apply lemma 2.6 in the displayed orientation; reversing orientation changes all determinant signs simultaneously.

Conversely, first take $\varepsilon = 1$. The inequalities force the points to be distinct and no three of them to be collinear. Fix i. For every $k \notin \{i, i+1\}$, the indices $i, i+1, k$ occur in positive cyclic order, so

$$\det(z_{i+1} - z_i, z_k - z_i) > 0.$$

Thus every other point lies strictly in the left half-plane of the directed line from z_i to z_{i+1}. Hence $[z_i, z_{i+1}]$ is an exposed edge of $\operatorname{conv}\{z_0, \ldots, z_{m-1}\}$. This holds for every i, so all displayed points are extreme, the displayed consecutive segments are exactly the boundary edges, and the resulting polygon is strict and positively oriented. The case $\varepsilon = -1$ is identical with left replaced by right, or follows by reversing the displayed orientation. ☐

Lemma 2.8 (Simultaneous convex admissibility under perturbation)**.** *Let $I_0 \subset \mathbb{R}$ be an open interval containing 0, and let*

$$z_0(\tau), \ldots, z_{N-1}(\tau) \qquad (\tau \in I_0)$$

be continuous point functions such that $z_0(0), \ldots, z_{N-1}(0)$ are the positively ordered vertices of a strict N-gon. Let $\mathcal{A}$ and $\mathcal{B}$ be finite sets. For every $a \in \mathcal{A}$, fix a side label $k(a)$ and a continuous point $w_a(\tau)$ such that, throughout I_0,

$$w_a(\tau) \in \operatorname{aff}\big(z_{k(a)-1}(\tau), z_{k(a)}(\tau)\big), \qquad w_a(0) \in \operatorname{relint}[z_{k(a)-1}(0), z_{k(a)}(0)].$$

For every $b \in \mathcal{B}$, fix a side label $k(b)$ and a continuous point $u_b(\tau)$ such that

$$\det\big(z_{k(b)}(0) - z_{k(b)-1}(0), u_b(0) - z_{k(b)-1}(0)\big) > 0. \tag{2.10}$$

Then there is an open interval $I \subseteq I_0$ containing 0 on which all of the following hold simultaneously:

(i) *the list $z_0(\tau), \ldots, z_{N-1}(\tau)$ consists of distinct extreme points in the same positive cyclic order and therefore defines a strict N-gon P_τ;*

(ii) *$w_a(\tau) \in \operatorname{relint}[z_{k(a)-1}(\tau), z_{k(a)}(\tau)]$ for every $a \in \mathcal{A}$;*

(iii) *the strict inequality obtained from (2.10) by replacing 0 with τ holds for every $b \in \mathcal{B}$.*

For each such P_τ, its interior is the intersection of the N open left half-planes determined by its positively oriented sides. In particular, a continuous test point satisfying all N strict side inequalities belongs to $\operatorname{int} P_\tau$.

At the unperturbed polygon, a point in the relative interior of one side satisfies the strict side inequality for every other side. Equivalently, if $w \in \operatorname{relint}[z_{c-1}(0), z_c(0)]$, then

$$\det\big(z_k(0) - z_{k-1}(0), w - z_{k-1}(0)\big) > 0 \qquad (k \neq c). \tag{2.11}$$

Proof. For a positively oriented strict polygon, the directed line of side $[z_{k-1}(0), z_k(0)]$ supports the polygon, the polygon lies in its closed left half-plane, and the contact face is exactly that side. If $w \in \operatorname{relint}[z_{c-1}(0), z_c(0)]$ and $k \neq c$, then w does not belong to the contact face of side k: distinct maximal sides of a strict polygon have disjoint relative interiors and adjacent sides meet only at their common endpoint. The supporting inequality is therefore strict, proving (2.11). The same support description shows that the polygon is the intersection of the closed left half-planes and its interior is the intersection of their open counterparts.

For every cyclically ordered triple of distinct labels, the corresponding oriented determinant is positive at $\tau = 0$. There are finitely many such determinants, so continuity gives a common neighborhood on which they all remain positive. The triple-sign criterion lemma 2.7 proves item (i).

Fix $a \in \mathcal{A}$. Choose a real linear functional f_a satisfying

$$f_a\big(z_{k(a)}(0) - z_{k(a)-1}(0)\big) \neq 0.$$

After another common shrinking, the corresponding denominator remains nonzero, and the collinearity hypothesis gives the unique scalar

$$\alpha_a(\tau) = \frac{f_a\big(w_a(\tau) - z_{k(a)-1}(\tau)\big)}{f_a\big(z_{k(a)}(\tau) - z_{k(a)-1}(\tau)\big)}$$

with

$$w_a(\tau) = (1 - \alpha_a(\tau))z_{k(a)-1}(\tau) + \alpha_a(\tau)z_{k(a)}(\tau).$$

This scalar is continuous and belongs to $(0,1)$ at 0; finiteness of $\mathcal{A}$ gives item (ii) on one common interval. Finally, the determinants in item (iii) are finitely many continuous functions that are positive at 0, so a final common shrinking preserves all their signs. The interior assertion follows from the open half-plane description proved in the first paragraph. $\square$

Lemma 2.9 (Support-face test for strict polygons)**.** *Let P be a strict polygon with cyclic vertices x_i, and let L be a supporting line of P through x_i. Then L is a strict supporting line at x_i if and only if neither adjacent vertex x_{i-1} nor x_{i+1} lies on L. Equivalently, once support is known, strictness is certified by the two nonzero determinants*

$$\det(v, x_{i-1} - x_i) \neq 0, \qquad \det(v, x_{i+1} - x_i) \neq 0,$$

where v is any nonzero direction vector of L.

Proof. The contact set $P \cap L$ is a face of the polygon. If it is more than the single point x_i, it contains a nondegenerate boundary segment having x_i as one endpoint. In a strict polygon the maximal nondegenerate faces are exactly the displayed sides, so this segment lies in either $[x_{i-1}, x_i]$ or $[x_i, x_{i+1}]$, and the corresponding adjacent vertex is on L. Conversely, if an adjacent vertex lies on L, then the contact face contains the side joining it to x_i and the support is not strict. The determinant formulation is the line-membership test for the two adjacent vertices. □

Lemma 2.10 (Angular monotonicity along a strict polygon)**.** *Let P be a strict polygon with $0 \in \operatorname{int} P$, and list its vertices x_i in positive cyclic order. There are lifted arguments Θ_i with $\Theta_{i+N} = \Theta_i + 2\pi$ such that*

$$0 < \Theta_i - \Theta_{i-1} < \pi \qquad (i \in \mathbb{Z}). \tag{2.12}$$

Moreover the polar argument is strictly increasing along every positively oriented side $[x_{i-1}, x_i]$.

Proof. The interior of a positively oriented polygon lies strictly to the left of every oriented side. Applying this to the interior point 0 gives

$$\det(x_{i-1}, x_i) > 0.$$

Thus the positive angular increment from x_{i-1} to x_i belongs to $(0, \pi)$, which yields the lifted sequence and (2.12). For $z(t) = (1-t)x_{i-1} + tx_i$, $0 \le t \le 1$, the segment avoids 0 and

$$\frac{d}{dt} \operatorname{Arg} z(t) = \frac{\det(z(t), x_i - x_{i-1})}{|z(t)|^2} = \frac{\det(x_{i-1}, x_i)}{|z(t)|^2} > 0.$$

This proves the sidewise assertion. □

3 Hereditary saturation

Let P be a strict invariant polygon containing 0 in its interior. Write its outward unit normals in cyclic order as

$$u_i = e^{i\phi_i}, \qquad \phi_1 < \cdots < \phi_m < \phi_1 + 2\pi,$$

and let $h_i = h_P(u_i) > 0$ be its support numbers. For each i, express $e^{-i\theta} u_i$ in the cone generated by the two adjacent fan rays that contain it:

$$e^{-i\theta} u_i = a_i u_j + b_i u_{j+1}, \qquad a_i, b_i \ge 0. \tag{3.1}$$

The coefficients are unique, including the boundary case, and define a nonnegative matrix $B_\Phi(\theta)$.

Proposition 3.1 (Normal-fan transfer)**.** *With $h = (h_i)$,*

$$\lambda P \subseteq P \quad \Longleftrightarrow \quad \rho B_\Phi(\theta) h \le h. \tag{3.2}$$

Moreover, for $u = (u_i)^T$,

$$B_\Phi(\theta) u = e^{-i\theta} u. \tag{3.3}$$

The support vectors defining strict polygons with this fixed normal fan form an open subset of $\mathbb{R}^m$.

Proof. If $v = au_j + bu_{j+1}$ belongs to the cone generated by two adjacent normal rays, with $a, b \ge 0$, the two adjacent support inequalities are maximized at their common polygon vertex. For every $z \in P$ one therefore has

$$\langle v, z \rangle \le a h_P(u_j) + b h_P(u_{j+1}),$$

with equality at that common vertex. Hence $h_P(v) = ah_P(u_j) + bh_P(u_{j+1})$. Applying this to $v = e^{-i\theta}u_i$ gives

$$h_P(e^{-i\theta}u_i) = a_i h_j + b_i h_{j+1} = (B_\Phi h)_i.$$

The support function of λP in direction u_i is $\rho h_P(e^{-i\theta}u_i)$, proving the equivalence. The second identity is the vector decomposition itself.

For openness, the two boundary lines with normals u_i, u_{i+1} meet in a vertex depending linearly on (h_i, h_{i+1}). At a strict polygon each such vertex satisfies all nonincident side inequalities strictly, and all cyclic turn determinants have the convex nonzero sign. These are finitely many strict inequalities in continuous functions of h and therefore persist under a sufficiently small perturbation. □

Theorem 3.2 (Hereditary saturation theorem)**.** *Let T be N-critical and let R be a nondegenerate polygon with at most N vertices such that $TR \subseteq R$. Then R has exactly N vertices and is strict. Every side of R meets TR, and every vertex of TR lies on ∂R.*

Proof. By $\nu_{\text{poly}}(T) = N$, the polygon cannot have fewer than N vertices. Hence it has exactly N. By lemma 2.5, it has nonempty interior and contains 0 in its interior; listing its extreme points makes it a strict N-gon.

Write R in its fixed normal fan and put $B = B_\Phi(\theta)$. From (3.2) and $h > 0$, the weighted-norm bound in lemma A.1 gives $\operatorname{spr}(\rho B) \leq 1$. Suppose the inequality were strict. Then

$$g = (I - \rho B)^{-1}\mathbf{1} = \sum_{k \geq 0} (\rho B)^k \mathbf{1} > 0$$

satisfies $g - \rho B g = \mathbf{1}$. For sufficiently small $s > 0$, the support vector $h_s = (1-s)h + sg$ defines a strict polygon with the same fan, and

$$h_s - \rho B h_s = (1-s)(h - \rho B h) + s\mathbf{1} > 0.$$

The strict inequality persists after replacing ρ by $(1+\eta)\rho$ for some $\eta > 0$, producing an invariant N-gon for $(1+\eta)T$. This contradicts radial criticality. Therefore

$$\operatorname{spr}(\rho B) = 1. \tag{3.4}$$

The nonnegative Perron theorem in lemma A.1 supplies a nonzero $w \geq 0$ such that

$$B^T w = \rho^{-1} w. \tag{3.5}$$

Since $h - \rho B h \geq 0$,

$$w^T(h - \rho B h) = w^T h - \rho (B^T w)^T h = 0.$$

Every summand is nonnegative, so

$$w_i\big(h_i - \rho(Bh)_i\big) = 0 \qquad (1 \leq i \leq N). \tag{3.6}$$

We prove that every coordinate of w is positive. Let $S = \{i : w_i > 0\}$. Pairing (3.5) with (3.3) gives

$$w^T u = 0,$$

because $\rho^{-1} \neq e^{-i\theta}$. Because every coefficient w_i with $i \in S$ is positive, this relation has the following dichotomy: either the normals indexed by S are contained in one line, or their positive cone is all of $\mathbb{R}^2$. Indeed, if the positive cone were proper, a nonzero separating functional v would satisfy $\langle v, u_i \rangle \geq 0$ for every $i \in S$; pairing with $\sum_{i \in S} w_i u_i = 0$ would force every one of these scalar products to vanish, so all the normals would lie in $v^\perp$. We exclude this collinear alternative. If S is the full index set and the normals were contained in $\{v, -v\}$, then the intersection of the side half-planes would be a strip, a half-plane, or the whole plane, never the bounded polygon R. Thus collinearity is impossible when $S = \{1, \ldots, N\}$. Suppose instead that S is proper. If its normals were contained in $\{v, -v\}$, then for every $k \notin S$ the equality

$$0 = (B^T w)_k = \sum_{i \in S} B_{ik} w_i$$

would force $B_{ik} = 0$ for every $i \in S$, because all summands are nonnegative and every w_i with $i \in S$ is positive. Each row indexed by S would then express $(Bu)_i$ as a real multiple of v, whereas $(Bu)_i = e^{-i\theta} u_i$ is not collinear with v. This contradiction excludes the remaining case. The normals indexed by S are therefore not collinear, and the positive relation $w^T u = 0$ now implies that they positively span the plane.

Consider the retained half-plane intersection

$$R_S = \bigcap_{i \in S} \{z : \langle u_i, z \rangle \le h_i\}.$$

Positive spanning makes its recession cone trivial, so R_S is bounded. It is an intersection of finitely many closed half-planes and is therefore closed and compact. Because every h_i is positive, a sufficiently small disk about 0 satisfies all retained inequalities strictly; hence $0 \in \operatorname{int} R_S$, so R_S is nondegenerate. After redundant inequalities are deleted, each remaining boundary line carries one maximal side. Thus R_S has at most $|S|$ sides and, being a nondegenerate planar polygon, at most $|S|$ extreme points. For $i \in S$, the same zero-column argument shows that row i of B uses only columns in S. Thus, for $z \in R_S$,

$$\langle u_i, \lambda z \rangle \le \rho \sum_{j \in S} B_{ij} h_j = \rho (Bh)_i = h_i,$$

where the last equality is complementarity. Hence $\lambda R_S \subseteq R_S$. If S were proper, this would contradict $\nu_{\text{poly}}(T) = N$. Consequently $w_i > 0$ for every i, and (3.6) gives

$$h = \rho B h. \tag{3.7}$$

For each side, equality of support values means that the side meets TR. This proves full side touching.

It remains to prove full vertex touching. The polar polygon

$$R^\circ = \{y : \langle y, z \rangle \le 1 \text{ for all } z \in R\}$$

is a strict N-gon; its vertices correspond to the maximal sides of R and its maximal sides correspond to the vertices of R. Polarity and the real adjoint give

$$TR \subseteq R \quad \Longrightarrow \quad T^* R^\circ \subseteq R^\circ.$$

By proposition 2.2, T^* is also N-critical. The side argument just proved therefore applies to R°: every side of R° meets $T^* R^\circ$.

Fix a vertex x of R and its dual side

$$F_x = \{y \in R^\circ : \langle y, x \rangle = 1\}.$$

Choose $y \in R^\circ$ with $T^* y \in F_x$. Then

$$1 = \langle T^* y, x \rangle = \langle y, Tx \rangle.$$

In particular, $y \neq 0$. Since $y \in R^\circ$ and $Tx \in R$, equality for the nonzero supporting functional y is possible only on the boundary. Thus $Tx \in \partial R$. Invertibility of T identifies the vertices of R with the vertices of TR, completing the proof. □

Remark 3.3. The theorem applies to every replacement polygon produced later. This is the precise hereditary statement required by the no-skipping contradiction: once a deformation has independently been shown to remain a strict invariant N-gon, every image vertex must again lie on its boundary.

4 One-sided contact selection

Write the vertices of a strict polygon P in positive cyclic order as x_i ($i \in \mathbb{Z}/N\mathbb{Z}$), and put

$$E_i = [x_{i-1}, x_i], \qquad E_i^+ = (x_{i-1}, x_i], \qquad E_i^- = [x_{i-1}, x_i).$$

All cyclic intervals below are taken in this fixed orientation. We shall repeatedly use the following elementary consequence of full side touching.

Lemma 4.1 (Side contact is witnessed by an image vertex)**.** *Let $Q \subseteq P$ be polygons. If a side E of P meets Q, then E contains a vertex of Q.*

Proof. Let ℓ be a supporting functional of P whose maximum face is E. If $E \cap Q \neq \varnothing$, then the maximum of ℓ on Q equals its maximum on P. A linear functional on a polygon attains its maximum at a vertex, so a vertex of Q belongs to E. □

4.1 Symbolic endpoint ownership

The following finite model fixes the endpoint convention used in every cap and surgery argument. It is not an additional assumption; it is just the right-half-open side convention written as a cyclic word.

Definition 4.2 (Half-open ownership word)**.** For a positively oriented strict polygon, field i *owns* the half-open side

$$E_i^+ = (x_{i-1}, x_i].$$

Thus a polygon vertex x_i is owned by the incoming field i, not by the outgoing field $i+1$. In a one-sided contact system, the image vertex ξ_i is owned by field i:

$$\xi_i \in E_i^+.$$

Equivalently the local boundary word on E_i is exactly one of

$$x_{i-1} < \xi_i < x_i \quad \text{or} \quad x_{i-1} < x_i = \xi_i,$$

where $<$ denotes positive boundary order. For two consecutive image vertices y_j, y_{j+1}, the cap gap is always measured by the same ownership rule: the positive half-open gap $(y_j, y_{j+1}]_{\partial P}$ owns its right endpoint and excludes its left endpoint.

Lemma 4.3 (Half-open side atlas)**.** *Let P be a positively oriented strict N-gon with vertices x_i and oriented sides $E_i = [x_{i-1}, x_i]$. For*

$$D_i(z) = \det(x_i - x_{i-1}, z - x_{i-1}),$$

the following finite atlas holds.

(i) *The half-open sides $E_i^+ = (x_{i-1}, x_i]$ are pairwise disjoint and form a partition of ∂P.*

(ii) *If $z \in E_i^+$, then the zero-side set*

$$Z(z) = \{r : D_r(z) = 0\}$$

is

$$Z(z) = \{i\} \quad \textit{for } z \in \operatorname{relint} E_i, \qquad Z(x_i) = \{i, i+1\}.$$

(iii) *If $z \in P$ and $D_i(z) = 0$, then $z \in E_i$.*

(iv) *If $z \in E_i^+$ and $z \in E_j^+$, then $i = j$.*

Proof. The positively oriented side inequalities are exactly $D_i(z) \geq 0$. Their zero sets cut out the boundary faces. Since the polygon is strict, the nondegenerate boundary faces are precisely the displayed sides and the only sides incident with x_i are E_i and E_{i+1}. Thus a boundary point lies either in one relative side interior or at one displayed vertex, giving the zero-side list in (ii) and the implication in (iii). The half-open convention assigns every relative side interior to its side and assigns the vertex x_i only to the incoming side E_i^+, proving the partition and disjointness statements. $\square$

Lemma 4.4 (Boundary-face rigidity)**.** *Let R be a convex polygon with nonempty interior, let $A, B \in R$ with $A \neq B$, and let $0 < s < 1$. If*

$$(1-s)A + sB \in \partial R,$$

then A and B lie on one common side of R, and the whole segment $[A, B]$ is contained in that side.

Proof. Choose a supporting affine functional $f \leq c$ for R at $z = (1-s)A + sB$. Since

$$c = f(z) = (1-s)f(A) + sf(B), \qquad f(A), f(B) \leq c,$$

positivity of both coefficients forces $f(A) = f(B) = c$. Thus A and B belong to the exposed face $R \cap \{f = c\}$. This face is nondegenerate because $A \neq B$, so it is a side of the polygon and contains $[A, B]$. $\square$

Lemma 4.5 (Boundary segment locator)**.** *Let P be a positively oriented strict polygon. Suppose*

$$A \in E_{j-1}^+, \qquad B \in E_j^+, \qquad A \neq B,$$

and suppose that a strict convex combination $(1-s)A + sB$, $0 < s < 1$, lies on ∂P. Then

$$A = x_{j-1}, \qquad [A, B] \subseteq E_j.$$

In particular there is a unique $t \in (0, 1]$ such that

$$B = (1-t)x_{j-1} + tx_j,$$

and every strict convex combination of A and B lies in $\operatorname{relint} E_j$*.*

Proof. By lemma 4.4, the two endpoints A, B lie on one common side of P. Hence $Z(A) \cap Z(B)$ is nonempty. The half-open side atlas gives

$$Z(A) \subseteq \{j-1, j\}, \qquad Z(B) \subseteq \{j, j+1\},$$

and the common index can only be j. The index j belongs to $Z(A)$ only when A is the right endpoint of E_{j-1}^+, namely $A = x_{j-1}$. The common side is therefore E_j, so $B \in E_j$ as well. Since $B \in E_j^+$ and $B \neq A = x_{j-1}$, the displayed representation with $t \in (0, 1]$ is unique. Strict convex combinations of A and B omit both endpoints unless $B = A$, which has been excluded; hence they lie in $\operatorname{relint} E_j$. $\square$

Lemma 4.6 (Labeled side-matrix certificate)**.** *Let P be a positively oriented strict N-gon with displayed vertices x_i and half-open sides E_i^+. Let η_j be N points such that, for every j, either*

$$\eta_j = x_j,$$

or

$$\eta_j = (1-t_j)x_{j-1} + t_j x_j, \qquad 0 < t_j < 1.$$

Put

$$C_{r,k} = \det(x_r - x_{r-1}, x_k - x_{r-1}), \qquad C_{r,r-1} = C_{r,r} = 0,$$

and

$$D_{r,j} = \det(x_r - x_{r-1}, \eta_j - x_{r-1}).$$

Then $\eta_j \in E_j^+$ *for every* j *and*

$$D_{r,j} \geq 0 \qquad (0 \leq r, j < N).$$

More precisely, if $\eta_j = x_j$*, then* $D_{r,j} = C_{r,j}$*; if* $\eta_j = (1-t_j)x_{j-1} + t_j x_j$*, then*

$$D_{r,j} = (1-t_j)C_{r,j-1} + t_j C_{r,j}.$$

In the second case $D_{j,j} = 0$ *and* $D_{r,j} > 0$ *for every* $r \neq j$*. Thus the side ownership is collision-free by the half-open partition, and* $\operatorname{conv}\{\eta_j\} \subseteq P$*.*

Proof. The endpoint x_j belongs to E_j^+ and to no other half-open side, while a strict barycentric point of $[x_{j-1}, x_j]$ belongs to $\operatorname{relint} E_j \subset E_j^+$. This proves the asserted ownership statements. The determinant identities follow by bilinearity. Strict convexity gives $C_{r,k} > 0$ for every nonincident vertex x_k and $C_{r,r-1} = C_{r,r} = 0$ for incident vertices. Hence all entries of the displayed matrix are nonnegative, with strict positivity in the stated nonincident cases. The inequalities $D_{r,j} \geq 0$ are exactly the oriented side inequalities of P, so every η_j belongs to P and therefore their convex hull belongs to P. □

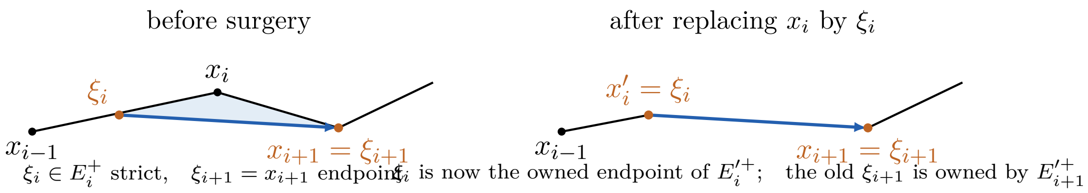

Figure 1: Half-open ownership for the unchanged image vertices in the elementary surgery pattern. The shaded cap is the only discarded part of P; the blue edge is an actual edge of $Q = \lambda P$. When $\kappa = 1$, field $i+1$ has a changed source and is treated separately in proposition 5.1.

Lemma 4.7 (Local ownership of unchanged image vertices)**.** *Suppose* $Q = \lambda P \subseteq P$*, the image vertices* ξ_i, ξ_{i+1} *are consecutive on* Q*,* $\xi_i \in \operatorname{relint} E_i$*, and* $\xi_{i+1} = x_{i+1}$*. After replacing* x_i *by* $x'_i = \xi_i$*, the local ownership word becomes*

$$x_{i-1} < x'_i = \xi_i < x_{i+1} = \xi_{i+1}.$$

Thus contact i *becomes an endpoint contact owned by* $E_i'^+$*, and the unchanged old image vertex* ξ_{i+1} *is owned by* $E_{i+1}'^+$*. In a shifted contact system, this says that contact* $i+1$ *remains an endpoint whenever its source is unchanged. If* $\kappa = 1$*, its source is the replaced vertex, so its new image is* $\lambda\xi_i$ *and its status is not asserted here; it is the changed contact treated in proposition 5.1.*

Proof. Before the replacement the local word is

$$x_{i-1} < \xi_i < x_i < x_{i+1} = \xi_{i+1}.$$

The clipping edge is $[\xi_i, \xi_{i+1}]$, so the only displayed vertex in the discarded open cap is x_i. The new sides adjacent to x'_i are $E'_i = [x_{i-1}, \xi_i]$ and $E'_{i+1} = [\xi_i, x_{i+1}]$. Since the right endpoint is included in $E_i'^+$ and the left endpoint is excluded from $E_{i+1}'^+$, the point ξ_i is owned only by field i. The point $x_{i+1} = \xi_{i+1}$ is the right endpoint of $E_{i+1}'^+$; the next half-open side excludes that same point as its left endpoint. Hence it is owned only by field $i+1$. These are exactly the two ownership statements claimed. The final distinction follows because the source of field $i+1$ is $x_{i+1-\kappa}$, which equals the replaced vertex precisely when $\kappa = 1$. $\square$

4.2 Edge caps and one-sided interlacing

The cap argument will only cut along an actual edge of $Q = \lambda P$. This removes all ambiguities associated with coincident cap endpoints, shared boundary segments, or a chord that happens to be supporting.

Lemma 4.8 (Edge-cap clipping with exact vertex count)**.** *Let P be a strict N-gon, let $\lambda \neq 0$, and put $Q = \lambda P$. Assume $Q \subseteq P$ and that every vertex of Q lies on ∂P. List the vertices of Q in their positive cyclic order as $y_0, \ldots, y_{N-1}$. Their order on ∂P is the same cyclic order.*

For the edge $[y_j, y_{j+1}]$ of Q, let H_j be the closed half-plane bounded by its line and containing Q, and put

$$P_j = P \cap H_j. \tag{4.1}$$

If $P_j \neq P$, let A_j be the closed boundary arc of P cut off by the opposite open half-plane. It joins y_j to y_{j+1} and contains no other vertex of Q. If k_j is the number of vertices of P on A_j, with either endpoint counted when it is a vertex of P, then

$$Q \subseteq P_j \subsetneq P, \qquad \lambda P_j \subseteq P_j, \qquad |\operatorname{Ext} P_j| \leq N + 2 - k_j. \tag{4.2}$$

If $P_j = P$, the edge is called a shared-side edge*; its boundary gap contains at most one vertex of P after the left endpoint is excluded.*

Proof. Because multiplication by $\lambda \neq 0$ is an invertible orientation- preserving real-linear map, $Q = \lambda P$ is a strict polygon. All of its vertices lie on ∂P, so lemma 2.6 shows that the positive cyclic order on ∂Q is exactly the order induced from ∂P. Thus consecutive Q-vertices bound a unique P-boundary gap containing no other Q-vertex.

The line of a Q-edge supports Q, so $Q \subseteq H_j$. Hence $Q \subseteq P_j \subseteq P$. Since $P_j \subseteq P$,

$$\lambda P_j \subseteq \lambda P = Q \subseteq P_j.$$

If $P_j \neq P$, the edge line is not supporting for P. We first check that y_j and y_{j+1} are the two endpoints of P on that line. Let $I = P \cap \operatorname{aff}(y_j, y_{j+1})$. This is a compact segment containing $[y_j, y_{j+1}]$. Since P has points on both sides of the line exactly when the clip is nontrivial, no point of $\operatorname{relint} I$ can lie on ∂P: a small segment transverse to the line and contained in P on both sides would make such a point interior to P. Hence the boundary points y_j and y_{j+1} must be the endpoints of I. The portion of ∂P in the discarded half-plane is therefore the claimed arc A_j, and it cannot contain another Q-vertex because all of Q lies in H_j.

Clipping removes every P-vertex in the relative interior of A_j. Each endpoint y_j, y_{j+1} is retained, and it is a new candidate vertex precisely when it was not already a P-vertex. Thus the number of candidate vertices is

$$N - \#\{\text{internal } P\text{-vertices of } A_j\} + \#\{\text{endpoints not already } P\text{-vertices}\} = N + 2 - k_j.$$

Deleting collinear candidates can only lower the number of extreme vertices.

If $P_j = P$, the edge line supports P as well. Its intersection with a strict polygon is one side, so the subarc from y_j to y_{j+1} lies in a single side and contains at most one P-vertex after its left endpoint is excluded. $\square$

Lemma 4.9 (Area-minimal cap bound)**.** *Among the strict invariant N-gons for the N-critical multiplier λ, impose the circumradius normalization*

$$\max_{z\in P}|z| = 1 \tag{4.3}$$

and choose one having least area within that normalized class. Let $Q = \lambda P$, and use the notation of lemma 4.8. Then every nontrivial edge cap satisfies

$$k_j \le 2, \tag{4.4}$$

and at most one nontrivial edge cap has $k_j = 2$.

Proof. The normalized class is compact in the Hausdorff metric. To see this without invoking a compactness theorem for polytopes, display every member as $\operatorname{conv}(z_1,\dots,z_N)$ with z_i in the closed unit disk, allowing repeated points. A subsequence of the N-tuples converges, and convex hull commutes with this finite limit; hence the Hausdorff limit is a convex polygon with at most N vertices. The conditions $\max|z| = 1$ and $\lambda P \subseteq P$ are closed. The limit cannot be a singleton: if it were $\{v\}$, normalization would give $|v| = 1$, while closed invariance would give $\lambda v = v$, impossible because $0 < |\lambda| < 1$. It cannot be a nondegenerate segment either, because multiplication by a nonreal number maps its supporting line to a different line. Thus every limit is a non-singleton polygon of positive area. If it had fewer than N extreme points, the resulting invariant polygon would have fewer than N vertices, contrary to $\nu_{\mathrm{poly}}(T) = N$. Hence every limit is again a strict invariant N-gon of positive area. Area is continuous here by lemma A.4, so a least-area normalized representative exists.

Choose a vertex v of P with $|v| = 1$. Since $Q \subseteq \rho\overline{\mathbb{D}}$ and $\rho < 1$, the point v is not a vertex of Q. If a nontrivial cap had $k_j \ge 3$, then (4.2) would give an invariant polygon with at most $N-1$ vertices, contradicting $\nu_{\mathrm{poly}}(T) = N$.

Suppose a nontrivial cap has $k_j = 2$ and its relative interior does not contain v. The clipped polygon P_j is a proper subset of P, contains v, remains normalized, and has at most N vertices. It cannot have fewer than N vertices, so it is another strict invariant N-gon, but its area is strictly smaller. This contradicts the choice of P.

The relative interiors of the N boundary gaps between consecutive Q-vertices are disjoint and cover $\partial P \setminus \operatorname{Ext} Q$. Since v is not a Q-vertex, it belongs to exactly one such relative interior. Therefore at most one nontrivial $k_j = 2$ cap can escape the preceding clipping contradiction. □

Lemma 4.10 (Finite cyclic endpoint ledger)**.** *Let $(r_j)_{j\in\mathbb{Z}/N\mathbb{Z}}$ take values in $\{0,1,2\}$, satisfy $\sum_j r_j = N$, and contain at most one entry equal to 2. Let $(c_j)_{j\in\mathbb{Z}/N\mathbb{Z}}$ be a binary cyclic word. If an entry $r_s = 2$ occurs, assume*

$$c_s = 0, \qquad c_{s+1} = 1, \tag{4.5}$$

and assume that every transition $c_j = 1$, $c_{j+1} = 0$ can occur only at an index j for which $r_j = 0$. Then either $r_j = 1$ for every j, or there are unique indices s,t such that

$$r_s = 2, \qquad r_t = 0, \qquad r_j = 1 \quad (j \notin \{s,t\}),$$

and in the latter case

$$\ell_j := r_j + c_j - c_{j+1} = 1 \qquad (j \in \mathbb{Z}/N\mathbb{Z}). \tag{4.6}$$

Proof. If no entry is 2, then every $r_j \le 1$ and the sum of the N entries is N, so all entries equal 1. Suppose $r_s = 2$. Its surplus over 1 is one. Since no second entry can exceed 1, the equality $\sum_j r_j = N$ forces a unique zero $r_t = 0$ and makes every remaining entry equal to 1.

The transition $0 \to 1$ at s in (4.5) forces at least one transition $1 \to 0$ in the cyclic binary word. By hypothesis it can occur only at the unique zero index t. In a cyclic binary word the number of $0 \to 1$ transitions equals the number of $1 \to 0$ transitions, because

$$\sum_j (c_{j+1} - c_j) = 0.$$

Consequently the transitions at s and t are the unique transitions of their respective types. Hence $c_j = c_{j+1}$ for $j \notin \{s, t\}$, while

$$(c_s, c_{s+1}) = (0, 1), \qquad (c_t, c_{t+1}) = (1, 0).$$

Substitution gives $\ell_s = 2 + 0 - 1 = 1$, $\ell_t = 0 + 1 - 0 = 1$, and $\ell_j = 1$ at every remaining index. □

Lemma 4.11 (Cyclic interlacing with endpoint bookkeeping). *Let P be the least-area representative of lemma 4.9. Then one of the following two alternatives holds:*

(i) *every arc $(y_j, y_{j+1}]_{\partial P}$ contains exactly one vertex of P;*

(ii) *every arc $[y_j, y_{j+1})_{\partial P}$ contains exactly one vertex of P.*

In alternative (ii)*, every right-half-open side of P contains exactly one vertex of Q. In alternative* (i)*, the conjugate configuration*

$$\widetilde{P} = \overline{P}, \qquad \widetilde{Q} = \overline{Q} = \overline{\lambda}\,\widetilde{P}, \qquad \widetilde{x}_i = \overline{x}_{-i} \tag{4.7}$$

has the same property when its vertices are read in positive cyclic order. Thus a right-half-open convention is obtained without ever reading a convex polygon in the wrong orientation.

Proof. Put

$$c_j = \begin{cases} 1, & y_j \in \operatorname{Ext} P, \\ 0, & y_j \notin \operatorname{Ext} P, \end{cases} \qquad r_j = \#(\operatorname{Ext} P \cap (y_j, y_{j+1}]_{\partial P}).$$

The half-open gaps partition the N vertices of P, so

$$\sum_{j=0}^{N-1} r_j = N. \tag{4.8}$$

For a nontrivial cap, the closed-arc count is

$$k_j = r_j + c_j. \tag{4.9}$$

For a shared-side edge, lemma 4.8 gives $r_j \le 1$. Hence lemma 4.9 implies

$$0 \le r_j \le 2, \qquad \text{and at most one } r_j \text{ equals } 2. \tag{4.10}$$

Indeed, if $r_j = 2$, then $c_j = 0$ by (4.9), and the gap cannot lie in one side of a strict polygon; it is therefore the unique possible nontrivial $k_j = 2$ cap.

If no r_j equals 2, equations (4.8)–(4.10) force $r_j = 1$ for every j, which is alternative (i). Otherwise there are unique indices s, t with

$$r_s = 2, \qquad r_t = 0, \qquad r_j = 1 \quad (j \ne s, t). \tag{4.11}$$

We determine the endpoint statuses at the two exceptional gaps.

First, $c_s = 0$ by (4.9). Also $c_{s+1} = 1$. For if y_{s+1} were not a P-vertex, the two P-vertices in the open gap would be consecutive vertices of P, and their joining side would contain no

Q-vertex. This contradicts lemma 4.1 and full side touching. Thus the binary cyclic word (c_j) has a transition $0 \to 1$ at s.

Now consider a transition $c_j = 1$, $c_{j+1} = 0$ with $j \neq t$. Then $r_j = 1$, so (4.9) gives $k_j = 2$. This cap is nontrivial. Indeed, on a shared side whose positive gap starts at the P-vertex y_j and ends at a nonvertex y_{j+1}, the gap must stop before the next P-vertex, so it would have $r_j = 0$, contrary to $r_j = 1$. It would therefore be a second nontrivial $k = 2$ cap, impossible. Hence every transition $1 \to 0$ can occur only at the unique zero index t. The hypotheses of lemma 4.10 are now verified: (4.11) gives the unique two and zero, while the preceding paragraph gives the required $0 \to 1$ transition at s. Therefore the opposite half-open counts

$$\ell_j = \#(\operatorname{Ext} P \cap [y_j, y_{j+1})_{\partial P}) = r_j + c_j - c_{j+1}$$

all equal 1. This is alternative (ii).

It remains to pass from gaps between Q-vertices to sides between P-vertices. In alternative (ii), let x_j be the unique P-vertex in $[y_j, y_{j+1})$. Then the cyclic order is

$$x_{j-1} < y_j \leq x_j,$$

so $y_j \in (x_{j-1}, x_j] = E_j^+$. No other Q-vertex lies there.

In alternative (i), let x_j be the unique vertex in $(y_j, y_{j+1}]$. Then $y_{j+1} \in [x_j, x_{j+1})$. Apply complex conjugation and index the conjugate polygon positively by $\widetilde{x}_i = \overline{x}_{-i}$. The conjugate of $[x_j, x_{j+1})$ is, with its order reversed,

$$(\widetilde{x}_{-j-1}, \widetilde{x}_{-j}];$$

therefore it is a right-half-open side of $\widetilde{P}$. Conjugation is a bijection on vertices and takes Q to $\widetilde{Q} = \overline{\lambda}\,\widetilde{P}$, so every such side contains exactly one vertex of $\widetilde{Q}$. This proves the final assertion with all endpoint inclusions explicit. □

Corollary 4.12 (Collision-free global half-open ownership)**.** *In either orientation supplied by lemma 4.11, the owned contacts form a genuine labeled bijection*

$$\{\textit{vertices of } Q\} \longleftrightarrow \{E_i^+ : i \in \mathbb{Z}/N\mathbb{Z}\}.$$

Equivalently, no image vertex can be owned by two fields and no field can own two image vertices. After the possible conjugation in (4.7), the labels may be chosen so that the unique owned image in E_i^+ is denoted by ξ_i.

Proof. The half-open sides $E_i^+ = (x_{i-1}, x_i]$ are pairwise disjoint and their union is ∂P. In alternative (ii) of lemma 4.11, each interval $[y_j, y_{j+1})_{\partial P}$ contains exactly one vertex of P, and the proof there gives $y_j \in E_j^+$. Thus every side owns at least one image vertex. There are exactly N sides and exactly N image vertices, while the half-open partition forbids one image vertex from being owned twice; hence the ownership relation is a bijection. In alternative (i), the same argument applies after the conjugate reindexing (4.7). This proves the stated collision-free labeling. □

The next step has two logically distinct parts. Half-open ownership and cyclic order make the vertex-to-side assignment an orientation-preserving degree-one correspondence of two cyclic N-point sets, hence a shift modulo N. Passing that correspondence to the universal cover of the circle introduces a possible extra full turn at each field. The interval estimate in the proof rules out those turns separately; summation then gives a strict rational bracket for the multiplier angle.

Lemma 4.13 (One-sided contact representative)**.** *After replacing (P,λ) by the conjugate configuration (4.7) if required, there is an invariant N-gon representative, a positive cyclic indexing, and an integer κ with $1 \le \kappa \le N$ such that*

$$\xi_i := \mu x_{i-\kappa} = \beta_i x_{i-1} + \alpha_i x_i, \qquad \alpha_i > 0, \quad \beta_i \ge 0, \quad \alpha_i + \beta_i = 1. \tag{4.12}$$

Here μ is either λ or $\overline{\lambda}$, indices are modulo N, and $\vartheta \in (0, 2\pi)$ denotes the positive lifted argument of μ. If $y = \vartheta/(2\pi)$, then

$$\frac{\kappa - 1}{N} < y < \frac{\kappa}{N}. \tag{4.13}$$

If $\kappa = N$, every contact is strict.

Proof. Choose the least-area normalized invariant N-gon from lemma 4.9. In alternative (ii) of lemma 4.11, put $\mu = \lambda$. In alternative (i), replace the configuration by (4.7) and put $\mu = \overline{\lambda}$. In either case the vertices are now in positive cyclic order and every E_i^+ contains exactly one vertex ξ_i of $Q = \mu P$.

Multiplication by the nonzero complex number μ preserves cyclic order. Hence the order-preserving incidence bijection between source vertices and right-half-open sides is a single cyclic shift. Indeed, after identifying both cyclic N-sets with $\mathbb{Z}/N\mathbb{Z}$, a cyclic-order-preserving bijection sends each successor to the successor of its image and therefore has the form $j \mapsto j+\kappa$. Choose its integer lift $\kappa \in \{1, \dots, N\}$ and write $\xi_i = \mu x_{i-\kappa}$. Because $\xi_i \in (x_{i-1}, x_i]$, it has the unique barycentric form (4.12) with $\alpha_i > 0$ and $\beta_i \ge 0$.

Choose lifted polar angles Θ_i with $\Theta_{i+N} = \Theta_i + 2\pi$, and choose the lifted multiplier argument $\vartheta \in (0, 2\pi)$. Since $0 \in \operatorname{int} P$, polar angle is strictly increasing along every positively oriented side. The side incidence first gives, for some integer m_i,

$$\Theta_{i-1} < \Theta_{i-\kappa} + \vartheta + 2\pi m_i \le \Theta_i. \tag{4.14}$$

The winding integer is necessarily zero. If $1 \le \kappa < N$, then

$$0 \le \Theta_{i-1} - \Theta_{i-\kappa} < \Theta_i - \Theta_{i-\kappa} < 2\pi,$$

because this interval spans κ consecutive angular gaps and omits at least one positive gap. If $\kappa = N$, then

$$\Theta_{i-1} - \Theta_{i-N} = 2\pi - (\Theta_i - \Theta_{i-1}), \qquad \Theta_i - \Theta_{i-N} = 2\pi.$$

Thus the possible difference interval in (4.14) is contained in $[0, 2\pi)$ in the first case and in $(0, 2\pi]$ in the second. Since $\vartheta \in (0, 2\pi)$, the only possible integer is $m_i = 0$. Hence (4.12) gives the genuine lifted inequality

$$\Theta_{i-1} < \Theta_{i-\kappa} + \vartheta \le \Theta_i.$$

Summing for one complete period, say over $i = 1, \dots, N$, and using $\sum_{i=1}^N \Theta_{i-\kappa} = \sum_{i=1}^N \Theta_i - 2\pi\kappa$ and $\sum_{i=1}^N \Theta_{i-1} = \sum_{i=1}^N \Theta_i - 2\pi$ yields

$$2\pi(\kappa - 1) < N\vartheta \le 2\pi\kappa.$$

The final equality cannot occur. Equality in the sum would force equality in every right-hand inequality, so every contact would be an endpoint and $\mu x_{i-\kappa} = x_i$ for all i. Iteration around a shift orbit gives $x_i = \mu^{N/\gcd(N,\kappa)} x_i$, impossible because $|\mu| < 1$ and no vertex is 0. This proves (4.13).

If $\kappa = N$, then the source of contact i is x_i itself. An endpoint contact would give $\mu x_i = x_i$, again impossible. Hence every contact is strict. ☐

From this point through the end of the contact-return analysis, fix the right-admissible contact system supplied by lemma 4.13 and denote its complex multiplier by λ. Put

$$\theta = \arg_+(\lambda) \in (0, 2\pi), \qquad x = \frac{\theta}{2\pi}.$$

This is the *contact eigenvalue*. The change from the symbol μ in lemma 4.13 to λ is only a local renaming of this already fixed right-admissible system; it is never undone. Its orientation, half-open ownership, contact rotation, mutation class, return section, and projective deformation are not changed anywhere in the mutation, finite-rotation, or no-skipping arguments. If the final monodromy strip is more naturally read in the opposite orientation, that reversal is performed only after those arguments have ended, by the explicit strip-reflection lemma lemma 8.3; the resulting output eigenvalue then receives the new symbol μ.

A tuple

$$(P, \lambda, \kappa; (\alpha_i, \beta_i)_{i\in\mathbb{Z}/N\mathbb{Z}})$$

satisfying (4.12), with P a strict invariant N-gon for an N-critical map and one image vertex in every E_i^+, will be called a *right-admissible contact system.* Thus admissibility includes all side and vertex touching conclusions proved above.

Lemma 4.14 (Exact lifted endpoint paths)**.** *In a right-admissible contact system, extend the vertex angles by $\Theta_{i+N} = \Theta_i + 2\pi$. For every integer field j,*

$$\Theta_{j-1} < \Theta_{j-\kappa} + \theta \leq \Theta_j. \tag{4.15}$$

If field j is an endpoint contact, equality holds on the right. Hence, if the destination fields $a+\kappa, \dots, a+t\kappa$ are all endpoint contacts, then

$$\Theta_a + t\theta = \Theta_{a+t\kappa}. \tag{4.16}$$

If the first $t-1$ destination fields are endpoint contacts and the field $a+t\kappa$ is strict, then instead

$$\Theta_{a+t\kappa-1} < \Theta_a + t\theta < \Theta_{a+t\kappa}. \tag{4.17}$$

Proof. The side incidence gives an integer m_j such that

$$\Theta_{j-1} < \Theta_{j-\kappa} + \theta + 2\pi m_j \leq \Theta_j.$$

Exactly as in the lift check in lemma 4.13, the interval of possible differences is contained in $[0, 2\pi)$ when $1 \leq \kappa < N$ and in $(0, 2\pi]$ when $\kappa = N$. Since the chosen multiplier argument satisfies $0 < \theta < 2\pi$, this forces $m_j = 0$ and proves (4.15). At an endpoint contact the image is x_j, forcing equality with Θ_j. Iteration gives (4.16); replacing the final endpoint by a strict contact gives (4.17). $\square$

For the mutation and return theory we now assume

$$1 \leq \kappa < N. \tag{4.18}$$

The identity contact rotation $\kappa = N$ is treated directly in lemma 8.4; it requires no mutation or projective deformation.

5 Contact mutations and interval reduction

Call contact i *strict* if $\beta_i > 0$ and an *endpoint contact* if $\beta_i = 0$.

Encode the strict fields by chips on $\mathbb{Z}/N\mathbb{Z}$. When field i is strict and $i+1$ is empty, clipping the corner x_i moves that chip from i to $i+\kappa$; a collision coalesces two strict-status occupancies. No geometric image vertices coalesce: the old target image is replaced when x_i is removed, and multiplication by $\lambda \neq 0$ keeps the N images of the N primed vertices distinct. The geometric content of the next lemma is that this combinatorial update preserves strict convexity, invariance, half-open ownership, and every unchanged contact. Consequently the later lexicographic reduction is an operation on actual invariant polygons, not merely on a binary word.

Shift boundary register. The mutation theorem is used only in the proper-shift range (4.18). Put $j_0 = i + \kappa$ in $\mathbb{Z}/N\mathbb{Z}$. The complete coincidence register is

$$j_0 = i+1 \iff \kappa = 1, \qquad j_0 - 1 = i+1 \iff \kappa = 2, \qquad j_0 = i-1 \iff \kappa = N-1.$$

The first equality is the only overlap between the changed source field and the changed target-side field. The second is the only case in which the preceding image A below is the unchanged endpoint x_{i+1} of a changed side. In the third case the changed image is assigned to the preceding field $i-1$, but neither of its target-side endpoints is the replaced vertex x_i; hence it belongs to the generic incidence row. For $N \geq 4$, the cases $\kappa = 1$, $\kappa = 2$, $\kappa = N-1$, and $3 \leq \kappa \leq N-2$ are disjoint and exhaustive, with the last two sharing the same algebraic formula. When $N = 3$, the cases $\kappa = 2$ and $\kappa = N-1$ coincide, and the $\kappa = 2$ row applies. All equalities and side labels in the certificate below are interpreted modulo N, so wrap-around creates no additional case.

Proposition 5.1 (Contact-surgery certificate: the exact local move)**.** *Let*

$$(P, \lambda, \kappa; (\alpha_j, \beta_j)_{j \in \mathbb{Z}/N\mathbb{Z}})$$

be a right-admissible contact system with $1 \leq \kappa < N$, and put

$$S = \{j : \beta_j > 0\}.$$

Suppose $i \in S$ and $i+1 \notin S$. Thus

$$\xi_i = \beta_i x_{i-1} + \alpha_i x_i \in \operatorname{relint} E_i, \qquad \xi_{i+1} = x_{i+1}.$$

Set

$$x'_i = \xi_i, \qquad x'_j = x_j \quad (j \neq i), \qquad P' = \operatorname{conv}\{x'_0, \ldots, x'_{N-1}\},$$

and put $\eta_j = \lambda x'_{j-\kappa}$. Then, with the uniquely induced coefficients (α'_j, β'_j), the primed tuple

$$(P', \lambda, \kappa; (\alpha'_j, \beta'_j)_{j \in \mathbb{Z}/N\mathbb{Z}})$$

is another right-admissible contact system. In particular, P' is a strict invariant N-gon, every η_j belongs to E'^{+}_j and the integer lift remains κ. More precisely, for some $t \in (0, 1]$ the new coefficients are

$$\alpha'_i = 1, \quad \beta'_i = 0, \qquad \alpha'_{i+\kappa} = \alpha_i t, \quad \beta'_{i+\kappa} = 1 - \alpha_i t, \tag{5.1}$$

and $(\alpha'_j, \beta'_j) = (\alpha_j, \beta_j)$ for $j \notin \{i, i+\kappa\}$. The new strict-contact set is

$$S' = (S \setminus \{i\}) \cup \{i + \kappa\}. \tag{5.2}$$

In particular, if $i + \kappa \in S$, the number of strict contacts decreases by one.

Proof. All indices are modulo N. Write

$$D(a,b,c) = \det(b-a, c-a).$$

For distinct vertices of a positively oriented strict convex polygon, $D(x_a, x_b, x_c) > 0$ whenever x_a, x_b, x_c occur in positive cyclic order.

Put $\alpha = \alpha_i$ and $\beta = \beta_i$. Strictness of contact i gives

$$\xi_i = \beta x_{i-1} + \alpha x_i, \qquad 0 < \alpha, \beta < 1, \qquad \alpha + \beta = 1.$$

Step 1: the replacement is exactly one edge-cap clip. The half-open ownership partition puts no image vertex in the boundary arc from ξ_i to $\xi_{i+1} = x_{i+1}$, apart from its endpoints. Since the cyclic order of the vertices of $Q = \lambda P$ agrees with their order on ∂P, the two points ξ_i and ξ_{i+1} are consecutive vertices of Q. Hence, for

$$H = \{z : D(\xi_i, x_{i+1}, z) \geq 0\},$$

one has $Q \subseteq H$. The sign of every displayed vertex of P relative to the cutting line is explicit:

$$D(\xi_i, x_{i+1}, x_i) = -\beta D(x_{i-1}, x_i, x_{i+1}) < 0, \tag{5.3}$$
$$D(\xi_i, x_{i+1}, x_k) = \beta D(x_{i-1}, x_{i+1}, x_k) + \alpha D(x_i, x_{i+1}, x_k) > 0 \tag{5.4}$$

for $k \notin \{i, i+1\}$, while $D(\xi_i, x_{i+1}, x_{i+1}) = 0$. In (5.4), convexity makes the first summand nonnegative and the second positive; this also covers $k = i-1$. Thus the cutting line meets ∂P exactly at ξ_i and x_{i+1}, and x_i is the only discarded displayed vertex. Consequently

$$P' = P \cap H = \operatorname{conv}(x_0, \dots, x_{i-1}, \xi_i, x_{i+1}, \dots, x_{N-1}). \tag{5.5}$$

Since $Q \subseteq P' \subseteq P$,

$$\lambda P' \subseteq \lambda P = Q \subseteq P'. \tag{5.6}$$

For completeness, the only changed consecutive-turn determinants in the displayed boundary list are

$$\begin{aligned}
D(x_{i-2}, x_{i-1}, \xi_i) &= \alpha D(x_{i-2}, x_{i-1}, x_i) > 0, \\
D(x_{i-1}, \xi_i, x_{i+1}) &= \alpha D(x_{i-1}, x_i, x_{i+1}) > 0, \\
D(\xi_i, x_{i+1}, x_{i+2}) &= \beta D(x_{i-1}, x_{i+1}, x_{i+2}) + \alpha D(x_i, x_{i+1}, x_{i+2}) > 0.
\end{aligned}$$

All other consecutive-turn determinants are unchanged. Together with the convex intersection description (5.5), these inequalities show that the displayed list consists of N distinct extreme points in positive cyclic order. Hence P' is a strict invariant N-gon.

The preceding calculation already proves that P' is a strict invariant N-gon. Because the multiplier is the same N-critical map, theorem 3.2 additionally gives full side and vertex touching for P'. Thus the only remaining right-admissibility item is the labeled right-half-open incidence.

Step 2: all unchanged sources and target sides are accounted for. Put $j_0 = i + \kappa$. Apart from the location of the changed image η_{j_0}, the complete incidence ledger is

field j	new image	target side	status
i	$\eta_i = \xi_i = x'_i$	$E'_i = [x_{i-1}, \xi_i]$	endpoint
$i+1$, $\kappa \neq 1$	$\eta_{i+1} = \xi_{i+1} = x_{i+1}$	$E'_{i+1} = [\xi_i, x_{i+1}]$	endpoint
j_0	$\eta_{j_0} = \lambda \xi_i$	E'_{j_0}	checked below
$j \notin \{i, i+1, j_0\}$	$\eta_j = \xi_j$	$E'_j = E_j$	unchanged

The rows are disjoint: if $\kappa = 1$, the second row is absent and $j_0 = i + 1$; otherwise $i, i+1, j_0$ are distinct. Thus only field j_0 remains to be located.

Step 3: the changed image has exactly one possible side. Set

$$A = \lambda x_{i-1} = \xi_{j_0-1}, \qquad B = \lambda x_i = \xi_{j_0}.$$

Then

$$\eta_{j_0} = \lambda \xi_i = \beta A + \alpha B \in \operatorname{relint}[A, B]. \tag{5.7}$$

The endpoint incidences after surgery are exhausted by

κ	A	B
1	$A = x'_i \in E'^+_i = E'^+_{j_0-1}$	$B = x_{i+1} \in E'^+_{i+1} = E'^+_{j_0}$
2	$A = x_{i+1} \in E'^+_{i+1} = E'^+_{j_0-1}$	$B = \xi_{i+2} \in E^+_{i+2} = E'^+_{j_0}$
$\kappa \notin \{1, 2\}$	$A = \xi_{j_0-1} \in E^+_{j_0-1} = E'^+_{j_0-1}$	$B = \xi_{j_0} \in E^+_{j_0} = E'^+_{j_0}$

Thus, uniformly in κ,

$$A \in E'^+_{j_0-1}, \qquad B \in E'^+_{j_0}. \tag{5.8}$$

The point x'_i is an extreme point of P', and multiplication by the nonzero scalar λ maps extreme points bijectively to extreme points of $\lambda P'$. Thus $\eta_{j_0} = \lambda x'_i$ is a vertex of $\lambda P'$. Full vertex touching gives $\eta_{j_0} \in \partial P'$. Since $\eta_{j_0} = \beta A + \alpha B$ with $A \neq B$ and $0 < \alpha, \beta < 1$, the hypotheses of lemma 4.5 apply to the primed polygon, using (5.8). Hence

$$A = x'_{j_0-1}, \qquad [A, B] \subseteq E'_{j_0},$$

Notice that when $\kappa \geq 2$, field $j_0 - 1 = i + \kappa - 1$ was already an endpoint contact before the surgery. For $\kappa = 2$, this is the legality hypothesis $i + 1 \notin S$. For $\kappa \geq 3$, the vertex in question is unchanged, and the identities

$$A = \xi_{j_0-1} = x'_{j_0-1} = x_{j_0-1}$$

imply $\beta_{j_0-1} = 0$. Thus the "status unchanged" entry for this field in Step 2 is consistent; the endpoint property is a consequence of the existing hypotheses, not an additional assumption.

Because $B \in E'^+_{j_0}$ and $B \neq A$, there is a unique $t \in (0, 1]$ such that

$$B = A + t(x'_{j_0} - A).$$

Equation (5.7) now yields

$$\eta_{j_0} = A + \alpha t(x'_{j_0} - A), \qquad 0 < \alpha t < 1.$$

Therefore $\eta_{j_0} \in \operatorname{relint} E'_{j_0}$; equivalently, its new contact coefficients are

$$\alpha'_{j_0} = \alpha_i t \in (0, 1), \qquad \beta'_{j_0} = 1 - \alpha_i t \in (0, 1).$$

Step 4: complete ownership and side-inequality audit. Thus field i changes from strict to endpoint, field $i + \kappa$ is strict after surgery, and every other field retains its status. This proves equations (5.1) and (5.2). We now make explicit why this is a global labeled contact system, not merely a local picture.

For the strict polygon P' put

$$C'_{r,k} = \det(x'_r - x'_{r-1}, x'_k - x'_{r-1})$$

with cyclic indices, and extend this notation by $C'_{r,r-1} = C'_{r,r} = 0$. Since P' is strict and positively oriented,

$$P' = \{z : \det(x'_r - x'_{r-1}, z - x'_{r-1}) \geq 0 \text{ for every } r\},$$

and the half-open sides E'^+_r form a disjoint partition of $\partial P'$. The incidence ledger gives the following exhaustive list.

(a) $\eta_i = x'_i \in E_i'^{+}$.
(b) If $\kappa \neq 1$, then $\eta_{i+1} = x'_{i+1} \in E_{i+1}'^{+}$.
(c) If $j \notin \{i, i+1, i+\kappa\}$, then the source and the target side are unchanged, so $\eta_j = \xi_j \in E_j^+ = E_j'^{+}$.
(d) For $j_0 = i+\kappa$, the preceding step proved $\eta_{j_0} \in \operatorname{relint} E'_{j_0}$.

These alternatives are disjoint and cover all $j \in \mathbb{Z}/N\mathbb{Z}$; when $\kappa = 1$ the second item is intentionally absent and the field $i+1$ is covered by item (d). Hence each image vertex is owned by its specified half-open side, and because the half-open partition is disjoint there is no possible global ownership collision.

Equivalently, lemma 4.6, applied to the primed vertices and the just-listed points η_j, gives the complete side matrix. In expanded form: if $\eta_j = x'_j$, then

$$\det(x'_r - x'_{r-1}, \eta_j - x'_{r-1}) = C'_{r,j} \geq 0;$$

if $\eta_j = (1-t_j)x'_{j-1} + t_j x'_j$ with $0 < t_j < 1$, then for every r

$$\det(x'_r - x'_{r-1}, \eta_j - x'_{r-1}) = (1-t_j)C'_{r,j-1} + t_j C'_{r,j} \geq 0.$$

The formula gives the exact zero on the assigned side and strict positivity away from incident sides. Thus the labeled incidences independently imply $\lambda P' \subseteq P'$; the earlier inclusion (5.6) was not being used to infer side ownership. Finally, the algebraic identities remain

$$\eta_j = \lambda x'_{j-\kappa} \qquad (j \in \mathbb{Z}/N\mathbb{Z}),$$

with the same integer $\kappa \in \{1, \ldots, N-1\}$. Since the target field of the source $j-\kappa$ is still exactly j, the integer lift κ, not only its residue class, is preserved. Together with the heredity conclusion above, this proves that the primed tuple is right-admissible. □

Corollary 5.2 (Intrinsic form of the exact mutation law)**.** *Let (P, T, χ) be the one-sided contact representation obtained in the contact orientation, and write its positive indexed form as in lemma 4.13. Thus*

$$E_i = [x_{i-1}, x_i], \qquad \chi(x_{i-\kappa}) = E_i.$$

Then

$$s(E_i) = E_{i+1}, \qquad \sigma(E_i) = E_{i+\kappa}. \tag{5.9}$$

A legal intrinsic mutation at $e = E_i$ is therefore exactly the indexed surgery condition $i \in S$, $i+1 \notin S$, and the surgery certificate gives

$$I' = (I \setminus \{e\}) \cup \{\sigma(e)\} \tag{5.10}$$

while preserving the same contact rotation σ.

Under an invertible real-linear conjugacy, with the transported boundary orientation, this statement is carried verbatim by the induced side bijection. In particular, the earlier choice between the two ambient orientations introduces no additional sign or index convention into (5.10)*.*

Proof. The successor identity in (5.9) is the definition of the positive side indexing. Since the head of E_i is x_i, its image under the intrinsic contact rotation is

$$\sigma(E_i) = \chi(h(E_i)) = \chi(x_i) = E_{i+\kappa},$$

because $x_i = x_{(i+\kappa)-\kappa}$. Strict fields are precisely the sides in I, and the intrinsic legality condition $s(E_i) \notin I$ is precisely $i+1 \notin S$. Equation (5.2) now becomes (5.10); preservation of the integer lift κ in proposition 5.1 preserves the same map σ. Real-linear covariance follows from (2.3) in proposition 2.3. The contact orientation is already the positively indexed orientation used above, including when it arose from the conjugate alternative in lemma 4.11; no subsequent orientation reversal occurs in the mutation module. □

Corollary 5.3 (Geometric realization of every legal chip sequence)**.** *Starting from a one-sided representative, every finite sequence of moves $i \mapsto i + \kappa$ for which field i is occupied and field $i + 1$ is empty in the current configuration is realized by successive right-admissible invariant strict N-gons with the same integer lift κ. The combinatorial strict-contact set after each step is exactly the set produced by the chip rule* (5.2)*.*

Proof. Apply proposition 5.1 inductively. Its conclusion includes preservation of the one-sided convention and of κ, so the hypotheses and the chip description remain valid at every later step. □

Corollary 5.4 (Boolean sweeps are geometrically reachable)**.** *Every Boolean chip sequence used in the group-sweep argument of lemma 5.5 is a sequence of actual polygon surgeries. More precisely, if a finite sequence of fields $i_1, \dots, i_m$ is legal for the current Boolean strict-contact set at each step, then there are strict invariant N-gons*

$$P = P^{(0)}, P^{(1)}, \dots, P^{(m)}$$

with the same multiplier and the same lifted shift κ whose strict fields are exactly the Boolean states produced after the corresponding updates.

Proof. This is a finite induction on the length of the sequence. At each stage legality means precisely that the current field is strict and the next field is endpoint, which is the hypothesis of proposition 5.1. That proposition returns a right-admissible strict invariant polygon with the strict set updated by (5.2). Since right-admissibility and the integer lift κ are part of the conclusion, the next Boolean legal move is again a geometrically legal surgery. No separate geometric-reachability assumption enters lemma 5.5. □

5.1 Reduction to one strict block

Put

$$\delta = \gcd(N, \kappa). \tag{5.11}$$

A *group* is a maximal cyclic interval of occupied fields.

The reduction minimizes first the number of chips and then the number of groups. A collision would improve the first coordinate and a merger the second, so neither can occur from a minimizing pattern. Because the κ-orbits are exactly the residue classes modulo $\delta = \gcd(N, \kappa)$, every such class must contain a chip; otherwise a full orbit of endpoint equalities would imply $x_j = \lambda^{N/\delta} x_j$. These two facts force a single block of length at least δ.

Lemma 5.5 (Reduced strict block and its first record)**.** *Starting from the one-sided representative fixed in lemma 4.13, consider only contact systems obtainable by a finite sequence of legal surgeries. Choose one whose contact score is lexicographically least:*

$$(\textit{number of strict contacts},\ \textit{number of strict groups}). \tag{5.12}$$

This is minimization only within the surgery-reachable class; it is neither a new vertex-count minimization nor an area minimization. After a cyclic relabelling, its strict contacts form one block

$$\{1, 2, \dots, \varphi\}, \qquad \varphi \geq \delta. \tag{5.13}$$

If $\varphi < N$, then for some $h > 0$

$$[h\kappa]_N = N - \varphi, \qquad [m\kappa]_N < N - \varphi \quad (0 \leq m < h), \tag{5.14}$$

where $[\cdot]_N$ denotes the representative in $\{0, \dots, N-1\}$. For $\varphi = N$ we use the initial record $[0\kappa]_N = 0$.

Proof. Let $\mathcal{R}$ be the family of strict-contact sets of reachable representatives. It is a nonempty subset of the finite set $2^{\mathbb{Z}/N\mathbb{Z}}$, so a lexicographic minimizer $S \in \mathcal{R}$ exists. Write

$$c(T) = |T|, \qquad g(T) = \text{the number of cyclic groups of } T.$$

If $S = \mathbb{Z}/N\mathbb{Z}$, the conclusion holds with $\varphi = N$ and the declared time-zero record. Assume henceforth that $S \neq \mathbb{Z}/N\mathbb{Z}$; every group of S is then proper.

We first record the complete transition ledger for a group sweep. Let $G = \{a, a+1, \dots, b\}$ be a group of S, let $\ell = |G| < N$, and put $R = S \setminus G$. Starting from $S^{(0)} = S$, perform the legal moves at

$$j_r = b - r, \qquad 0 \le r < \ell,$$

and denote the resulting sets by

$$S^{(r+1)} = (S^{(r)} \setminus \{j_r\}) \cup \{j_r + \kappa\}.$$

Up to the first collision, every source j_r is occupied when used: it belonged to G initially, has not yet appeared as a source, and earlier moves remove only their distinct sources. For $r = 0$, its right neighbor $b+1$ is empty by maximality of G. For $r > 0$, the right neighbor $j_r + 1 = j_{r-1}$ was removed in the preceding move. Before the first collision it cannot have been reinserted by an earlier target: if $j_s + \kappa = j_{r-1}$ for some $s < r-1$, then j_{r-1} was still occupied when the move at j_s was made, so that move would already have been a collision; the case $s = r-1$ would force $\kappa \equiv 0 \pmod N$, contrary to $1 \le \kappa < N$. Hence every move up to and including the first collision is legal. Until that collision, direct induction gives the disjoint-union invariant

$$S^{(r)} = R \,\dot\cup\, \{a, \dots, b-r\} \,\dot\cup\, \big(\{b-r+1, \dots, b\} + \kappa\big). \tag{5.15}$$

The score at the first exceptional event, or at the end of a collision-free sweep, is therefore exactly as follows.

event	chip count	group count
$j_r + \kappa \in S^{(r)}$ at the first collision	$c(S^{(r+1)}) = c(S) - 1$	irrelevant
no collision and $G + \kappa$ is not adjacent to R	$c(S^{(\ell)}) = c(S)$	$g(S^{(\ell)}) = g(S)$
no collision and $G + \kappa$ is adjacent to R	$c(S^{(\ell)}) = c(S)$	$g(S^{(\ell)}) \le g(S) - 1$

Indeed, in the last two rows

$$S^{(\ell)} = R \,\dot\cup\, (G + \kappa). \tag{5.16}$$

This ledger also makes the minimality consequence and the repeatability precise. During a partial sweep, an accidental adjacency of the translated suffix to a component of R is harmless for legality: the next source still has its right neighbor empty by the right-to-left order. If such a partial adjacency ever lowers the group count, the reachable intermediate state already contradicts lexicographic minimality; otherwise it is ignored, and only the completed translate is used in the repeatability invariant. For later use, after m completed sweeps write

$$G_m = G + m\kappa, \qquad T_m = R \,\dot\cup\, G_m.$$

The repeatability invariant is that T_m is reachable with score $(c(S), g(S))$, while G_m is a proper maximal group of T_m, disjoint from and nonadjacent to every group contained in R. It holds at $m = 0$: G is a proper maximal group of S and $R = S \setminus G$.

Suppose it holds at m. Apply the same transition ledger to the group G_m in T_m. A collision would produce a reachable set with fewer chips, and a collision-free sweep whose terminal translate is adjacent to R would produce one with the same number of chips and fewer groups. Both contradict the lexicographic minimality of S. Thus the sweep is collision-free and its completed translate is nonadjacent to R, and

$$T_{m+1} = R \,\dot{\cup}\, G_{m+1}.$$

Translation preserves $|G_{m+1}| = |G| < N$, so G_{m+1} is proper; its nonadjacency to the unchanged set R makes it a maximal group of T_{m+1}. The score is unchanged, and hence the invariant holds at $m+1$. Therefore the sweep may be repeated for as many complete sweeps as required. No assertion about the number of groups during a partial sweep is being used.

Every residue class modulo δ meets S. If one did not, all $L = N/\delta$ fields in the corresponding κ-orbit would be endpoint contacts. Iterating their identities $\lambda x_{j-\kappa} = x_j$ gives

$$x_j = \lambda^L x_j,$$

contrary to $x_j \neq 0$ and $|\lambda| < 1$.

Distinct groups of S have disjoint residue sets modulo δ. For if $p \in G$ and $q \in H$, with $G \neq H$ and $p \equiv q \pmod{\delta}$, there is $t \in \{1, \ldots, L-1\}$ such that

$$p + t\kappa = q \pmod{N}.$$

Translate G through t complete sweeps, leaving all other groups fixed. The preceding ledger permits each repetition unless it has already produced a forbidden collision or a complete-sweep endpoint merger; during the t-th sweep the chip starting at p otherwise reaches the occupied field q, which is itself a forbidden collision. This proves residue disjointness.

If some group has at least δ consecutive fields, its residue set is all of $\mathbb{Z}/\delta\mathbb{Z}$, and residue disjointness excludes every other group. Suppose instead that every group has length less than δ. The residue set of each group is then a proper cyclic interval in $\mathbb{Z}/\delta\mathbb{Z}$. These intervals are pairwise disjoint and, by residue coverage, cover the residue circle. Two of them are therefore consecutive. Write the corresponding groups as $G = \{a, \ldots, b\}$ and $H = \{c, \ldots, d\}$, choosing their terminal and initial fields so that

$$\delta \mid (c - b - 1).$$

Because the subgroup generated by κ is $\delta\mathbb{Z}/N\mathbb{Z}$, there is $t \in \{0, \ldots, L-1\}$ with

$$t\kappa \equiv c - b - 1 \pmod{N}. \tag{5.17}$$

Here $t \neq 0$, since $t = 0$ would make b and c adjacent fields and hence members of the same group. Translate G through t complete sweeps. A collision is forbidden; if none occurs, the terminal field of the final translate is $c - 1$, so the final translate is adjacent to H. The third row of the ledger then gives the same chip count and fewer groups, again impossible. Hence S has one group. Since it meets every residue class, its length φ is at least δ. A cyclic relabelling gives (5.13).

Assume now that $\varphi < N$. Relabel temporarily so that

$$S = F \,\dot{\cup}\, \{0\}, \qquad F = \{N - \varphi + 1, \ldots, N-1\}.$$

Put

$$a_m = [m\kappa]_N, \qquad I = \{N - \varphi, \ldots, N-1\}.$$

The orbit of 0 consists of the multiples of δ modulo N. Because $|I| = \varphi \geq \delta$, the interval I meets this orbit; because $0 \notin I$, there is a least $h > 0$ such that $a_h \in I$.

The exact moving-chip induction is

$$S_m = F \,\dot{\cup}\, \{a_m\} \qquad (0 \leq m \leq h), \tag{5.18}$$

where S_m is reachable from S by moving only the displayed chip. For $m = 0$ the formula reads $S_0 = F \dot\cup \{0\} = S$, so it holds. If $m < h$, minimality of h gives $a_m < N - \varphi$, and therefore

$$a_m + 1 \notin F \cup \{a_m\};$$

the move at a_m is legal. If its target a_{m+1} belonged to F, the move would be a collision and would produce only $\varphi - 1$ chips, contrary to minimality. Thus the target is empty and (5.18) follows at $m + 1$. At the first entrance time one consequently has

$$a_h \in I \setminus F = \{N - \varphi\},$$

whereas $a_m \notin I$, hence $a_m < N - \varphi$, for every $m < h$. This is exactly (5.14). The computation used displacements from the terminal chip, so the final common cyclic relabelling to (5.13) changes neither κ nor the residue statement. For $\varphi = N$ the declared time-zero record applies. □

6 Finite rotation sections from the lattice sail

The cyclic arithmetic is isolated in this section. It depends only on the rotation of a finite cyclic set; no polygon is used until the final tower identities are read through endpoint contacts.

Fix integers

$$N \geq 2, \qquad 1 \leq \kappa < N, \qquad \delta = \gcd(N, \kappa),$$

and write $[a]_N$ for the representative of a in $\{0, \dots, N-1\}$. Put

$$L(a, b) = a\kappa - bN. \tag{6.1}$$

A time $h \geq 0$ is an *upper-record time* when

$$[h\kappa]_N > [t\kappa]_N \qquad (0 \leq t < h).$$

Its deficit is $\nu = N - [h\kappa]_N$, with the convention that time zero has deficit N. Equivalently, with

$$b = \begin{cases} 1, & h = 0, \\ \lceil h\kappa/N \rceil, & h > 0, \end{cases} \qquad V = (h, b),$$

one has

$$L(V) = -\nu, \qquad [t\kappa]_N \leq N - \nu \quad (0 \leq t < h). \tag{6.2}$$

The terminal record has deficit δ, because the attainable residues are exactly the multiples of δ and $N - \delta$ is the largest one.

Theorem 6.1 (Finite rotation return-section theorem)**.** *Let $V = (h, b)$ be an upper-record vector with deficit $\nu > \delta$, and let $V' = (h', b')$ be the next upper-record vector. Define*

$$U = V' - V = (q, p), \qquad \Delta = \nu - \nu'.$$

Then

$$q\kappa - pN = \Delta > 0, \qquad h\kappa - bN = -\nu, \tag{6.3}$$

$$0 < \Delta < \nu, \qquad \det(U, V) = 1, \tag{6.4}$$

$$q\nu + h\Delta = N, \qquad \gcd(\Delta, \nu) = \delta. \tag{6.5}$$

The initial record $\nu = N$ has $V = (0, 1)$. Moreover $h = 0$ if and only if $\nu = N$.

Let

$$q\kappa - pN = \Delta, \qquad h\kappa - bN = -\nu, \qquad q\nu + h\Delta = N, \tag{6.6}$$

and define

$$H_i = \begin{cases} q, & 1 \le i \le \nu - \Delta, \\ q+h, & \nu - \Delta < i \le \nu. \end{cases}$$

Then

$$\mathcal{D} = \{(t,i) : 1 \le i \le \nu,\ 0 \le t < H_i\} \longrightarrow \mathbb{Z}/N\mathbb{Z}, \qquad F(t,i) = [i + t\kappa]_N \tag{6.7}$$

is a bijection. The first-return map on the base interval $\{1,\dots,\nu\}$ *is addition of* Δ *modulo* ν *and has the two return times displayed above.*

If $\Delta = 1$*, put*

$$w = \lfloor h/q \rfloor, \qquad E = V - wU = (e,c), \qquad d = -L(E) = \nu + w.$$

Then E *is an upper-record vector, and*

$$q\kappa - pN = 1, \qquad e\kappa - cN = -d, \qquad qd + e = N, \qquad 0 \le e < q. \tag{6.8}$$

In particular $d = \lfloor N/q \rfloor \ge \nu$*, and*

$$E, E+U, \dots, E+(w+1)U$$

are consecutive upper-record vectors.

Finally, suppose a right-admissible polygonal contact system has no strict field outside the base interval $\{1,\dots,\nu\}$ *and has the record data above. Then every internal tower state is an endpoint contact and*

$$\lambda^t x_i = x_{F(t,i)} \qquad (0 \le t < H_i). \tag{6.9}$$

The top and closure relations are

$$\lambda^q x_i = \xi_{i+\Delta} \qquad (1 \le i \le \nu - \Delta), \tag{6.10}$$

$$\lambda^{q+h} x_i = \xi_{i+\Delta-\nu} \qquad (\nu - \Delta < i \le \nu), \tag{6.11}$$

$$\lambda^h x_\nu = x_0, \tag{6.12}$$

$$\lambda^q x_0 = \xi_\Delta. \tag{6.13}$$

Proof. We proceed in four stages.

1. Consecutive record vectors are unimodular. List the upper-record times chronologically. The list is finite and ends at deficit δ. Indeed,

$$\gcd(\kappa/\delta, N/\delta) = 1,$$

so multiplication by κ/δ permutes the residue classes modulo N/δ. Hence there is an h such that

$$h\kappa/\delta \equiv -1 \pmod{N/\delta},$$

equivalently $h\kappa \equiv N - \delta \pmod N$. Thus the attainable residue $N-\delta$ occurs, and the upper-record list ends at deficit δ. Every record vector $V = (h,b)$ is primitive. If $V = gW$ with $g > 1$, then $L(W) = -\nu/g < 0$ and the time coordinate of W is smaller than h; its residue is $N - \nu/g > N - \nu$, contradicting the record property.

Let $V = (h,b)$ and $V' = (h',b')$ be consecutive records, with deficits $\nu > \nu' > 0$. Since

$$b = \frac{h\kappa + \nu}{N}, \qquad b' = \frac{h'\kappa + \nu'}{N},$$

we have

$$bh' - hb' = \frac{h'\nu - h\nu'}{N} > 0.$$

Thus $\det(V, V') < 0$. Suppose $D = -\det(V, V') > 1$. By lemma A.6, the half-open fundamental parallelogram generated by V, V' contains a nonzero lattice point $W = \alpha V + \beta V'$ with $0 \leq \alpha, \beta < 1$. Primitivity excludes a nonzero point on either radial edge. Replacing W by $V + V' - W$ when $\alpha + \beta > 1$, we obtain a lattice point in the triangle $\operatorname{conv}\{0, V, V'\}$ different from its three vertices. Hence

$$W = \alpha V + \beta V', \qquad \alpha, \beta > 0, \qquad \alpha + \beta \leq 1.$$

Its time $t = \alpha h + \beta h'$ is an integer with $0 < t < h'$, while

$$-L(W) = \alpha\nu + \beta\nu' < \nu.$$

Thus $[t\kappa]_N > N - \nu$ before the next record time h', a contradiction. Therefore $D = 1$ and

$$\det(V, V') = -1.$$

For $U = V' - V$, this is $\det(U, V) = 1$. Linearity of L gives $L(U) = \nu - \nu' = \Delta$, so $0 < \Delta < \nu$. Expanding the determinant,

$$q\nu + h\Delta = q(bN - h\kappa) + h(q\kappa - pN) = N(qb - ph) = N.$$

Because U, V are a unimodular basis, the subgroup generated by $L(U) = \Delta$ and $L(V) = -\nu$ equals $L(\mathbb{Z}^2)$. The latter is $\delta\mathbb{Z}$. Hence $\gcd(\Delta, \nu) = \delta$. For time zero, $V = (0, 1)$ by definition, so $L(V) = -N$ and the deficit is N; conversely $h = 0$ forces this vector and this deficit.

2. The tower map is bijective. Give $\mathcal{D}$ the successor map

$$\sigma(t, i) = \begin{cases} (t+1, i), & t + 1 < H_i, \\ (0, i + \Delta), & t = q - 1,\ i \leq \nu - \Delta, \\ (0, i + \Delta - \nu), & t = q + h - 1,\ i > \nu - \Delta. \end{cases}$$

The congruences in (6.6) give

$$F(\sigma(t, i)) = F(t, i) + \kappa \pmod{N}. \tag{6.14}$$

On the bases, σ induces addition of Δ modulo ν. It has $\delta = \gcd(\Delta, \nu)$ cycles. Each base cycle contains ν/δ bases and Δ/δ long bases, so its total number of states is

$$q\frac{\nu}{\delta} + h\frac{\Delta}{\delta} = \frac{N}{\delta}.$$

Addition of κ on $\mathbb{Z}/N\mathbb{Z}$ also has δ cycles of length N/δ. Since $F(t, i) \equiv i \pmod{\delta}$, distinct base cycles map into distinct κ-orbits. Equation (6.14) makes the restriction to each cycle an equivariant map between two cycles of the same finite length, hence a bijection. This proves (6.7) and the return-time statement.

3. A unit record edge propagates backwards without an Euclidean index chain. Assume $\Delta = 1$. By construction $E = V - wU = (e, c)$ has $0 \leq e < q$ and $L(E) = -(\nu + w) = -d$. The determinant remains $\det(U, E) = 1$, so

$$qd + e = N.$$

It remains to prove that E and the displayed arithmetic progression are exactly consecutive record vectors.

Because U, E are a basis, write any lattice vector $Z = (t, z)$ uniquely as $Z = AE + BU$. Suppose first that $0 \le t < e$ and $-d < L(Z) < 0$. The inequalities are

$$-d < -Ad + B < 0.$$

If $A \le 0$, then $B < Ad \le 0$ and $t = Ae + Bq < 0$. If $A = 1$, then $B \ge 1$ and $t \ge e + q$. If $A \ge 2$, then $B > (A-1)d \ge 1$ and again $t = Ae + Bq > e$. All alternatives contradict $0 \le t < e$. Hence E is an upper record.

More generally, fix $j \ge 0$ with $d - j > 0$ and suppose $0 \le t < e + (j+1)q$ while

$$-(d-j) < L(Z) < 0.$$

Equivalently,

$$-(d-j) < -Ad + B < 0.$$

If $A \le 0$, then $t < 0$. If $A = 1$, the left inequality gives $B > j$, so $B \ge j+1$ and $t \ge e+(j+1)q$. If $A \ge 2$, it gives $B > (A-1)d + j$, hence $B \ge j+1$ and $t = Ae + Bq \ge e + (j+1)q$. Thus no improved residue occurs before the time of $E + (j+1)U$, whose deficit is one less than that of $E + jU$. Induction shows that the displayed vectors are consecutive records. Since $V = E + wU$, this includes the given pair $V, V + U$. Finally $N = qd + e$ with $0 \le e < q$ gives $d = \lfloor N/q \rfloor$.

4. Geometric tower identities. Assume now that every strict field belongs to $\{1, \ldots, \nu\}$. If an internal state $F(t, i)$ with $1 \le t < H_i$ belonged to the base block, then it and the corresponding level-zero state would have the same image under the bijection F, impossible. Hence every internal destination field is an endpoint contact. If $j = F(t, i)$, the endpoint identity at field j is

$$\lambda x_{F(t-1,i)} = x_{F(t,i)}.$$

Induction gives (6.9). One further multiplication at a short or long top gives (6.10) and (6.11). Since $F(h, \nu) \equiv \nu + h\kappa \equiv 0$ and $h < H_\nu$, the internal identity gives (6.12); combining it with the long top at base ν gives (6.13). □

Corollary 6.2 (Endpoint-padded upper-record sections)**.** *Let a right-admissible contact system have strict-field set S. Suppose $U = (q, p)$ and $E = (e, c)$ are integer vectors for which E and $E + U$ are consecutive upper-record vectors and*

$$q\kappa - pN = 1, \qquad e\kappa - cN = -d, \qquad qd + e = N, \qquad 0 \le e < q, \qquad d > 1. \tag{6.15}$$

Assume

$$S \subseteq J := \{1, \ldots, d\}. \tag{6.16}$$

Then the return section with base interval J has height q at bases $1, \ldots, d-1$, height $q + e$ at base d, and first return equal to the successor. Its tower states form a bijection with the N polygon labels, every internal destination field lies outside J and is therefore an endpoint contact, and

$$\lambda^q x_{j-1} = \xi_j \qquad (1 \le j \le d), \tag{6.17}$$

$$\lambda^e x_d = x_0. \tag{6.18}$$

For every $j \in J \setminus S$, one has $\xi_j = x_j$. Thus enlarging an actual strict block to such an upper-record interval inserts only endpoint cells; in the scalar return product these are exactly the factors with $\beta_j = 0$ and $\alpha_j = 1$.

Proof. Apply theorem 6.1 to the consecutive record pair $V = E$, $V' = E + U$. In the notation of that theorem,

$$\nu = d, \qquad \Delta = 1, \qquad h = e.$$

The arithmetic assumptions in (6.15) are precisely (6.6); hence the tower map is bijective, its return map is addition of one modulo d, and its heights are the stated two values.

The geometric conclusion of theorem 6.1 requires only that there be no strict field outside the chosen base interval. This is exactly (6.16). More explicitly, if an internal tower state had destination label $j \in J$, then it would have the same image under the tower bijection as the level-zero state $(0, j)$, which is impossible. Thus every internal destination lies outside J and, since all strict fields belong to J, is an endpoint contact. The short-return identities for bases $1, \ldots, d-1$ give $\lambda^q x_i = \xi_{i+1}$; the virtual short return gives $\lambda^q x_0 = \xi_1$; and the tower closure gives $\lambda^e x_d = x_0$. These are (6.17)–(6.18). Finally, in a right-admissible system a nonstrict field has $\beta_j = 0$ in (4.12), so $\xi_j = x_j$ and the asserted padding interpretation follows. □

Remark 6.3. The record vectors are precisely the primitive vertices of the lattice sail between the vertical ray and the rational ray of slope κ/N. Unimodularity is forced by the absence of a lattice point in the triangle between consecutive records. This geometric formulation is equivalent to continued fractions, but it avoids parity conventions, semiconvergent ownership rules, and a terminal-remainder exception.

7 Projective no-skipping

This step proves that a nonminimal strict block cannot skip over more than one base at its first positive return. The proof is separated into two parts. First, the return towers produce a shortest boundary corridor whose nonclosing incidences are exact identities. Second, an abstract projective corridor theorem shows that the single unconstrained closing contact can be opened inward. The contradiction is then only the local full-vertex-touching property of replacement polygons from theorem 3.2.

Throughout this section assume temporarily that $\varphi > \delta$. By theorem 6.1, there are integers q, h, p, b, Δ such that

$$q\kappa - pN = \Delta, \qquad h\kappa - bN = -\varphi, \qquad q\varphi + h\Delta = N, \qquad 0 < \Delta < \varphi. \tag{7.1}$$

We shall prove that the alternative $\Delta > 1$ is impossible. Apply theorem 6.1 with $\nu = \varphi$. For $1 \le i \le \varphi$, define the return height and return field by

$$H_i = \begin{cases} q, & 1 \le i \le \varphi - \Delta, \\ q + h, & \varphi - \Delta < i \le \varphi, \end{cases} \qquad r(i) = \begin{cases} i + \Delta, & 1 \le i \le \varphi - \Delta, \\ i + \Delta - \varphi, & \varphi - \Delta < i \le \varphi. \end{cases} \tag{7.2}$$

Thus the two top relations (6.10)–(6.11) are the single identity

$$\lambda^{H_i} x_i = \xi_{r(i)} \in \operatorname{relint} E_{r(i)} \qquad (1 \le i \le \varphi). \tag{7.3}$$

The virtual short relation is

$$\lambda^q x_0 = \xi_\Delta. \tag{7.4}$$

We also record the transported terminal side. From (6.12) and the same internal-tower calculation applied to $\varphi - 1$,

$$\lambda^h x_\varphi = x_0, \qquad \lambda^h x_{\varphi-1} = x_{-1}, \qquad \lambda^h \operatorname{relint} E_\varphi = \operatorname{relint} E_0. \tag{7.5}$$

If $h = 0$, then theorem 6.1 gives $\varphi = N$, and the same display is just the cyclic identity $E_\varphi = E_0$.

For a base i, put

$$L_i := \lambda^{-H_i} \operatorname{aff} E_{r(i)}. \tag{7.6}$$

Since $\lambda^{H_i} P \subseteq P$ and (7.3) places $\lambda^{H_i} x_i$ in the relative interior of the side $E_{r(i)}$, the line L_i supports P at x_i.

Lemma 7.1 (Strict supports on the short return corridor)**.** *Assume $\Delta > 1$, and set*

$$m_+ = \Delta, \qquad m_- = \varphi - \Delta + 1, \qquad m = \min\{m_+, m_-\}. \tag{7.7}$$

If $m = m_+$, then L_i is a strict support of P at x_i for $2 \le i \le \Delta$. If $m = m_-$, then L_i is a strict support of P at x_i for $\Delta - 1 \le i \le \varphi - 2$.

Proof. By the support-face test lemma 2.9, it is enough to show that neither neighboring vertex of x_i lies on L_i. Since multiplication by λ^{H_i} is invertible, this is equivalent to showing that $\lambda^{H_i} x_{i-1}$ and $\lambda^{H_i} x_{i+1}$ avoid $\operatorname{aff} E_{r(i)}$.

If a neighboring base has the same height as i, its image is the strict contact on the adjacent return side:

$$\lambda^{H_i} x_{i-1} = \xi_{r(i)-1}, \qquad \lambda^{H_i} x_{i+1} = \xi_{r(i)+1},$$

with cyclic side labels. A relative-interior point of an adjacent side of a strict polygon is not on the line of $E_{r(i)}$. Thus only the interface between the last short base

$$i_0 = \varphi - \Delta$$

and the first long base $i_0 + 1$ needs separate checking.

For the last short base i_0, the return side is E_φ. Its outgoing neighbor is the first long base. If $h > 0$, then level q in that long tower is internal, and

$$\lambda^q x_{i_0+1} = x_{\varphi+1} \notin \operatorname{aff} E_\varphi.$$

If $h = 0$, then $\varphi = N$ and the long top relation gives $\lambda^q x_{i_0+1} = \xi_1 \in \operatorname{relint} E_1$, again not on the line of E_φ.

For the first long base $i_0 + 1$, the return side is E_1. The incoming neighbor is the last short base, and (7.5) gives

$$\lambda^{q+h} x_{i_0} = \lambda^h \xi_\varphi \in \operatorname{relint} E_0.$$

The outgoing neighbor returns to $\xi_2 \in \operatorname{relint} E_2$. Neither point is on the line of E_1.

The two index ranges in the statement meet the short–long interface at most once and contain no other height transition. Hence both adjacent vertices avoid every displayed pulled-back line, and the supports are strict. $\square$

The shorter of the two cyclic arcs is now read as a single boundary chain. The following table is the whole branch dictionary; after this point the projective argument is orientation-free.

	forward corridor	reverse corridor
condition	$2\Delta \le \varphi + 1$	$2\Delta > \varphi + 1$
length	$m = \Delta$	$m = \varphi - \Delta + 1$
chain vertices	$X_i = x_i$	$X_i = x_{\varphi - i}$
chain contacts	$C_i = \xi_i$	$C_i = \xi_{\varphi - i + 1}$
transport exponent	$a = q$	$a = q + h$
moved bases	$\{1, \dots, \Delta\}$	$\{\Delta - 1, \dots, \varphi - 1\}$
closing field	$\Delta + 1$	$\Delta - 1$

Here $0 \le i \le m+1$ for the chain vertices, while the contact formula is used only for $2 \le i \le m+1$. In either column,

$$C_i \in \operatorname{relint}[X_{i-1}, X_i] \qquad (2 \le i \le m+1), \tag{7.8}$$

and the seed side is transported by

$$\lambda^a X_1 = C_{m+1}, \qquad \lambda^a X_0 = C_m. \tag{7.9}$$

In the forward corridor the first identity is the short-top relation for base 1, and the second is (7.4). In the reverse corridor they are the long-top relations for bases $\varphi - 1$ and φ.

For $2 \le i \le m$, define

$$\mathcal{L}_i = L_{b_i}, \qquad b_i = \begin{cases} i, & \text{in the forward corridor,} \\ \varphi - i, & \text{in the reverse corridor.} \end{cases} \tag{7.10}$$

By lemma 7.1, $\mathcal{L}_i$ is a strict support of P at X_i.

Lemma 7.2 (Proper corridor chain, including boundary values)**.** *Assume $N \ge 4$, $1 < \Delta < \varphi \le N$, and choose the branch in the preceding table. Then $m \ge 2$ and*

$$m + 1 < N. \tag{7.11}$$

Consequently the displayed vertices $X_0, \dots, X_{m+1}$ are pairwise distinct consecutive vertices in the chosen cyclic orientation and the chain omits at least one side of P.

This conclusion includes all boundary values used later: if $\Delta = 2$, the forward branch has $m = 2$; if $2\Delta = \varphi + 1$, the equality belongs to the forward branch and does not identify two chain labels; and if $h = 0$, then $\varphi = N$ but the same strict inequality (7.11) *still holds.*

Proof. In the forward branch, $m = \Delta$. If $\varphi < N$, then

$$m + 1 = \Delta + 1 \le \varphi < N$$

because $\Delta < \varphi$. If $\varphi = N$ and equality $m+1 = N$ were to hold, then $\Delta = N - 1$; the branch inequality $2\Delta \le \varphi + 1 = N + 1$ would give $2N - 2 \le N + 1$, hence $N \le 3$, a contradiction. Thus (7.11) holds. The labels $0, 1, \dots, \Delta + 1$ therefore represent distinct vertices modulo N. The special equality $2\Delta = \varphi + 1$ changes the return-edge wrap but not this label interval. When $\Delta = 2$, the same calculation gives $m = 2$ and $m + 1 = 3 < N$.

In the reverse branch, $m = \varphi - \Delta + 1 \ge 2$. If $\varphi < N$, then $\Delta \ge 2$ gives

$$m + 1 = \varphi - \Delta + 2 \le \varphi < N.$$

If $\varphi = N$, the strict branch inequality $2\Delta > N + 1$ forces $\Delta \ge 3$ because $N \ge 4$. Hence

$$m + 1 = N - \Delta + 2 < N.$$

The reverse labels run through the integer interval $\varphi, \varphi - 1, \dots, \Delta - 2$, whose length is $m + 2$ and whose total index variation is $m + 1 < N$; they are therefore distinct modulo N. When $\varphi = N$, the first label $x_\varphi = x_0$ is not repeated because $\Delta - 2 \ge 1$. Finally, theorem 6.1 gives $h = 0 \iff \varphi = N$, so the preceding two $\varphi = N$ arguments also cover the zero-height case. In either branch, $m + 1 < N$ means that the consecutive chain uses fewer than all N sides and is proper. $\square$

Proposition 7.3 (Exact return-edge ledger). *Let*

$$\mathcal{B} = \{1, \dots, \varphi\}, \qquad s = r^{-1},$$

where r is addition of Δ modulo φ in the representative set $\mathcal{B}$. Assume $\Delta > 1$ and choose the forward or reverse corridor from the preceding table. Define the moved-base set M, the seed base b_*, the supported moved bases M°, the controlled target fields D, the closing field c, the supported-return fields R, and the fixed remainder A as follows.

An integer interval $\{u, \dots, v\}$ is understood to be empty when $u > v$. In the forward branch $2\Delta \le \varphi + 1$, put

$$M = \{1, \dots, \Delta\}, \quad b_* = 1, \quad M^\circ = \{2, \dots, \Delta\}, \tag{7.12}$$

$$D = \{2, \dots, \Delta\}, \quad c = \Delta + 1, \quad R = r(M^\circ), \quad A = \mathcal{B} \setminus (D \cup R \cup \{c\}). \tag{7.13}$$

More explicitly,

$$R = \begin{cases} \{\Delta + 2, \dots, 2\Delta\}, & 2\Delta \le \varphi, \\ \{\Delta + 2, \dots, \varphi\} \cup \{1\}, & 2\Delta = \varphi + 1. \end{cases} \tag{7.14}$$

In the reverse branch $2\Delta > \varphi + 1$, equivalently $2\Delta \ge \varphi + 2$, put

$$M = \{\Delta - 1, \dots, \varphi - 1\}, \quad b_* = \varphi - 1, \quad M^\circ = \{\Delta - 1, \dots, \varphi - 2\}, \tag{7.15}$$

$$D = \{\Delta, \dots, \varphi - 1\}, \quad c = \Delta - 1, \quad R = r(M^\circ), \quad A = \mathcal{B} \setminus (D \cup R \cup \{c\}). \tag{7.16}$$

In this branch

$$R = \{2\Delta - \varphi - 1, \dots, \Delta - 2\}. \tag{7.17}$$

For either branch the following assertions hold.

(i) *The four sets*

$$\mathcal{B} = D \,\dot\cup\, R \,\dot\cup\, \{c\} \,\dot\cup\, A \tag{7.18}$$

are pairwise disjoint, and

$$r(M) = R \,\dot\cup\, \{c\}, \qquad r(b_*) = c, \qquad r(M^\circ) = R. \tag{7.19}$$

Consequently

$$s(D) \cap M = \varnothing, \qquad s(R) = M^\circ, \qquad s(c) = b_*, \qquad s(A) \cap M = \varnothing. \tag{7.20}$$

Thus every return edge belongs to exactly one of four classes: fixed source to controlled line, supported moved source to fixed line, seed to closing line, or fixed source to fixed line.

(ii) *Suppose the tower deformation fixes every tower whose base is outside M, transports every moved tower by the powers of λ, and moves the seed on its original anchor line: $\operatorname{aff} E_1$ in the forward branch and $\operatorname{aff} E_\varphi$ in the reverse branch. Then, among the strict fields $\mathcal{B}$, the fields in D are exactly the nonclosing side lines controlled by the corridor recursion, the field c is the closing side, and every side line with label in $R \cup A$ is fixed. No strict side field is omitted or listed twice.*

Proof. Because r is a translation of the finite cyclic set $\mathcal{B}$, it is a bijection. It is therefore enough to compute the images of the displayed integer intervals and to verify the stated disjointness.

Forward branch. For $2 \le i \le \Delta - 1$ one has

$$i + \Delta \le 2\Delta - 1 \le \varphi,$$

so $r(i) = i + \Delta$. For $i = \Delta$, either $2\Delta \le \varphi$ and $r(\Delta) = 2\Delta$, or $2\Delta = \varphi + 1$ and $r(\Delta) = 1$. This proves (7.14). In particular,

$$R \subseteq \{\Delta + 2, \dots, \varphi\} \cup \{1\},$$

which is disjoint from $D = \{2, \dots, \Delta\}$ and from $c = \Delta + 1$. Since $r(1) = \Delta + 1 = c$ and r is injective, (7.18) and (7.19) follow.

For $j \in D$, the inverse formula is

$$s(j) = \varphi - \Delta + j.$$

The branch inequality gives

$$s(j) \ge \varphi - \Delta + 2 \ge \Delta + 1,$$

so $s(D) \cap M = \varnothing$. The assertions for R, c, and A now follow from bijectivity and (7.19).

Only the base labels $1, \dots, \Delta$ move. The vertex x_0 belongs to the tower based at φ, which is outside M, so it is fixed; the line of E_1 is therefore fixed because the seed moves on its original line. For $\Delta + 2 \le j \le \varphi$, the endpoint x_{j-1} is an unmoved base. The other endpoint is also an unmoved base when $j < N$; if $j = \varphi = N$, it is $x_j = x_0$, already shown to be fixed. Hence the line of every such E_j is fixed. The remaining strict fields are exactly $D = \{2, \dots, \Delta\}$ and $c = \Delta + 1$: the former are the controlled chain sides and the latter is the closing side. This proves part (ii) in the forward branch, including the boundary case $2\Delta = \varphi + 1$, where the supported return field 1 is still on the fixed anchor line.

Reverse branch. For every $i \in M^\circ$,

$$i + \Delta \ge 2\Delta - 1 \ge \varphi + 1, \qquad i + \Delta \le \varphi + \Delta - 2 < 2\varphi,$$

so there is exactly one wrap and

$$r(i) = i + \Delta - \varphi.$$

As i runs from $\Delta - 1$ to $\varphi - 2$, this gives precisely (7.17). Its lower endpoint is at least 1 because $2\Delta \ge \varphi + 2$. Hence

$$R \subseteq \{1, \dots, \Delta - 2\},$$

which is disjoint from $c = \Delta - 1$ and from $D = \{\Delta, \dots, \varphi - 1\}$. Also $r(\varphi - 1) = \Delta - 1 = c$. Injectivity of r again proves (7.18) and (7.19).

For $j = \Delta$ one has $s(j) = \varphi$. For $\Delta < j \le \varphi - 1$ one has $s(j) = j - \Delta$, and

$$1 \le j - \Delta \le \varphi - \Delta - 1 \le \Delta - 3.$$

Thus $s(D) \cap M = \varnothing$, and the remaining assertions in (7.20) again follow from bijectivity.

Only the base labels $\Delta - 1, \dots, \varphi - 1$ move. The line of E_φ remains fixed because x_φ is fixed and the seed $x_{\varphi-1}$ moves on its original line. For $2 \le j \le \Delta - 2$, both endpoint labels of E_j lie outside M, so those lines are fixed. For $j = 1$, the endpoint x_1 is an unmoved base, while $x_0 = \lambda^h x_\varphi$ by (7.5); the base φ is outside M, so x_0 is fixed as well. Hence the line of E_1 is fixed. The remaining strict fields are exactly $D = \{\Delta, \dots, \varphi - 1\}$ and $c = \Delta - 1$, with the asserted roles. This completes the reverse branch and the proof. □

7.1 The projective corridor theorem

The next theorem is purely projective. It contains no stochastic matrices, no Euclidean algorithm, and no modular labels.

Definition 7.4 (Projective corridor and holonomy)**.** Let P be a strict convex polygon. A *projective corridor* consists of a proper consecutive boundary chain—meaning that the displayed vertices are pairwise distinct and the chain omits at least one side of P—

$$X_0, X_1, \dots, X_{m+1}, \qquad m \geq 2,$$

read in either cyclic orientation, points

$$C_i \in \operatorname{relint}[X_{i-1}, X_i] \qquad (2 \leq i \leq m+1),$$

and strict supporting lines $\mathcal{L}_i$ of P at X_i for $2 \leq i \leq m$. Starting with a point on $\operatorname{aff}(X_0, X_1)$, project successively through the centers C_i onto $\mathcal{L}_i$, and finally project through X_{m+1} onto $\operatorname{aff}(C_m, C_{m+1})$. This composition of perspectivities is the *corridor holonomy.*

Proposition 7.5 (Admissible projective chart)**.** *For every projective corridor there is a projective affine chart in which P is bounded, strict convexity and all incidences are preserved, the endpoint strict supports at X_0 and X_{m+1} are parallel, no side of the displayed chain and no line $\mathcal{L}_i$ is parallel to them, and the displayed chain is the graph of a convex piecewise-linear function whose consecutive edge slopes are strictly increasing over a strictly increasing coordinate.*

Proof. At a vertex of a strict polygon, the supporting lines that meet the polygon only at that vertex form a nonempty open interval in the pencil of all lines through the vertex. Let $\mathcal{F}$ be the finite family consisting of all side lines of P and all lines $\mathcal{L}_i$. Choose any strict support M_0 at X_0. It is distinct from every line in $\mathcal{F}$: a line in $\mathcal{F}$ that is a side line is not a strict support at X_0, while a strict support at another displayed vertex cannot contain X_0.

Now vary a strict support M_1 at X_{m+1}. Exclude first the single direction parallel to M_0, so that $O = M_0 \cap M_1$ is a finite point. For a fixed $F \in \mathcal{F}$, the condition $O \in F$ excludes at most one further member of the pencil. Indeed, if M_0 is not parallel to F, put $K_F = M_0 \cap F$; then $O \in F$ is equivalent to $M_1 = \operatorname{aff}(X_{m+1}, K_F)$. If M_0 is parallel to F, no finite point of M_0 lies on F. Removing these finitely many exceptional lines from the open interval of strict supports leaves a valid choice of M_1. For this choice,

$$O = M_0 \cap M_1 \notin F \qquad (F \in \mathcal{F}). \tag{7.21}$$

The point O lies outside P. Otherwise strictness of M_0 would give $O = X_0$, while strictness of M_1 would give $O = X_{m+1}$, contradicting that the boundary chain is proper. Since P is compact and convex, strict separation of O from P gives an affine functional ℓ such that

$$\ell(z) < \ell(O) \qquad (z \in P).$$

Hence the line $J = \{z : \ell(z) = \ell(O)\}$ passes through O and is disjoint from P. Apply a projective automorphism T sending J to the line at infinity. In affine coordinates it has the form

$$T(z) = \frac{Az + b}{d(z)},$$

where d is affine and vanishes precisely on J. After multiplying the homogeneous matrix by -1 if necessary, $d > 0$ on the compact convex set P. For $x, y \in P$ and $0 \leq t \leq 1$, direct substitution gives

$$T((1-t)x + ty) = \frac{(1-t)d(x)}{(1-t)d(x) + td(y)} T(x) + \frac{td(y)}{(1-t)d(x) + td(y)} T(y). \tag{7.22}$$

The two coefficients are nonnegative and sum to one, and they are both strictly positive when $0 < t < 1$. Thus T maps every segment in P onto the segment joining the endpoint images and maps relative interiors to relative interiors. Applying the same formula to T^{-1} in the image chart shows that no additional points enter those segments. Consequently $T(P)$ is a compact, hence bounded, strict convex polygon; faces, strict supports, incidences, and boundary order are preserved, up to a simultaneous reversal of orientation.

The images of M_0 and M_1 are distinct parallel supporting lines, because their projective closures meet the new line at infinity at the common point which is the image of O. By (7.21), no side line of P and no $\mathcal{L}_i$ has that point at infinity; equivalently, none is parallel to the endpoint supports.

Choose an affine coordinate t constant on lines parallel to the endpoint supports, with $t(X_0) < t(X_{m+1})$. The endpoint supports are the two opposite supporting lines of P in this direction and, being strict, meet P only at X_0 and X_{m+1}. For every intermediate value of t, the corresponding transverse line cuts the convex polygon in a nondegenerate segment; its two endpoints lie on the two boundary arcs joining X_0 to X_{m+1}. Thus each arc is a graph over t. Since no side of the displayed arc is parallel to the endpoint supports, no graph segment is vertical and t is strictly increasing along the displayed chain. After reflecting the second affine coordinate if necessary, this chain is the lower graph of the polygon. Convexity makes its consecutive slopes nondecreasing, and strictness rules out equality of consecutive slopes; hence they are strictly increasing. □

Lemma 7.6 (Convex-chain calibration)**.** *Regard every affine line below as its projective completion, and write $\overline{AB}$ for the projective line through two distinct points A, B. Set $Z_1 = X_0$ and recursively define the projective points*

$$Z_i = \mathcal{L}_i \cap \overline{Z_{i-1}C_i} \qquad (2 \le i \le m). \tag{7.23}$$

There exists an admissible affine chart as in proposition 7.5 in which all the points Z_i are finite. The two projective lines in the final intersection are distinct, and the projectively defined point

$$W_* := \overline{Z_m X_{m+1}} \cap \overline{C_m C_{m+1}} \in \operatorname{relint}[C_{m+1}, C_m] \tag{7.24}$$

in the original affine chart. The same assertion holds when the original boundary chain is read in the opposite cyclic orientation.

Proof. Apply a projective transformation supplied by proposition 7.5, and in its target affine chart use the same symbols for the transformed points and lines. The recursion (7.23) is projectively natural, so the transformed points satisfy the same incidence relations. Write

$$X_i = (t_i, f_i), \qquad t_0 < t_1 < \cdots < t_{m+1},$$

and let s_i be the slope of the edge $[X_{i-1}, X_i]$. Since the chain is the lower graph of a strict convex polygon,

$$s_1 < s_2 < \cdots < s_{m+1}. \tag{7.25}$$

If ℓ_i is the slope of the strict support $\mathcal{L}_i$ at X_i, the supporting-slope interval at that vertex is $[s_i, s_{i+1}]$, and strictness of the support excludes both endpoint slopes. Hence

$$s_i < \ell_i < s_{i+1} \qquad (2 \le i \le m). \tag{7.26}$$

Write $C_i = (c_i, \chi_i)$. Because $C_i \in \operatorname{relint}[X_{i-1}, X_i]$,

$$t_{i-1} < c_i < t_i, \qquad \chi_i = f_i + s_i(c_i - t_i). \tag{7.27}$$

We prove inductively that every projectively defined point Z_i is affine, say $Z_i = (z_i, g_i)$, and that, for $2 \le i \le m$,

$$c_i < z_i < t_i, \qquad r_i < s_i, \tag{7.28}$$

where r_i is the slope of $R_i = \overline{Z_{i-1}C_i}$. When $i < m$, we also prove that Z_i lies strictly above the backward extension of the next edge $[X_i, X_{i+1}]$.

For $i = 2$, the point $Z_1 = X_0$, and the slope of R_2 is

$$r_2 = \frac{(t_1 - t_0)s_1 + (c_2 - t_1)s_2}{c_2 - t_0}. \tag{7.29}$$

Both weights are positive, so $s_1 < r_2 < s_2$, in particular $r_2 < s_2$.

Assume now that $i \geq 3$ and the assertion has been proved at $i - 1$. The previous "above the next edge" statement says

$$g_{i-1} > f_{i-1} + s_i(z_{i-1} - t_{i-1}).$$

Since C_i lies on that same edge line, $\chi_i = f_{i-1} + s_i(c_i - t_{i-1})$. Also $z_{i-1} < t_{i-1} < c_i$. Subtracting the two displayed relations gives

$$\chi_i - g_{i-1} < s_i(c_i - z_{i-1}),$$

and division by the positive denominator $c_i - z_{i-1}$ yields

$$r_i = \frac{\chi_i - g_{i-1}}{c_i - z_{i-1}} < s_i. \tag{7.30}$$

Regard R_i and $\mathcal{L}_i$ as affine functions of the coordinate t. At $t = c_i$, using (7.27),

$$\begin{aligned} R_i(c_i) - \mathcal{L}_i(c_i) &= \chi_i - (f_i + \ell_i(c_i - t_i)) \\ &= (\ell_i - s_i)(t_i - c_i) > 0. \end{aligned} \tag{7.31}$$

At $t = t_i$,

$$R_i(t_i) - \mathcal{L}_i(t_i) = (r_i - s_i)(t_i - c_i) < 0. \tag{7.32}$$

Since $r_i < \ell_i$, the two lines have a unique intersection, and the opposite signs at c_i and t_i locate it at $c_i < z_i < t_i$. Because $Z_i \in \mathcal{L}_i$, its vertical difference from the backward extension of $[X_i, X_{i+1}]$ equals

$$g_i - (f_i + s_{i+1}(z_i - t_i)) = (\ell_i - s_{i+1})(z_i - t_i) > 0. \tag{7.33}$$

Here both factors on the right are negative. This completes the induction.

Let $H = \operatorname{aff}(Z_m, X_{m+1})$, and denote its slope by η. Since $Z_m \in \mathcal{L}_m$, direct subtraction of the endpoint ordinates gives

$$\eta = \frac{(t_m - z_m)\ell_m + (t_{m+1} - t_m)s_{m+1}}{t_{m+1} - z_m}. \tag{7.34}$$

The two coefficients in the numerator are positive and sum to the denominator. Therefore

$$\ell_m < \eta < s_{m+1}. \tag{7.35}$$

The line $R_m = \operatorname{aff}(C_m, Z_m)$ has slope $r_m < s_m < \ell_m < \eta$. Since $c_m < z_m$,

$$H(c_m) - \chi_m = (\eta - r_m)(c_m - z_m) < 0. \tag{7.36}$$

On the other hand, H passes through X_{m+1}, while C_{m+1} lies on the last chain edge of slope s_{m+1}; hence

$$H(c_{m+1}) - \chi_{m+1} = (\eta - s_{m+1})(c_{m+1} - t_{m+1}) > 0. \tag{7.37}$$

Let $K = \operatorname{aff}(C_m, C_{m+1})$. The affine function $H - K$ is negative at c_m and positive at c_{m+1}. Because $c_m < t_m < c_{m+1}$, it has a zero at an abscissa strictly between c_m and c_{m+1}. This zero is exactly the transformed point corresponding to W_*, and it lies in $\operatorname{relint}[C_m, C_{m+1}]$ in the admissible chart. The projective transformation used above and its inverse map every segment contained in the polygon onto the corresponding segment and preserve relative interiors, as proved in proposition 7.5. Applying the inverse transformation therefore gives (7.24) in the original affine chart. Distinctness of the two final projective lines is likewise preserved.

If the boundary chain is read in the opposite cyclic orientation, apply proposition 7.5 to that oriented chain and repeat the same calculation; no sign or index change is required inside the argument. □

Lemma 7.7 (Fixed-point escape for a projectivity)**.** *Let u be a nonconstant real projectivity, written in an affine coordinate near 0, such that*

$$u(0) = 0, \qquad 0 < u(1) < 1.$$

Then there are arbitrarily small nonzero real τ for which

$$\tau - u(\tau) > 0. \tag{7.38}$$

Proof. A nonconstant projectivity fixing 0 has, on a pole-free neighborhood of 0, the normal form

$$u(\tau) = \frac{a\tau}{1 + c\tau}, \qquad a \neq 0. \tag{7.39}$$

Consequently

$$\tau - u(\tau) = \frac{\tau(1 - a + c\tau)}{1 + c\tau}. \tag{7.40}$$

If $a \neq 1$, choose the sign of τ so that $\tau(1 - a) > 0$, and then choose $|\tau|$ small enough that $1 + c\tau > 0$ and $1 - a + c\tau$ has the sign of $1 - a$. The right side of (7.40) is then positive. If $a = 1$, the condition $0 < u(1) < 1$ gives $0 < 1/(1 + c) < 1$, hence $c > 0$; for every sufficiently small nonzero τ,

$$\tau - u(\tau) = \frac{c\tau^2}{1 + c\tau} > 0.$$

In either case the admissible τ's can be chosen arbitrarily close to 0. □

Theorem 7.8 (Projective corridor escape)**.** *Let a projective corridor be given, and define the moving chain by*

$$X_1(\tau) = (1 - \tau)X_1 + \tau X_0, \qquad X_i(\tau) = \mathcal{L}_i \cap \operatorname{aff}(X_{i-1}(\tau), C_i) \quad (2 \leq i \leq m). \tag{7.41}$$

Put

$$Y(\tau) = (1 - \tau)C_{m+1} + \tau C_m.$$

The recursive intersections are finite and continuous on some open interval about 0. Every sufficiently small neighborhood of 0 contains a nonzero signed parameter τ for which $Y(\tau)$ lies in the open half-plane bounded by $\operatorname{aff}(X_m(\tau), X_{m+1})$ that contains $X_{m-1}(\tau)$. For all parameters in a sufficiently small such interval, the displayed vertices remain in strict convex position with their original cyclic order.

Proof. Put

$$A_1 = \overline{\operatorname{aff}(X_0, X_1)}, \qquad A_i = \overline{\mathcal{L}_i} \quad (2 \leq i \leq m), \qquad K = \overline{\operatorname{aff}(C_m, C_{m+1})},$$

where the bars denote projective completion. We first verify every perspectivity in the construction.

For $i = 2$, the center $C_2 \in \operatorname{relint}[X_1, X_2]$ is not on A_1, and strictness of $\mathcal{L}_2$ gives $C_2 \notin A_2$. For $3 \le i \le m$, the point $C_i \in \operatorname{relint}[X_{i-1}, X_i]$ lies on neither the strict support A_{i-1} at X_{i-1} nor the strict support A_i at X_i. Hence projection from center C_i defines a projective isomorphism $A_{i-1} \to A_i$. For the closing projection, strictness gives $X_{m+1} \notin A_m$. Also $X_{m+1} \notin K$: if it were, then K, which already contains $C_{m+1} \in \operatorname{relint}[X_m, X_{m+1}]$, would equal the side line $\operatorname{aff}(X_m, X_{m+1})$; this would put $C_m \in \operatorname{relint}[X_{m-1}, X_m]$ on that adjacent side line, contradicting strictness. Thus projection from X_{m+1} is a projective isomorphism $A_m \to K$.

The affine seed parametrization

$$\tau \longmapsto X_1(\tau) = (1-\tau)X_1 + \tau X_0$$

extends to a projective isomorphism from the compactified parameter line to A_1. Composing it with the verified perspectivities gives a single global projectivity

$$\mathcal{H} : \mathbb{P}^1(\mathbb{R}) \longrightarrow K. \tag{7.42}$$

Let

$$\upsilon : K \longrightarrow \mathbb{P}^1(\mathbb{R})$$

be the projective coordinate induced by the original affine line $\operatorname{aff}(C_m, C_{m+1})$, normalized by $\upsilon(C_{m+1}) = 0$ and $\upsilon(C_m) = 1$, and define the global real projectivity

$$u = \upsilon \circ \mathcal{H}.$$

Both $\mathcal{H}$ and υ are projective isomorphisms; hence u is a nonconstant real projectivity, as required by lemma 7.7. At $\tau = 0$, the projective recursion gives $\mathcal{H}(0) = C_{m+1}$, so $u(0) = 0$. At $\tau = 1$, the seed is X_0; projective naturality of the recursion gives $\mathcal{H}(1) = W_*$, where W_* is the point of lemma 7.6. Since $W_* \in \operatorname{relint}[C_{m+1}, C_m]$,

$$0 < u(1) < 1. \tag{7.43}$$

At $\tau = 0$, every recursive intersection is the original vertex X_i, and the closing intersection is C_{m+1}. Therefore none of the intermediate affine coordinate maps has a pole at 0. Since each has at most one pole, there is an open interval I about 0 on which all points $X_i(\tau)$ and the closing intersection

$$W(\tau) = \operatorname{aff}(X_m(\tau), X_{m+1}) \cap \operatorname{aff}(C_m, C_{m+1}) \tag{7.44}$$

are finite and continuous. In particular, the two lines in (7.44) are distinct throughout I. On this interval the local affine recursion agrees with the global holonomy: $W(\tau) = \mathcal{H}(\tau)$, and $u(\tau)$ is precisely the normalized affine coordinate of $W(\tau)$.

By lemma 7.7, the set of nonzero parameters satisfying $\tau - u(\tau) > 0$ accumulates at 0. We postpone the choice of such a parameter until all remaining open conditions have been fixed.

Set

$$z(t) = (1-t)C_{m+1} + tC_m.$$

Choose $\varepsilon \in \{1, -1\}$ so that

$$\varepsilon \det(X_{m+1} - X_m, X_{m-1} - X_m) > 0,$$

and define

$$D(t, \tau) = \varepsilon \det\big(X_{m+1} - X_m(\tau), z(t) - X_m(\tau)\big). \tag{7.45}$$

Thus the positive sign is calibrated by the preceding chain vertex, without reference to any undeclared moving polygon. For $\tau \in I$, the point $z(u(\tau)) = W(\tau)$ is the unique intersection of that closing line with $\operatorname{aff}(C_m, C_{m+1})$. Since D is affine in t,

$$D(t, \tau) = \gamma(\tau)(t - u(\tau)), \qquad \gamma(\tau) = \varepsilon \det\big(X_{m+1} - X_m(\tau), C_m - C_{m+1}\big). \tag{7.46}$$

The coefficient γ is continuous. Write $C_m = (1-\sigma)X_{m-1} + \sigma X_m$ with $0 < \sigma < 1$. The calibration of ε gives

$$\gamma(0) = D(1,0) = (1-\sigma)\varepsilon \det(X_{m+1} - X_m, X_{m-1} - X_m) > 0. \tag{7.47}$$

After shrinking I, if necessary, $\gamma(\tau) > 0$ throughout I.

Fix the orientation of the original displayed chain. By lemma 2.6, every cyclically ordered triple of its displayed vertices has a nonzero determinant with one common sign at $\tau = 0$. There are finitely many such determinants, and all vertices depend continuously on τ on I. Shrink I once more so that every one of these determinants keeps its sign throughout I. The triple-sign criterion, lemma 2.7, then shows that the displayed vertices remain distinct, in the same cyclic order, and in strict convex position for every $\tau \in I$. In particular,

$$\varepsilon \det\big(X_{m+1} - X_m(\tau), X_{m-1}(\tau) - X_m(\tau)\big) > 0$$

throughout I. Hence the half-plane described by $D(t,\tau) > 0$ is exactly the open half-plane of the moving closing line that contains the preceding chain vertex.

Now use the accumulation statement from lemma 7.7 to choose a nonzero $\tau \in I$ with $\tau - u(\tau) > 0$. Evaluating (7.46) at $t = \tau$ gives

$$D(\tau,\tau) = \gamma(\tau)(\tau - u(\tau)) > 0.$$

Since $Y(\tau) = z(\tau)$, this says precisely that the closing image lies in the open half-plane of the moving closing line that contains $X_{m-1}(\tau)$. Because the escape parameters accumulate at 0, the chosen parameter can be made arbitrarily small. □

7.2 Global realization of the corridor motion

The local projective motion is useful only after it has been transported to all N polygon vertices and all return incidences. The next lemma makes that passage a single closed contract. In particular, no incidence or vertex label is left to an implicit convention.

Lemma 7.9 (Deformation admissibility for the global return corridor)**.** *Assume $\varphi > \delta$ and $\Delta > 1$, choose the forward or reverse corridor from the branch table, and suppose that its displayed boundary chain is proper. Let $M, b_*, M^\circ, D, R, c, A$ be the sets of proposition 7.3. For $0 \le i \le m+1$, define the chain label*

$$\iota_i = \begin{cases} i, & \text{in the forward branch,} \\ \varphi - i, & \text{in the reverse branch,} \end{cases}$$

so that $X_i = x_{\iota_i}$. For $2 \le i \le m+1$, define the contact-field label

$$\gamma_i = \begin{cases} i, & \text{in the forward branch,} \\ \varphi - i + 1, & \text{in the reverse branch,} \end{cases}$$

so that $C_i = \xi_{\gamma_i}$. Then the map $i \mapsto \iota_i$ is a bijection $\{1,\dots,m\} \to M$, the map $i \mapsto \gamma_i$ is a bijection $\{2,\dots,m\} \to D$, and

$$b_* = \iota_1, \qquad c = \gamma_{m+1} = r(b_*), \qquad a = H_{b_*}.$$

In particular, the chain and contact labels have no repetitions. Moreover, $a = q$ in the forward branch while $a = q + h$ in the reverse branch.

Define the moving chain by

$$X_1(\tau) = (1-\tau)X_1 + \tau X_0, \qquad X_i(\tau) = \mathcal{L}_i \cap \operatorname{aff}\big(X_{i-1}(\tau), C_i\big) \quad (2 \le i \le m). \tag{7.48}$$

There is a pole-free open interval U_0 about 0 on which all these points are finite and continuous. On U_0, define one base function for each $j \in \mathcal{B} = \{1, \dots, \varphi\}$ by

$$B_j(\tau) = \begin{cases} X_i(\tau), & j = \iota_i \text{ for some } 1 \le i \le m, \\ x_j, & j \notin M, \end{cases} \tag{7.49}$$

and define all N polygon-vertex functions by the tower formula

$$\widehat{x}_{F(t,j)}(\tau) = \lambda^t B_j(\tau) \qquad (1 \le j \le \varphi,\ 0 \le t < H_j). \tag{7.50}$$

For $k \in \mathbb{Z}/N\mathbb{Z}$, put

$$\widehat{E}_k(\tau) = [\widehat{x}_{k-1}(\tau), \widehat{x}_k(\tau)], \qquad P_\tau = \operatorname{conv}\{\widehat{x}_k(\tau) : k \in \mathbb{Z}/N\mathbb{Z}\}.$$

Then, after shrinking U_0 to an open interval U about 0, the following statements hold for every $\tau \in U$.

(i) Unique global assignment and cyclic order. *Equation (7.50) assigns exactly one function to each polygon label, every function is continuous, $\widehat{x}_k(0) = x_k$, and the N points $\widehat{x}_k(\tau)$ are distinct extreme points in their original positive cyclic order. Hence P_τ is a strict N-gon with sides $\widehat{E}_k(\tau)$.*

(ii) Exact internal propagation. *For every tower state with $t + 1 < H_j$,*

$$\lambda \widehat{x}_{F(t,j)}(\tau) = \widehat{x}_{F(t+1,j)}(\tau). \tag{7.51}$$

Thus every internal tower image is exactly a polygon vertex, with no side or endpoint convention involved.

(iii) Complete return-edge accounting. *For each strict field $k \in \mathcal{B}$, let*

$$\widehat{V}_k(\tau) = \lambda^{H_{s(k)}} B_{s(k)}(\tau). \tag{7.52}$$

For every $k \in \mathcal{B} \setminus \{c\}$,

$$\widehat{V}_k(\tau) \in \operatorname{relint} \widehat{E}_k(\tau). \tag{7.53}$$

The sole return incidence not constrained to its side is the seed return in field c:

$$Y(\tau) := \widehat{V}_c(\tau) = \lambda^a X_1(\tau) = (1 - \tau) C_{m+1} + \tau C_m. \tag{7.54}$$

Its assigned side is exactly

$$\widehat{E}_c(\tau) = [X_m(\tau), X_{m+1}], \tag{7.55}$$

where the endpoints may appear in the reverse order relative to the positive boundary orientation in the reverse branch.

(iv) All nonclosing side inequalities for the closing image. *For every polygon side label k, define*

$$G_k(\tau) = \det\big(\widehat{x}_k(\tau) - \widehat{x}_{k-1}(\tau), Y(\tau) - \widehat{x}_{k-1}(\tau)\big). \tag{7.56}$$

Then

$$G_k(\tau) > 0 \qquad (k \ne c,\ \tau \in U). \tag{7.57}$$

Thus the closing side is the only side inequality not fixed by openness.

(v) Exact admissibility criterion and uniqueness of the defect. *For every $\tau \in U$, all image vertices other than $Y(\tau)$ lie on ∂P_τ, while*

$$Y(\tau) \in \operatorname{Ext}(\lambda P_\tau).$$

If, in addition, $G_c(\tau) > 0$, then

$$Y(\tau) \in \operatorname{int} P_\tau, \qquad \lambda P_\tau \subseteq P_\tau, \qquad \operatorname{Ext}(\lambda P_\tau) \cap \operatorname{int} P_\tau = \{Y(\tau)\}. \tag{7.58}$$

Proof. We prove the five conclusions in six explicit stages.

1. The local chain and the fixed chain endpoints. By lemma 7.1, every line $\mathcal{L}_i$ in (7.48) is a strict support of P at X_i. Together with (7.8), the displayed data form a projective corridor in the sense of definition 7.4. The first assertion of theorem 7.8 therefore supplies a pole-free open interval U_0 on which the recursion is finite and continuous. At $\tau = 0$, induction gives $X_i(0) = X_i$: for $i = 1$ this is immediate, and for $i \geq 2$ both lines in the defining intersection pass through X_i and are distinct because $\mathcal{L}_i$ is a strict support while $C_i \neq X_i$ lies on the incident side.

The endpoints X_0 and X_{m+1} are fixed global tower vertices. In the forward branch, $X_{m+1} = x_{\Delta+1}$ is an unmoved base. Also $F(h, \varphi) = 0$ and $h < H_\varphi = q + h$; when $h = 0$, this statement is the same identity with $F(0, \varphi) = 0$ and $\varphi = N$. Hence

$$\widehat{x}_0(\tau) = \lambda^h B_\varphi(\tau) = \lambda^h x_\varphi = x_0 = X_0.$$

In the reverse branch, $X_0 = x_\varphi$ and $X_{m+1} = x_{\Delta-2}$ are unmoved bases. The reverse inequality forces $\Delta \geq 3$, so the latter label belongs to $\mathcal{B}$. Thus in both branches the constants appearing in the local corridor are exactly the corresponding vertices in the global tower assignment.

2. The tower formula is a unique continuous labeling. The branch formulas show directly that $i \mapsto \iota_i$ is a bijection from $\{1, \ldots, m\}$ onto M. Consequently (7.49) assigns exactly one formula to every base label: a moved formula for $j \in M$ and the fixed formula x_j for $j \notin M$.

The map F in (6.7) is a bijection from the finite state set

$$\mathcal{D} = \{(t, j) : 1 \leq j \leq \varphi,\ 0 \leq t < H_j\}$$

onto $\mathbb{Z}/N\mathbb{Z}$. Therefore every polygon label has exactly one preimage (t, j), so (7.50) gives one and only one definition for every $\widehat{x}_k(\tau)$. In particular, no moved formula and no fixed formula can assign different points to the same polygon label. Each base function is continuous and multiplication by the fixed linear map λ^t is continuous, hence all global vertex functions are continuous.

At $\tau = 0$, the geometric tower identity (6.9) gives

$$\widehat{x}_{F(t,j)}(0) = \lambda^t B_j(0) = \lambda^t x_j = x_{F(t,j)}.$$

Since F is onto, $\widehat{x}_k(0) = x_k$ for every polygon label k.

Apply the simultaneous convex-admissibility lemma lemma 2.8 first with no auxiliary test points. It gives an open interval $U_1 \subseteq U_0$ on which the global vertex list is distinct, retains its positive cyclic order, and is in strict convex position. This proves part (i) on U_1.

Equation (7.51) is now an identity:

$$\lambda \widehat{x}_{F(t,j)}(\tau) = \lambda^{t+1} B_j(\tau) = \widehat{x}_{F(t+1,j)}(\tau).$$

This proves part (ii).

3. Side-line register for the strict fields. Write

$$\ell_k(\tau) = \operatorname{aff}\big(\widehat{x}_{k-1}(\tau), \widehat{x}_k(\tau)\big) \qquad (k \in \mathcal{B}).$$

We identify every one of these lines.

The branch formulas also show that $i \mapsto \gamma_i$ maps $\{2, \ldots, m\}$ bijectively onto D and sends $m + 1$ to c.

If $k \in D$, there is therefore a unique chain index

$$\psi(k) = \begin{cases} k, & \text{forward branch,} \\ \varphi - k + 1, & \text{reverse branch,} \end{cases} \qquad 2 \leq \psi(k) \leq m,$$

for which $C_{\psi(k)} = \xi_k$. The two endpoints of $\widehat{E}_k(\tau)$ are $X_{\psi(k)-1}(\tau)$ and $X_{\psi(k)}(\tau)$, possibly in reverse order. By the recursion,

$$\ell_k(\tau) = \operatorname{aff}\bigl(X_{\psi(k)-1}(\tau), C_{\psi(k)}\bigr). \tag{7.59}$$

If $k \in R \cup A$, the line $\ell_k(\tau)$ is fixed. This is exactly part (ii) of proposition 7.3, applied to the present base motions: the seed remains on the fixed anchor line, and the only moved base labels are those in M. Finally, the one remaining strict side is the closing field c, and direct reading of the branch table gives (7.55). Thus the line register

$$\mathcal{B} = D \mathbin{\dot\cup} R \mathbin{\dot\cup} \{c\} \mathbin{\dot\cup} A$$

accounts for every strict side exactly once.

4. Return-edge register and exact collinearity. First partition the entire tower-state set into internal and top sources:

$$\begin{aligned} \mathcal{D}_{\text{int}} &= \{(t,j) \in \mathcal{D} : t+1 < H_j\}, \\ \mathcal{D}_{\text{top}} &= \{(H_j - 1, j) : j \in \mathcal{B}\}. \end{aligned} \tag{7.60}$$

These sets are disjoint, their union is $\mathcal{D}$, and

$$|\mathcal{D}_{\text{int}}| = N - \varphi, \qquad |\mathcal{D}_{\text{top}}| = \varphi.$$

Because F is bijective, they index disjoint sets of polygon vertices whose union is the complete set of N source vertices. Part (ii) has already accounted for every source in $\mathcal{D}_{\text{int}}$. The image of the top source $(H_j - 1, j)$ is $\lambda^{H_j} B_j(\tau) = \widehat{V}_{r(j)}(\tau)$. Since r is a bijection of $\mathcal{B}$, every top edge is therefore indexed exactly once by its target field k, with unique source base $s(k)$.

We now apply the inverse-source partition (7.20) to those φ top edges field by field. The complete ledger is

field class	source base	target side line	exact mechanism
D	fixed, since $s(D) \cap M = \varnothing$	controlled by the corridor recursion	the fixed top image $\xi_k = C_{\psi(k)}$ is built into that line
R	the supported moved base $s(k) \in M^\circ$	fixed	the base remains on the pulled-back support $L_{s(k)}$
A	fixed	fixed	the original incidence is unchanged
$\{c\}$	the seed b_*	the closing line	the unique unconstrained return $Y(\tau)$

The row sets are disjoint and exhaustive by (7.18); the source assertions are exactly (7.20).

If $k \in D$, then $s(k) \notin M$, so its source base is fixed. The original top relation gives

$$\widehat{V}_k(\tau) = \lambda^{H_{s(k)}} x_{s(k)} = \xi_k = C_{\psi(k)}.$$

Equation (7.59) puts this fixed point on the moving line $\ell_k(\tau)$.

If $k \in R$, then $s(k) \in M^\circ$. Let i be the unique chain index with $\iota_i = s(k)$; necessarily $2 \le i \le m$. The recursion puts $B_{s(k)}(\tau) = X_i(\tau)$ on

$$\mathcal{L}_i = L_{s(k)} = \lambda^{-H_{s(k)}} \ell_k(0).$$

The target line is fixed, so

$$\widehat{V}_k(\tau) = \lambda^{H_{s(k)}} B_{s(k)}(\tau) \in \ell_k(0) = \ell_k(\tau).$$

If $k \in A$, both its source base and its target side line are fixed. Hence

$$\widehat{V}_k(\tau) = \xi_k \in \ell_k(0) = \ell_k(\tau).$$

The only remaining field is c. Here $s(c) = b_*$, so

$$\widehat{V}_c(\tau) = \lambda^{H_{b_*}} B_{b_*}(\tau) = \lambda^a X_1(\tau).$$

Using the seed interpolation and (7.9),

$$\lambda^a X_1(\tau) = (1-\tau)\lambda^a X_1 + \tau\lambda^a X_0 = (1-\tau)C_{m+1} + \tau C_m.$$

This proves (7.54). The partition from proposition 7.3 proves simultaneously that no return edge is missing and that none has received two mechanisms.

For every nonclosing field k, exact collinearity has now been proved, and at $\tau = 0$ the point $\widehat{V}_k(0) = \xi_k$ belongs to relint E_k. Apply lemma 2.8 to the finite family

$$w_k(\tau) = \widehat{V}_k(\tau), \qquad k \in \mathcal{B} \setminus \{c\}.$$

After shrinking U_1 to an interval U_2, every one of these points lies in the relative interior of its assigned moving side. This proves (7.53) and part (iii).

5. All nonclosing inequalities for the seed image. At $\tau = 0$,

$$Y(0) = C_{m+1} = \xi_c \in \operatorname{relint} E_c.$$

By (2.11) in lemma 2.8, this point satisfies the strict side inequality for every $k \neq c$. Apply the same lemma to the finite family of test pairs

$$(Y(\tau), k), \qquad k \in \mathbb{Z}/N\mathbb{Z} \setminus \{c\}.$$

After shrinking U_2 to an interval $U \ni 0$, the functions (7.56) satisfy (7.57). This is a global statement: it includes sides whose endpoints are internal tower vertices, not merely the displayed corridor sides. This proves part (iv).

6. Exact admissibility and uniqueness of the closing defect. Let $k \in \mathbb{Z}/N\mathbb{Z}$ and let (t, j) be its unique tower preimage under F. If $t + 1 < H_j$, then (7.51) gives $\lambda\widehat{x}_k(\tau)$ as another polygon vertex. If $t = H_j - 1$ and $r(j) \neq c$, then

$$\lambda\widehat{x}_k(\tau) = \lambda^{H_j} B_j(\tau) = \widehat{V}_{r(j)}(\tau) \in \operatorname{relint} \widehat{E}_{r(j)}(\tau).$$

Thus every image vertex other than the closing image lies on ∂P_τ for every $\tau \in U$.

Consecutive record times satisfy $q > 0$ and $h \geq 0$, so $a = H_{b_*} \geq 1$. Put

$$k_* = F(a-1, b_*).$$

Then

$$\widehat{x}_{k_*}(\tau) = \lambda^{a-1} B_{b_*}(\tau), \qquad \lambda\widehat{x}_{k_*}(\tau) = Y(\tau).$$

The point $\widehat{x}_{k_*}(\tau)$ is an extreme point of the strict polygon P_τ. Since multiplication by $\lambda \neq 0$ is invertible, it maps extreme points bijectively to extreme points; hence $Y(\tau) \in \operatorname{Ext}(\lambda P_\tau)$ for every $\tau \in U$.

Now fix any $\tau \in U$ for which $G_c(\tau) > 0$. Together with (7.57), this gives all N strict side inequalities for $Y(\tau)$. The final assertion of lemma 2.8 therefore yields $Y(\tau) \in \operatorname{int} P_\tau$. Every vertex image belongs to P_τ: internal images are polygon vertices, nonclosing top images lie in relative side interiors, and the closing top image is $Y(\tau)$. Convexity and linearity give

$$\lambda P_\tau = \operatorname{conv}\{\lambda\widehat{x}_k(\tau) : k \in \mathbb{Z}/N\mathbb{Z}\} \subseteq P_\tau.$$

All image vertices except $Y(\tau)$ lie on the boundary, while $Y(\tau)$ is interior. This proves (7.58) and part (v). ∎

Theorem 7.10 (Global return-corridor deformation). *Under the hypotheses and notation of lemma 7.9, every neighborhood of* 0 *contains a nonzero parameter* $\tau \in U$ *for which*

$$Y(\tau) \in \operatorname{int} P_\tau, \qquad \lambda P_\tau \subseteq P_\tau, \qquad \operatorname{Ext}(\lambda P_\tau) \cap \operatorname{int} P_\tau = \{Y(\tau)\}. \tag{7.61}$$

In particular, $Y(\tau) \in \operatorname{Ext}(\lambda P_\tau)$ *is the unique image vertex that leaves the boundary, and it leaves it toward the polygon interior.*

Proof. The projective corridor escape theorem theorem 7.8 supplies, arbitrarily close to 0, a nonzero parameter for which $Y(\tau)$ lies in the open half-plane of the moving closing line that contains $X_{m-1}(\tau)$. Choose that parameter inside the common admissibility interval U from lemma 7.9. By part (i) of that lemma, P_τ is a strict polygon with the original cyclic order. The nonincident vertex $X_{m-1}(\tau)$ therefore lies strictly in the interior half-plane of the closing side. The projective escape condition is exactly $G_c(\tau) > 0$. Part (v) of lemma 7.9 now gives (7.61). □

7.3 The no-skipping conclusion

Theorem 7.11 (Projective unit return). *If* $\varphi > \delta$*, then* $\Delta = 1$*.*

Proof. Assume $\Delta > 1$. By lemma 7.2, the selected branch has $m \geq 2$, its displayed vertices are pairwise distinct consecutive boundary vertices, and it omits at least one polygon side. Thus it is a proper projective corridor. The same lemma verifies this directly at $\Delta = 2$, $2\Delta = \varphi + 1$, $h = 0$, and $\varphi = N$. Together with lemma 7.1 and proposition 7.3, all hypotheses of lemma 7.9 and theorem 7.10 are satisfied.

Choose the nonzero parameter supplied by that theorem. Then P_τ is a strict invariant N-gon for the same N-critical map, while the vertex $Y(\tau)$ of λP_τ lies in $\operatorname{int} P_\tau$. This contradicts hereditary image-vertex saturation in theorem 3.2. Hence $\Delta > 1$ is impossible. Since Δ is a positive integer, $\Delta = 1$. □

Remark 7.12 (Boundary-value ledger for no-skipping). No exceptional parameter is suppressed in the preceding proof.

(a) If $\Delta = 2$, the reverse inequality is impossible because $\Delta < \varphi$; the forward corridor has $m = 2$, one strict support $\mathcal{L}_2$, one controlled field, and the same four-set ledger.

(b) The equality $2\Delta = \varphi + 1$ belongs to the forward branch. It is exactly the wrap $r(\Delta) = 1$ displayed in (7.14); field 1 remains on the fixed anchor line.

(c) The equality $h = 0$ is equivalent to $\varphi = N$ by theorem 6.1. Then $\lambda^h E_\varphi = E_0$ is the cyclic identity, and every formula in the corridor and global tower construction remains literal.

(d) The orbit number δ, including $\delta = 1$, enters the no-skipping module only through the hypothesis $\varphi > \delta$ and the existence of the return section. It creates no additional deformation case.

Remark 7.13 (Protective invariant). The local invariant controlling the escape is the fixed-point germ of the normalized corridor holonomy in $\mathrm{PGL}_2(\mathbb{R})$. The global theorem above adds the missing combinatorial and convex-geometric contract: the tower bijection labels every moved and unmoved vertex exactly once, the four-set return-edge ledger leaves one and only one unconstrained return, and all N side inequalities are checked before the escaping parameter is selected.

Proof of theorem 1.3. Choose an adapted complex coordinate by proposition 2.1. Hereditary saturation is theorem 3.2. The least-area selection and cyclic interlacing in lemma 4.13 give a one-sided contact representation after choosing the appropriate orientation. Write the resulting positive indexing as in lemma 4.13, so that $E_i = [x_{i-1}, x_i]$ and $\chi(x_{i-\kappa}) = E_i$. If $\kappa = N$, then

$$\sigma(E_i) = \chi(h(E_i)) = \chi(x_i) = E_{i+\kappa} = E_i,$$

so σ is the identity. The final assertion of lemma 4.13 says that every contact is strict, and hence $I = \mathcal{E}(P)$. There are therefore no legal mutations, so assertion (ii) is vacuous; the only mutation-reachable strict set is I itself, which is a cyclic interval with $\varphi = N = \delta$. Assertions (iii) and (iv) follow immediately. This completes the identity case.

Henceforth assume $1 \leq \kappa < N$. The indexed surgery certificate is proposition 5.1, and corollary 5.2 translates its update $i \mapsto i + \kappa$ exactly into $e \mapsto \sigma(e)$, with the same contact rotation and with real-linear covariance. Apply lemma 5.5 to a lexicographically minimal mutation-reachable strict set. It is one interval of length $\varphi \geq \delta$, where δ is the number of contact-rotation orbits. If $\varphi = \delta$, the interval meets every orbit and has exactly one point in each; hence it is a complete transversal, and the common return time is the orbit length N/δ.

Suppose $\varphi > \delta$. The last assertion of lemma 5.5 makes $N - \varphi$ an upper-record residue. Apply theorem 6.1 with $\nu = \varphi$. It gives a two-height tower partition and a first-return rotation by Δ. The projective unit-return theorem, theorem 7.11, gives $\Delta = 1$. Thus the first return is the cyclic successor on the strict interval. All parts of the theorem have now been proved. □

8 Complex return monodromy and Farey arithmetic

Let F_n denote the Farey sequence of order n on $[0, 1]$.

Lemma 8.1 (Farey adjacency criterion)**.** *Two reduced fractions $a/b < c/d$ in F_n are consecutive if and only if*

$$bc - ad = 1, \qquad b + d > n. \tag{8.1}$$

Proof. Assume first that $bc - ad = 1$. If a reduced fraction h/k lies strictly between a/b and c/d, define

$$m = ck - dh, \qquad \ell = bh - ak.$$

Cross multiplication shows that m and ℓ are positive integers. The determinant-one identity gives the exact decomposition

$$(k, h) = m(b, a) + \ell(d, c).$$

Looking at the first coordinate yields $k = mb + \ell d \geq b + d$. Hence, if $b + d > n$, no fraction of denominator at most n can lie between the two endpoints, so they are consecutive in F_n.

Conversely, suppose the two fractions are consecutive in F_n and put $D = bc - ad$. The determinant is a positive integer. If $D > 1$, the half-open parallelogram spanned by the primitive integer vectors $u = (b, a)$ and $v = (d, c)$ contains a nonzero integer point

$$w = \alpha u + \beta v, \qquad 0 < \alpha, \beta < 1.$$

Indeed, lemma A.6 says that the half-open fundamental parallelogram contains exactly $D = |\det(u, v)|$ representatives of the quotient group $\mathbb{Z}^2/(\mathbb{Z}u + \mathbb{Z}v)$. When $D > 1$, one of these representatives is different from the origin. It cannot lie on either open coordinate edge of the parallelogram: such a point would be a nonintegral proper multiple of the primitive vector u or v. The strict inequalities follow because a primitive vector has no integer point in the relative interior of its first unit segment. Both w and $u + v - w$ are positive combinations of u, v, so their slopes lie strictly between a/b and c/d. Their first coordinates are positive integers k and $b + d - k$; one is at most $(b + d)/2 \leq n$ because $b, d \leq n$. After reducing that intermediate slope, its denominator can only decrease. This produces an element of F_n strictly between the alleged consecutive fractions, a contradiction. Therefore $D = 1$. Finally, if $b + d \leq n$, the reduced mediant $(a + c)/(b + d)$ has denominator at most n and lies strictly between the endpoints. Thus consecutiveness also forces $b + d > n$. □

Lemma 8.2 (Reflection of a Farey cell)**.** *Let $a/b < x < c/d$ be the open cell between two consecutive fractions of F_N. Then*

$$\frac{d-c}{d} < 1 - x < \frac{b-a}{b} \tag{8.2}$$

is also an open cell between consecutive fractions of F_N. The reflection exchanges the endpoint denominators: the left and right denominators in (8.2) are d and b, respectively. In particular, if $d \le b$, the reflected ordered cell has the smaller denominator first.

Proof. Reducedness is preserved because

$$\gcd(d-c, d) = \gcd(c, d) = 1, \qquad \gcd(b-a, b) = \gcd(a, b) = 1.$$

The inequalities follow by subtracting the original inequalities from 1. Moreover

$$(b-a)d - (d-c)b = bc - ad = 1,$$

and the sum of the reflected denominators is the unchanged number $b + d > N$. The Farey adjacency criterion, lemma 8.1, therefore proves consecutiveness. The denominator statement is immediate from the displayed fractions. $\square$

Lemma 8.3 (Reflection of a backward return strip)**.** *Let $\lambda \in \mathbb{C} \setminus \mathbb{R}$ satisfy $0 < |\lambda| < 1$, let $\theta = \arg_+(\lambda)$, and let $q, d \ge 1$ and $0 \le h < dq$. Suppose nonzero points $z_0, \dots, z_d$, numbers*

$$0 \le b_i < 1, \qquad a_i > 0 \qquad (1 \le i \le d),$$

and lifted arguments Θ_i satisfy

$$(\lambda^q - b_i) z_i = a_i z_{i-1} \qquad (1 \le i \le d), \tag{8.3}$$

$$\lambda^h z_d = z_0, \tag{8.4}$$

$$0 < \Theta_i - \Theta_{i-1} < \pi \qquad (1 \le i \le d), \tag{8.5}$$

$$\Theta_d + h\theta = \Theta_0 + 2\pi m \tag{8.6}$$

for an integer m.

Put

$$\mu = \overline{\lambda}, \qquad \vartheta = \arg_+(\mu) = 2\pi - \theta, \qquad e = -h, \qquad s = dq - h,$$

and, for $1 \le j \le d$, define

$$w_j = \overline{z}_{d-j}, \qquad w_0 = \overline{z}_d, \qquad \beta'_j = b_{d-j+1}, \qquad \alpha'_j = a_{d-j+1}.$$

Then

$$(\mu^q - \beta'_j) w_{j-1} = \alpha'_j w_j \qquad (1 \le j \le d), \tag{8.7}$$

$$\mu^e w_d = w_0. \tag{8.8}$$

Consequently

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta'_j) = \prod_{j=1}^{d} \alpha'_j, \tag{8.9}$$

$$\mu^s \prod_{j=1}^{d} (\mu^q - \beta'_j) = \mu^{dq} \prod_{j=1}^{d} \alpha'_j. \tag{8.10}$$

If

$$u_j = \Theta_{d-j+1} - \Theta_{d-j} \in (0, \pi),$$

then u_j is the argument selected by (8.7)*, and*

$$e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(m-h). \tag{8.11}$$

Proof. Conjugate (8.3) with $i = d - j + 1$. Since

$$\overline{z}_{d-j+1} = w_{j-1}, \qquad \overline{z}_{d-j} = w_j,$$

the conjugated equation is exactly (8.7). Conjugating (8.4) gives

$$\mu^h \overline{z}_d = \overline{z}_0.$$

Because $w_0 = \overline{z}_d$ and $w_d = \overline{z}_0$, this is equivalent, using $\mu \neq 0$, to (8.8) with $e = -h$. Multiplying the forward cell equations and cancelling the nonzero points w_0, w_d by the signed closure proves (8.9). Multiplication by μ^{dq} gives (8.10); the exponent on its left is $dq - h = s > 0$ by hypothesis.

Choose lifted arguments for the reflected points by

$$\Phi_j = -\Theta_{d-j} + C,$$

where the constant C is arbitrary. Then

$$\Phi_j - \Phi_{j-1} = \Theta_{d-j+1} - \Theta_{d-j} = u_j \in (0, \pi).$$

Since $\alpha'_j > 0$, equation (8.7) shows that u_j is congruent modulo 2π to the argument of $\mu^q - \beta'_j$. Its membership in $(0, \pi)$ selects the oriented branch uniquely. Finally,

$$\sum_{j=1}^{d} u_j = \Theta_d - \Theta_0 = 2\pi m - h\theta$$

by (8.6); therefore

$$e\vartheta + \sum_{j=1}^{d} u_j = -h(2\pi - \theta) + (2\pi m - h\theta) = 2\pi(m-h),$$

which is (8.11). □

Lemma 8.4 (The identity contact rotation closes after reflection)**.** *Assume the integer lift in lemma 4.13 is $\kappa = N$. For the monodromy output, take the eigenvalue in the opposite orientation:*

$$\mu = \overline{\lambda}, \qquad y = 1 - x.$$

For this eigenvalue the ordered Farey cell is

$$\frac{0}{1} < y < \frac{1}{N},$$

so $p = 0$, $q = 1$, $r = 1$, $s = N$, $d = N$, and $e = 0$. There are parameters satisfying (1.5) *for which* (1.6)*–*(1.9) *hold.*

Proof. Every contact is strict by lemma 4.13; hence $0 < \alpha_i, \beta_i < 1$. Extend the indices periodically and set $z_i = x_i$ for $0 \le i \le N$, with $z_N = z_0$ as a point and with lifted arguments $\Theta_N = \Theta_0 + 2\pi$. The contact equation for $\kappa = N$ is

$$\lambda x_i = \beta_i x_{i-1} + \alpha_i x_i.$$

After rearrangement it is the backward cell relation

$$(\lambda - \alpha_i) z_i = \beta_i z_{i-1} \qquad (1 \le i \le N),$$

where the coefficients at $i = N$ are the periodic coefficients at index 0. The inequalities required in (8.5) follow from lemma 2.10. Apply lemma 8.3 with

$$q = 1, \quad d = N, \quad h = 0, \quad m = 1, \quad b_i = \alpha_i, \quad a_i = \beta_i.$$

It produces the forward reflected cells with

$$\beta'_j = \alpha_{N-j+1}, \qquad \alpha'_j = \beta_{N-j+1}.$$

Because $\alpha_i + \beta_i = 1$, these coefficients satisfy $\alpha'_j = 1 - \beta'_j$ and hence (1.5). The reflection lemma also gives the polynomial and Laurent identities with $s = N$ and $e = 0$, and the phase identity

$$\sum_{j=1}^{N} u_j = 2\pi.$$

This is (1.9) because $r - dp = 1$.

It remains to identify the ordered Farey cell. The angle estimate (4.13) gives

$$\frac{N-1}{N} < x < 1.$$

The endpoints are consecutive in F_N, and lemma 8.2 gives

$$0 < 1 - x < \frac{1}{N}.$$

Thus the reflected orientation has exactly the Farey labels stated in the lemma, with the smaller denominator on the left. □

8.1 Return-strip monodromy

Proposition 8.5 (The nontransversal case $\varphi > \delta$)**.** *Assume $\varphi > \delta$. Then $\delta = 1$, and the monodromy output uses the contact eigenvalue*

$$\mu = \lambda.$$

There are integers p, q, d, e with

$$q\kappa - pN = 1, \qquad N = qd + e, \qquad 0 \le e < q, \qquad d = \lfloor N/q \rfloor, \tag{8.12}$$

and parameters satisfying (1.5) such that

$$(\lambda^q - \beta_j) x_{j-1} = \alpha_j x_j \qquad (1 \le j \le d), \tag{8.13}$$

$$\lambda^e x_d = x_0. \tag{8.14}$$

The ordered Farey cell is

$$\frac{p}{q} < x < \frac{\kappa}{N}. \tag{8.15}$$

Thus $r = \kappa$, $s = N$, and $q < s$. For these data (1.6)–(1.9) hold.

Proof. By theorem 7.11 and (6.5), $\Delta = 1$ and $\delta = \gcd(\Delta, \varphi) = 1$.

Suppose first that $\varphi = N$. The initial record pair is $U = (1,0)$, $V = (0,1)$, so $\Delta = \kappa$. The unit-return theorem gives $\kappa = 1$. Put

$$q = 1, \quad p = 0, \quad d = N, \quad e = 0.$$

Every field is strict and $\lambda x_{j-1} = \xi_j$, hence (8.13); the identity $x_N = x_0$ is (8.14).

Now assume $\varphi < N$. Use the unit-edge extension in theorem 6.1: let $E = (e, c)$ be the earliest record in the maximal backward arithmetic run with increment $U = (q,p)$, and put $d = -L(E)$. Equation (6.8) is exactly (8.12); in particular $d \geq \varphi$, and the proof of theorem 6.1 shows that E and $E+U$ are consecutive upper records. Since the actual strict block $\{1, \ldots, \varphi\}$ is contained in $J = \{1, \ldots, d\}$, the endpoint-padded section corollary corollary 6.2 applies. It gives

$$\lambda^q x_{j-1} = \xi_j \qquad (1 \leq j \leq d), \tag{8.16}$$

and its closure is (8.14). For $j \leq \varphi$, substitute the one-sided contact equation for ξ_j. For $j > \varphi$, the same corollary says that field j is an endpoint cell, so put $\beta_j = 0$, $\alpha_j = 1$ and again obtain (8.13).

Multiplying (8.13) gives

$$\left(\prod_{j=1}^{d} (\lambda^q - \beta_j) \right) x_0 = \left(\prod_{j=1}^{d} \alpha_j \right) x_d.$$

No vertex is zero because $0 \in \operatorname{int} P$. Using $x_0 = \lambda^e x_d$ proves (1.7). Multiplying by λ^{dq} and using $N = qd + e = s$ proves (1.6).

The return identity $q\varphi + h = N$ from (6.5) gives $q \leq N$. Equality is impossible, because $q = N$ would make $q\kappa - pN$ divisible by N, whereas (8.12) makes it equal to 1. Hence $q < N = s$. The determinant identity $q\kappa - pN = 1$ and the denominator sum $q + N > N$ now show that p/q and κ/N are consecutive in F_N. The upper inequality $x < \kappa/N$ follows from (4.13). To locate the lower endpoint, follow the q successive κ-steps beginning at the lifted vertex x_0. Their total index advance is $q\kappa = pN + 1$, and (8.16) ends strictly on the lifted side E_{pN+1}. By lemma 4.14,

$$\Theta_0 + 2\pi p < \Theta_0 + q\theta < \Theta_1 + 2\pi p,$$

or equivalently

$$0 < q\theta - 2\pi p < \Theta_1 - \Theta_0.$$

Thus $p/q < x$.

Finally, in each cell relation choose the factor argument u_j so that

$$u_j = \Theta_j - \Theta_{j-1}.$$

This is the continuous rooted branch selected by the return path. The arithmetic identity

$$d + e\kappa = (\kappa - dp)N \tag{8.17}$$

follows from $q\kappa - pN = 1$ and $N = qd + e$. The closure path consists entirely of endpoint contacts and ends at lifted index $d + e\kappa = (\kappa - dp)N$; therefore lemma 4.14 gives

$$\Theta_d + e\theta = \Theta_0 + 2\pi(\kappa - dp).$$

Telescoping the u_j gives

$$e\theta + \sum_{j=1}^{d} u_j = 2\pi(\kappa - dp),$$

which is (1.9) with $(r, s) = (\kappa, N)$. ∎

Proposition 8.6 (The transversal case $\varphi = \delta$)**.** *Assume $\varphi = \delta$. Put*

$$L = \frac{N}{\delta}, \qquad K = \frac{\kappa}{\delta},$$

choose the unique $h \in \{1, \ldots, L-1\}$ with $Kh \equiv -1 \pmod{L}$, and define

$$b = \frac{Kh+1}{L}, \qquad S = N - h, \qquad R = \kappa - b.$$

Then

$$KS - RL = 1, \qquad L + S > N, \qquad \frac{R}{S} < x < \frac{K}{L}. \tag{8.18}$$

The monodromy output is as follows.

(a) *If $\delta \geq 2$, use the opposite eigenvalue $\mu = \overline{\lambda}$. Then $y = 1 - x$ lies in the reflected Farey cell*

$$\frac{L-K}{L} < y < \frac{S-R}{S}, \tag{8.19}$$

whose denominators satisfy $q = L < s = S$. With

$$d = \delta, \qquad e = s - dq = -h,$$

there are parameters satisfying (1.5) for which (1.6)–(1.9) hold.

(b) *If $\delta = 1$, retain the contact eigenvalue $\mu = \lambda$. Then the ordered cell in (8.18) has*

$$\frac{p}{q} = \frac{R}{S}, \qquad \frac{r}{s} = \frac{K}{L}, \qquad q = S < s = L = N.$$

With $d = \lfloor N/q \rfloor$ and $e = s - dq$, there are parameters satisfying (1.5) for which the same identities hold.

Proof. The κ-orbits are the δ residue classes modulo δ, each of length L. The strict block $\{1, \ldots, \delta\}$ contains exactly one strict field in each orbit. Starting at x_i, the first $L-1$ destination fields are therefore endpoint fields and the last step reaches strict field i. Consequently

$$\lambda^L x_i = \xi_i = \beta_i x_{i-1} + \alpha_i x_i \qquad (1 \leq i \leq \delta). \tag{8.20}$$

Set

$$\widehat{\beta}_i = \alpha_i, \qquad \widehat{\alpha}_i = \beta_i.$$

Every contact in the block is strict, so both numbers belong to $(0, 1)$, and

$$(\lambda^L - \widehat{\beta}_i) x_i = \widehat{\alpha}_i x_{i-1}. \tag{8.21}$$

The congruence $h\kappa \equiv -\delta \pmod{N}$ moves x_δ to x_0. No intermediate destination field is strict: the orbit remains in residue class 0 modulo δ and meets the strict block again only after L steps. Hence

$$\lambda^h x_\delta = x_0. \tag{8.22}$$

Multiplying (8.21), cancelling the nonzero vertices, and using (8.22) gives

$$\lambda^{-h} \prod_{i=1}^{\delta} (\lambda^L - \widehat{\beta}_i) = \prod_{i=1}^{\delta} \widehat{\alpha}_i. \tag{8.23}$$

We next prove the Farey assertions and the lifted closure needed for both cases. The determinant computation is

$$KS - RL = K(N-h) - (\kappa - b)L = bL - Kh = 1,$$

and $L+S>N$ because $h<L$. Let $\gamma_i=\Theta_i-\Theta_{i-1}$ and $\Gamma=\sum_{i=1}^{\delta}\gamma_i$. The strict L-return in (8.20) lies in relint E_i. Since $L\kappa=KN$, lemma 4.14 places its lift strictly between $\Theta_{i-1}+2\pi K$ and $\Theta_i+2\pi K$. Thus

$$\varepsilon=2\pi K-L\theta \quad \text{satisfies} \quad 0<\varepsilon<\gamma_i.$$

Likewise $h\kappa=bN-\delta$. The endpoint closure (8.22) therefore has the exact lifted form

$$\Theta_\delta+h\theta=\Theta_0+2\pi b, \qquad \Gamma=2\pi b-h\theta. \tag{8.24}$$

Using $N=\delta L$ and $\kappa=\delta K$ gives

$$S\theta-2\pi R=(N-h)\theta-2\pi(\kappa-b)=\Gamma-\delta\varepsilon>0,$$

whereas $L\theta<2\pi K$. Hence $R/S<x<K/L$. Together with determinant one and $L+S>N$, the Farey adjacency criterion proves (8.18).

Assume first that $\delta\ge 2$. Since $1\le h<L$,

$$S=\delta L-h>L.$$

Apply lemma 8.2 to (8.18); this gives (8.19), with the smaller denominator L on the left. Set

$$\frac{p}{q}=\frac{L-K}{L}, \qquad \frac{r}{s}=\frac{S-R}{S}, \qquad d=\delta, \qquad e=-h.$$

Then $d=\lfloor N/q\rfloor$ because $N=\delta L$, and

$$s-dq=S-\delta L=-h=e.$$

The inequalities $0<\gamma_i<\pi$ required in (8.5) follow from lemma 2.10. Apply lemma 8.3 to (8.21)–(8.24) with

$$q=L, \quad d=\delta, \quad b_i=\widehat{\beta}_i, \quad a_i=\widehat{\alpha}_i, \quad m=b.$$

The hypothesis $h<dq$ is $h<N$, which is automatic. Because $\widehat{\alpha}_i+\widehat{\beta}_i=1$, the reflected coefficients satisfy (1.5). The lemma gives the forward recurrences for $\mu=\overline{\lambda}$, the Laurent identity with $e=-h$, and its polynomial form (1.6). It also gives

$$e\arg_+(\mu)+\sum_{j=1}^{d}u_j=2\pi(b-h).$$

Finally,

$$r-dp=(S-R)-\delta(L-K)=b-h,$$

so this is exactly (1.9). This proves part (a).

Now assume $\delta=1$. Then $L=N$ and $S=N-h<L$, so the original ordered cell already has the smaller denominator on the left. Put

$$q=S=N-h, \qquad s=N, \qquad p=R, \qquad r=K=\kappa.$$

Equation (8.23) becomes

$$\lambda^{-h}(\lambda^N-\widehat{\beta}_1)=\widehat{\alpha}_1,$$

or, after moving the second term,

$$\lambda^h(\lambda^{N-h}-\widehat{\alpha}_1)=\widehat{\beta}_1. \tag{8.25}$$

Let $d=\lfloor N/q\rfloor$ and $e=N-dq$. Define

$$\beta_1'=\widehat{\alpha}_1, \quad \alpha_1'=\widehat{\beta}_1, \qquad \beta_j'=0, \quad \alpha_j'=1 \quad (2\le j\le d).$$

Since $e+(d-1)q=h$, equation (8.25) is exactly

$$\lambda^e\prod_{j=1}^d(\lambda^q-\beta_j')=\prod_{j=1}^d\alpha_j'.$$

Multiplying by λ^{dq} gives (1.6).

The first factor has a genuine consecutive-vertex relation. Indeed, $\lambda^h x_1=x_0$ and $\lambda^N x_1=\widehat{\alpha}_1 x_0+\widehat{\beta}_1 x_1$ imply

$$(\lambda^q-\beta_1')x_0=\alpha_1' x_1. \tag{8.26}$$

Let $u_1=\Theta_1-\Theta_0\in(0,\pi)$ be its geometric factor argument. The lifted closure (8.24) is

$$h\theta+u_1=2\pi b. \tag{8.27}$$

For $2\le j\le d$, assign the padded zero factor the argument $u_j=q\theta-2\pi p$. The Farey inequalities and $q\kappa=pN+1$ show that this is the unique argument in $(0,\pi)$ of λ^q. Because $e+(d-1)q=h$,

$$e\theta+\sum_{j=1}^d u_j=2\pi\big(b-(d-1)p\big).$$

Here $p=R=\kappa-b$ and $r=K=\kappa$, so $b-(d-1)p=r-dp$. This proves (1.9) and part (b). ☐

The product identity alone determines factor arguments only modulo 2π. The vertex recurrences remove that ambiguity: consecutive lifted vertex angles select the upper-half-plane argument of each nonzero factor, and every padded zero factor is assigned the left endpoint argument A. The following lemma records this common sheet before convexity is applied.

Lemma 8.7 (The return factors lie on the Jensen sheet)**.** *For every monodromy output constructed in lemma 8.4 and propositions 8.5 and 8.6, let $\vartheta=\arg_+(\mu)$ and put*

$$A=q\vartheta-2\pi p.$$

Then $0<A<\pi$, and every recurrence argument u_j in (1.9) is the unique argument in $(0,\pi)$ of $\mu^q-\beta_j$. Equivalently, if $M=\arg_+(\mu^q-1)\in(A,\pi)$, then

$$u_j\in[A,M)\qquad(1\le j\le d). \tag{8.28}$$

Proof. Because this is an order-N Farey cell, $q+s>N\ge 4$; with $q\le s$ this gives $s\ge 3$. The determinant identity and $p/q<y<r/s$ therefore give

$$0<A<\frac{2\pi}{s}<\pi.$$

Thus

$$\operatorname{Im}(\mu^q-\beta)=|\mu|^q\sin A>0\qquad(0\le\beta<1),$$

so each factor has a unique argument in $(0,\pi)$. As β increases from 0 to 1, that argument increases continuously and strictly from A to M.

It remains only to identify the geometric lifts. In the nontransversal case, (8.13) has the form

$$(\mu^q-\beta_j)x_{j-1}=\alpha_j x_j,\qquad \alpha_j>0,$$

where x_{j-1},x_j are consecutive vertices. Their lifted angular gap lies in $(0,\pi)$ and is congruent modulo 2π to the factor argument; it is therefore the unique upper-half-plane argument. In both reflected cases—$\kappa = N$ and the transversal case $\delta \geq 2$—this conclusion is exactly the final assertion of lemma 8.3. In the transversal case $\delta = 1$ it applies to (8.26), while every padded zero factor was explicitly assigned the argument A. These cases exhaust the constructions listed in the statement, and the range assertion follows from the preceding monotonicity in β. ☐

Proof of theorem 1.4. Apply lemma 4.13; throughout the contact-return analysis its selected eigenvalue has been denoted by λ. Determine the output eigenvalue only after the return regime is known. The four mutually exclusive outputs are recorded before they are verified:

return regime	output	denominator order	signed e	source
$\kappa = N$	$\mu = \overline{\lambda}$	$1 < N$	0	lemma 8.4
$\kappa < N$, $\varphi > \delta$	$\mu = \lambda$	$q < N$	$0 \leq e < q$	proposition 8.5
$\kappa < N$, $\varphi = \delta \geq 2$	$\mu = \overline{\lambda}$	$L < S$	$e = -h < 0$	proposition 8.6, case (a)
$\kappa < N$, $\varphi = \delta = 1$	$\mu = \lambda$	$S < N$	$0 \leq e < S$	proposition 8.6, case (b)

If $\kappa = N$, choose $\mu = \overline{\lambda}$ and apply lemma 8.4. Its Farey cell has denominators $1 \leq N$, and it provides the polynomial identity, the equivalent Laurent identity, and the phase formula.

Assume $\kappa < N$ and pass to the reduced strict interval of lemma 5.5. If $\varphi > \delta$, choose $\mu = \lambda$ and apply proposition 8.5, using the unit return from theorem 7.11. Its right Farey endpoint is κ/N, so the ordered denominators satisfy $q \leq N = s$, and its Euclidean remainder e is nonnegative.

If $\varphi = \delta$, apply proposition 8.6. That proposition makes the orientation choice explicitly: it chooses $\mu = \overline{\lambda}$ when $\delta \geq 2$ and $\mu = \lambda$ when $\delta = 1$. In the first subcase $e = -h < 0$ and the homogeneous polynomial identity is primary; in the second subcase $e \geq 0$.

Thus in every case a specific eigenvalue $\mu \in \{\lambda, \overline{\lambda}\}$ has been selected, the ordered Farey cell has $q \leq s$, and (1.6)–(1.9) hold for that same μ. Lemma 8.7 verifies that the factor arguments appearing in the case constructions have precisely the normalized-factor description and Jensen-sheet range asserted in the theorem. No symbol is restored and no orientation is changed after this selection. ☐

9 Stochastic-matrix corollary

For $n \geq 1$, let

$$\Theta_n = \bigcup \Big\{ \operatorname{spec} A : A \in \mathbb{R}_{\geq 0}^{n \times n},\ A\mathbf{1} = \mathbf{1} \Big\}; \tag{9.1}$$

thus Θ_n is the union of the spectra of all real row-stochastic $n \times n$ matrices.

Proposition 9.1 (Compactness, conjugation, and disk bound)**.** *For every n, the set Θ_n is compact, invariant under complex conjugation, and contained in the closed unit disk.*

Proof. Let $\mathcal{S}_n$ be the compact set of row-stochastic matrices. First note that every eigenvalue lies in the closed unit disk: if $Av = \lambda v$, choose an index with $|v_i| = \|v\|_\infty$; then

$$|\lambda| \|v\|_\infty = |(Av)_i| \leq \|v\|_\infty.$$

Consequently the eigenpair set

$$\mathcal{E}_n = \{(A, \lambda) \in \mathcal{S}_n \times \overline{\mathbb{D}} : \det(\lambda I - A) = 0\}$$

is closed in a compact space and is therefore compact. Its projection to the λ-coordinate is exactly Θ_n, so Θ_n is compact. Real characteristic polynomials give conjugation symmetry. ☐

Theorem 9.2 (Invariant-polygon criterion)**.** *Let $n \geq 2$. For $\lambda \in \mathbb{C}$, the following are equivalent.*

(i) $\lambda \in \Theta_n$.

(ii) *There is a non-singleton convex polygon $P \subset \mathbb{C}$ with at most n vertices such that*

$$\lambda P \subseteq P. \tag{9.2}$$

Proof. If $Av = \lambda v$, put $P = \operatorname{conv}\{v_1, \ldots, v_n\}$. Every coordinate satisfies

$$\lambda v_i = \sum_j a_{ij} v_j \in P,$$

so (9.2) holds. If the hull is a singleton, all coordinates of the nonzero eigenvector are equal to the same nonzero number, so $\lambda = 1$; in that case any fixed segment may be used.

Conversely, let $x_1, \ldots, x_m$ be the vertices of an invariant polygon, $m \leq n$. Choose barycentric coordinates

$$\lambda x_i = \sum_{j=1}^{m} a_{ij} x_j, \qquad a_{ij} \geq 0, \qquad \sum_j a_{ij} = 1.$$

Then $A = (a_{ij})$ is row-stochastic and $Ax = \lambda x$. If $m < n$, append absorbing states and extend the eigenvector by zero coordinates. □

Corollary 9.3 (Radial filling)**.** *Let $n \geq 2$. If $\lambda \in \Theta_n$, then $t\lambda \in \Theta_n$ for every $0 \leq t \leq 1$.*

Proof. If $|\lambda| < 1$, take an invariant polygon P. Iteration gives $\lambda^k P \subseteq P$ and $\lambda^k x \to 0$, hence $0 \in P$; convexity then gives $t\lambda P \subseteq P$.

Suppose $|\lambda| = 1$. If $\lambda = 1$, the segment $[0, 1]$ satisfies $t[0, 1] \subseteq [0, 1]$. Otherwise take an invariant polygon with the least number of vertices. If it has nonzero area, area equality forces $\lambda P = P$, so multiplication by λ permutes its vertices and λ has finite order $k \leq n$. The regular orbit polygon

$$Q = \operatorname{conv}\{1, \lambda, \ldots, \lambda^{k-1}\}$$

contains 0, satisfies $\lambda Q = Q$, and therefore satisfies $t\lambda Q = tQ \subseteq Q$. If the minimal polygon is a segment, then $\lambda = -1$ and the same conclusion follows from $Q = [-1, 1]$. The invariant-polygon criterion completes the proof. □

Proposition 9.4 (Unit-circle points)**.** *Let $n \geq 2$. If $\lambda \in \Theta_n$ and $|\lambda| = 1$, then λ is a root of unity of order at most n. Conversely every such root belongs to Θ_n.*

Proof. Take an invariant polygon with the least number of vertices. If it has nonzero area, then $\lambda P \subseteq P$ and equality of areas force $\lambda P = P$ by lemma A.5. Multiplication by λ permutes the vertices, so a nonzero vertex has an orbit of length at most n and $\lambda^k = 1$. If the polygon is a segment, $\lambda = \pm 1$. Conversely, a cyclic permutation matrix realizes every root of unity of order at most n, after padding. □

For $0 < \theta < \pi$, define

$$R_n(\theta) = \max\left\{\rho \geq 0 : \rho e^{i\theta} \in \Theta_n\right\}. \tag{9.3}$$

The maximum exists by compactness and radial filling.

Corollary 9.5 (Representative-selection Farey compression for stochastic matrices)**.** *Let $N \geq 4$ and let*

$$\lambda = R_N(\theta) e^{i\theta} \in \Theta_N \setminus \Theta_{N-1}, \qquad 0 < |\lambda| < 1.$$

Then one of $\lambda, \overline{\lambda}$ satisfies the complete conclusion of theorem 1.4; equivalently, it has a Farey carrier, the polynomial identity (1.6), its equivalent Laurent form (1.7), normalized factor arguments (1.8) on the Jensen sheet $0 < A < M < \pi$ and $u_j \in [A, M)$, and the phase identity (1.9).

Proof. Let $T_\lambda z = \lambda z$. Since $0 < \theta < \pi$, we have $\Im\lambda > 0$, and hence

$$(\operatorname{tr} T_\lambda)^2 - 4 \det T_\lambda = -4(\Im\lambda)^2 < 0.$$

Together with $0 < |\lambda| < 1$, this shows that T_λ is an elliptic contraction. By the invariant-polygon criterion, membership in Θ_n is equivalent to the existence of a non-singleton invariant polygon with at most n vertices. By lemma 2.5, every such polygon for T_λ is nondegenerate, so its least possible number of vertices is exactly $\nu_{\mathrm{poly}}(T_\lambda)$. The assumption $\lambda \in \Theta_N \setminus \Theta_{N-1}$ therefore gives $\nu_{\mathrm{poly}}(T_\lambda) = N$. Radial maximality says that $t\lambda \notin \Theta_N$ for every $t > 1$; applying the same criterion to tT_λ gives $\nu_{\mathrm{poly}}(tT_\lambda) > N$. Thus T_λ is N-critical. Apply theorem 1.4. Reversing the orientation labels the conjugate eigenvalue, which explains the choice between λ and $\overline{\lambda}$. □

10 Conclusion

The parent theorem is a theorem about critical planar dynamics, not about stochastic matrices. Radial criticality forces hereditary side and vertex saturation; saturation makes contact mutation exact; mutation reduction produces a single cyclic return section; the lattice sail supplies its unimodular tower data; and projective holonomy rules out a nonadjacent return. Farey adjacency and the heterogeneous Ito product are scalar shadows of that adjacent return normal form. The stochastic theorem uses none of the internal deformation or index bookkeeping: it follows from the invariant-polygon criterion in one paragraph.

A Foundational finite-dimensional tools

This appendix proves the finite-dimensional ingredients invoked in the body of the paper. They are stated in the exact strength used there.

Lemma A.1 (Weighted spectral-radius bound and nonnegative Perron vectors)**.** *Let $B = (b_{ij}) \in \mathbb{R}^{m\times m}$ be entrywise nonnegative.*

(i) *If $x > 0$ coordinatewise and $Bx \le cx$, then* $\operatorname{spr}(B) \le c$.

(ii) *There are nonzero vectors $v, w \ge 0$ such that*

$$Bv = \operatorname{spr}(B)v, \qquad B^T w = \operatorname{spr}(B)w.$$

Proof. For $x > 0$, define the weighted supremum norm

$$\|z\|_x = \max_i \frac{|z_i|}{x_i} \qquad (z \in \mathbb{C}^m).$$

If $Bx \le cx$, then entrywise nonnegativity gives

$$|Bz| \le B|z| \le \|z\|_x Bx \le c\|z\|_x x.$$

Hence the operator norm of B in $\|\cdot\|_x$ is at most c. Every eigenvalue has modulus bounded by every operator norm, proving (i).

We first prove (ii) for a strictly positive matrix A. On the open probability simplex

$$\Sigma^\circ = \{x \in \mathbb{R}^m : x_i > 0,\ \sum_i x_i = 1\}$$

consider the scale-invariant function

$$\Phi(x) = \max_i \frac{(Ax)_i}{x_i}.$$

Let $a_0 = \min_{i,j} a_{ij} > 0$. If a sequence in Σ° approaches the boundary, some coordinate x_i tends to zero while $(Ax)_i \geq a_0 \sum_j x_j = a_0$; therefore $\Phi(x) \to \infty$. Fix one $x^{(0)} \in \Sigma^\circ$. The sublevel set $\{x \in \Sigma^\circ : \Phi(x) \leq \Phi(x^{(0)})\}$ stays a positive distance from the boundary and is closed in the compact probability simplex. It is therefore compact and nonempty, and continuity of Φ gives an interior minimizer x.

Put $M = \Phi(x)$. Then $Ax \leq Mx$. If the inequality were strict in at least one coordinate, the nonzero vector $Mx - Ax \geq 0$ would satisfy $A(Mx - Ax) > 0$ coordinatewise because every entry of A is positive. Consequently

$$A^2 x < MAx$$

coordinatewise. With $y = Ax > 0$, this says $\Phi(y) < M$, contradicting the minimality of x after normalizing y back to the simplex. Thus $Ax = Mx$. Part (i), applied with equality, shows $\operatorname{spr}(A) \leq M$, while M is itself an eigenvalue; hence $M = \operatorname{spr}(A)$. Applying the same argument to A^T gives a strictly positive left Perron vector.

Now let $B \geq 0$ and put

$$A_\varepsilon = B + \varepsilon \mathbf{1}\mathbf{1}^T \qquad (\varepsilon > 0).$$

The preceding paragraph gives positive right and left Perron vectors $v_\varepsilon, w_\varepsilon$, normalized by $\sum_i (v_\varepsilon)_i = \sum_i (w_\varepsilon)_i = 1$, and a positive Perron root $r_\varepsilon = \operatorname{spr}(A_\varepsilon)$. We show that $r_\varepsilon \to r := \operatorname{spr}(B)$.

First, $r \leq r_\varepsilon$. Choose an eigenvalue ζ of B with $|\zeta| = r$ and a corresponding nonzero complex eigenvector z. Since

$$r|z| = |Bz| \leq B|z| \leq A_\varepsilon |z|,$$

let $C = \max_i |z_i|/(v_\varepsilon)_i$ and choose an index attaining the maximum. At that index,

$$rC(v_\varepsilon)_i \leq (A_\varepsilon |z|)_i \leq C(A_\varepsilon v_\varepsilon)_i = Cr_\varepsilon (v_\varepsilon)_i,$$

so $r \leq r_\varepsilon$.

For the converse bound, fix $c > r$. Choose r_0 with $r < r_0 < c$. In Jordan normal form, the kth power of a block $J_\zeta = \zeta I + N$ of size m_ζ is

$$J_\zeta^k = \sum_{j=0}^{m_\zeta - 1} \binom{k}{j} \zeta^{k-j} N^j.$$

Because $|\zeta| \leq r < r_0$, this formula gives constants C, M such that $\|B^k\| \leq C k^M r_0^k$ for all k. Therefore the matrix series $\sum_{k\geq 0} c^{-k-1} B^k$ converges absolutely. Multiplying its partial sums by $cI - B$ and passing to the limit gives

$$(cI - B)^{-1} = \sum_{k=0}^{\infty} c^{-k-1} B^k.$$

Consequently

$$x_c = (cI - B)^{-1}\mathbf{1} = \sum_{k=0}^{\infty} c^{-k-1} B^k \mathbf{1} > 0, \qquad Bx_c = cx_c - \mathbf{1}.$$

Therefore

$$A_\varepsilon x_c = cx_c - \mathbf{1} + \varepsilon(\mathbf{1}^T x_c)\mathbf{1} \leq (c + \eta_\varepsilon) x_c,$$

where

$$\eta_\varepsilon = \varepsilon(\mathbf{1}^T x_c) \max_i (x_c)_i^{-1} \longrightarrow 0.$$

Part (i) gives $r_\varepsilon \le c + \eta_\varepsilon$. Hence $\limsup_{\varepsilon\downarrow 0} r_\varepsilon \le c$; letting $c \downarrow r$ proves $r_\varepsilon \to r$.

The closed probability simplex is compact. Along a sequence $\varepsilon_k \downarrow 0$, the normalized vectors converge to nonzero vectors $v, w \ge 0$. Passing to the limit in

$$A_{\varepsilon_k} v_{\varepsilon_k} = r_{\varepsilon_k} v_{\varepsilon_k}, \qquad A^T_{\varepsilon_k} w_{\varepsilon_k} = r_{\varepsilon_k} w_{\varepsilon_k}$$

gives $Bv = rv$ and $B^T w = rw$. $\square$

Lemma A.2 (Strict separation from a compact convex set)**.** *Let K be a nonempty compact convex subset of a Euclidean space and let $o \notin K$. There is a linear functional ℓ such that*

$$\ell(x) < \ell(o) \qquad (x \in K).$$

Proof. Choose $y \in K$ minimizing $|o - y|$ and put $v = o - y \ne 0$. For every $x \in K$, the segment $y + t(x - y)$, $0 \le t \le 1$, belongs to K. The function

$$f(t) = |o - y - t(x - y)|^2$$

has a minimum at $t = 0$, so $f'(0) \ge 0$. Thus $\langle v, x - y\rangle \le 0$. With $\ell(z) = \langle v, z\rangle$ we obtain

$$\ell(x) \le \ell(y) < \ell(o),$$

the last inequality being $\ell(o) - \ell(y) = |v|^2 > 0$. $\square$

Lemma A.3 (Polarity for planar polygons)**.** *Let $K \subset \mathbb{R}^2$ be a compact convex set with $0 \in \operatorname{int} K$, and define*

$$K^\circ = \{y : \langle y, x\rangle \le 1 \text{ for every } x \in K\}.$$

Then K° is compact, convex, and contains 0 in its interior. If A is linear and $AK \subseteq K$, then

$$A^* K^\circ \subseteq K^\circ.$$

If K is a strict polygon with N vertices, then K° is a strict polygon with N vertices. For every vertex x of K, the set

$$F_x = \{y \in K^\circ : \langle y, x\rangle = 1\}$$

is a side of K°, and every side arises uniquely in this way. Finally, if $x \in K$, $y \in K^\circ$, and $\langle y, x\rangle = 1$, then $x \in \partial K$ and $y \in \partial K^\circ$.

Proof. Choose radii $0 < r < R$ with $r\overline{\mathbb{D}} \subseteq K \subseteq R\overline{\mathbb{D}}$. The defining inequalities show

$$R^{-1}\overline{\mathbb{D}} \subseteq K^\circ \subseteq r^{-1}\overline{\mathbb{D}},$$

so K° has the asserted compactness and interior properties. If $y \in K^\circ$ and $x \in K$, then $Ax \in K$ and

$$\langle A^* y, x\rangle = \langle y, Ax\rangle \le 1;$$

therefore $A^* y \in K^\circ$.

Let the vertices of the strict polygon be $x_1, \ldots, x_N$. Since a linear functional reaches its maximum over a convex hull at a vertex,

$$K^\circ = \bigcap_{i=1}^{N} \{y : \langle y, x_i\rangle \le 1\}.$$

Every displayed inequality is irredundant. Indeed, at the vertex x_i choose a strict supporting functional u_i of K. Its support value $h_K(u_i)$ is positive because 0 is interior. After replacing u_i by $u_i/h_K(u_i)$, one has

$$\langle u_i, x_i\rangle = 1, \qquad \langle u_i, x_j\rangle < 1 \quad (j \ne i).$$

On the line $\langle y, x_i\rangle = 1$, all other inequalities remain strict in a small interval about u_i. Hence F_{x_i} is a nondegenerate side of K°. The intersection description has only the N displayed half-planes, so it has at most N sides, while the preceding argument has produced N distinct sides. It therefore has exactly N sides. Traversing the boundary of a bounded full-dimensional polygon alternates sides and vertices, so it also has exactly N vertices. Since no displayed side is redundant or degenerate, the polar polygon is strict. The same argument also proves the asserted bijection between vertices of K and sides of K°.

If x were interior to K and $y \neq 0$, then $x + \varepsilon y \in K$ for all sufficiently small $\varepsilon > 0$, giving

$$1 \geq \langle y, x + \varepsilon y\rangle = 1 + \varepsilon |y|^2,$$

a contradiction. Thus equality forces $x \in \partial K$. The defining inequality $\langle \cdot, x\rangle \leq 1$ for K° is attained at y, so $y \in \partial K^\circ$. $\square$

Lemma A.4 (Hausdorff limits of finitely generated polygons)**.** *Fix N and let*

$$P_k = \operatorname{conv}\{z_{k,1}, \dots, z_{k,N}\}$$

with all points in one compact subset of $\mathbb{R}^2$. If $z_{k,j} \to z_j$ for every j, then

$$P_k \longrightarrow P := \operatorname{conv}\{z_1, \dots, z_N\}$$

in the Hausdorff metric. If P has nonempty interior, then $\operatorname{area}(P_k) \to \operatorname{area}(P)$*. Moreover, for every fixed linear map A, the conditions $AP_k \subseteq P_k$ pass to the limit, and*

$$\max_{z \in P_k} |z| \longrightarrow \max_{z \in P} |z|.$$

Proof. Put $\varepsilon_k = \max_j |z_{k,j} - z_j|$. If $z = \sum_j t_j z_{k,j} \in P_k$, then $z' = \sum_j t_j z_j \in P$ and $|z - z'| \leq \varepsilon_k$; the converse estimate is identical. Hence the Hausdorff distance is at most ε_k.

All polygons lie in one large disk. We prove convergence of their indicator functions away from ∂P. If $z \notin P$, its positive distance from P and Hausdorff convergence imply $z \notin P_k$ for large k. If $z \in \operatorname{int} P$, choose $r > 0$ with $z + r\overline{\mathbb{D}} \subseteq P$. Hausdorff convergence implies uniform convergence of support functions:

$$|h_{P_k}(u) - h_P(u)| \leq d_H(P_k, P) \qquad (|u| = 1).$$

For large k this gives

$$h_{P_k}(u) > \langle u, z\rangle \qquad (|u| = 1).$$

If $z \notin P_k$, lemma A.2 would produce a unit vector u with $\langle u, z\rangle > h_{P_k}(u)$, a contradiction. Thus $z \in P_k$ eventually. Since the boundary of a polygon is a finite union of line segments and has planar measure zero, dominated convergence of the indicator functions gives area convergence.

If $AP_k \subseteq P_k$ and $x \in P$, choose $x_k \in P_k$ with $x_k \to x$. Then $Ax_k \in P_k$, and Hausdorff convergence gives $Ax \in P$. Finally, by convexity of the norm,

$$\max_{z \in P_k} |z| = \max_j |z_{k,j}|, \qquad \max_{z \in P} |z| = \max_j |z_j|,$$

which proves convergence of the maxima. $\square$

Lemma A.5 (Strict area monotonicity)**.** *If $K \subseteq L$ are compact convex subsets of $\mathbb{R}^2$, both with nonempty interior, and $K \neq L$, then*

$$\operatorname{area}(K) < \operatorname{area}(L).$$

Proof. Choose $y \in L \setminus K$. By lemma A.2, there is a linear functional ℓ with

$$\ell(y) > a := \max_{x \in K} \ell(x).$$

Choose a closed disk $D \subset \operatorname{int} K$. The convex hull $C = \operatorname{conv}(D \cup \{y\})$ belongs to L. Because D has nonempty interior and $\ell(y) > a$, the set $C \cap \{x : \ell(x) > a\}$ contains a nonempty open triangle. It is disjoint from K and has positive area. Hence $L \setminus K$ has positive area, which proves the strict inequality. □

Lemma A.6 (Lattice parallelogram count)**.** *Let $u, v \in \mathbb{Z}^2$ be linearly independent and put $D = |\det(u, v)|$. The half-open parallelogram*

$$\Pi = \{\alpha u + \beta v : 0 \leq \alpha, \beta < 1\}$$

contains exactly D integer points, one in each coset of $\mathbb{Z}^2/(\mathbb{Z}u + \mathbb{Z}v)$.

Proof. Every $z \in \mathbb{R}^2$ has a unique expression $z = au + bv$. Subtracting $\lfloor a \rfloor u + \lfloor b \rfloor v$ gives a representative in Π. Two integer points of Π cannot represent the same coset: their difference is $mu + nv$ with integers m, n, while both coefficient differences lie in $(-1, 1)$, forcing $m = n = 0$. Thus the integer points of Π are in bijection with the quotient group.

It remains to compute the order of that quotient. Integer elementary row and column operations, with determinant ± 1, do not change either the index of the generated sublattice or the absolute determinant. Applying the Euclidean algorithm to the 2×2 matrix with columns u, v reduces it to Smith normal form

$$\begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix}, \qquad d_1, d_2 > 0.$$

For this diagonal lattice the quotient has exactly $d_1 d_2$ elements, and $d_1 d_2$ is the absolute determinant. Reversing the unimodular operations gives the result for u, v. □

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Part II: Invariant Polygons and the Farey–Ito Boundary of Stochastic Spectra

Interface with Part I. Part II applies the structural results proved in Part I, in particular theorem 1.4. The signed closing exponent $e = s - dq$ follows the convention stated there: it may be negative, and in that case the homogeneous polynomial identity is the primary formulation.

II.1 Introduction and proof architecture

For $n \geq 1$, define

$$\Theta_n = \bigcup \left\{ \operatorname{spec} A : A \in \mathbb{R}^{n\times n}_{\geq 0},\ A\mathbf{1} = \mathbf{1} \right\}. \tag{II.1.1}$$

Throughout, we identify $\mathbb{C}$ with the oriented plane $\mathbb{R}^2$ and use

$$\langle z, w\rangle = \operatorname{Re}(\overline{z}w), \qquad \det(z, w) = \operatorname{Im}(\overline{z}w).$$

Positive cyclic order means counterclockwise order; oriented boundary intervals inherit that convention.

The problem is to determine the possible location of one eigenvalue, not an entire spectrum. The answer is a compact radial region: its unit-circle points are roots of unity of order at most n, and its noncircular boundary arcs are indexed by consecutive fractions in the Farey sequence of order n.

The stochastic eigenvalue-region problem was developed by Dmitriev and Dynkin [1] and solved in full by Karpelevič [4]. Ito's reformulation [2] expresses the boundary through one-parameter polynomial families indexed by Farey neighbors. Later matrix realizations and structural treatments include [3, 5, 6]. The present argument derives the Farey–Ito boundary directly from invariant polygons.

The proof passes from stochastic spectra to critical planar dynamics and then to a sharp one-dimensional inequality. Its logical spine is

$$\begin{aligned} &\text{stochastic eigenpair} \iff \text{invariant polygon} \implies \text{new-shell radial extremum} \\ &\implies N\text{-criticality} \implies \text{critical-polygon monodromy} \\ &\implies \text{heterogeneous Ito product and Jensen sheet} \\ &\implies \text{log-sine equalization} \implies \text{cellwise radial equality} \\ &\implies \text{explicit attainment and Farey nesting} \implies \text{Farey–Ito boundary.} \end{aligned} \tag{II.1.2}$$

The invariant-polygon criterion identifies a new-shell stochastic extremum with an intrinsic N-critical elliptic contraction. The monodromy theorem of Part I converts that criticality into the finite Farey product, phase, and argument-sheet data needed here. The remaining steps are the local scalar comparison, explicit realization, and global boundary ordering.

Dependency ledger. The upper bound uses, in order, the invariant-polygon criterion theorem II.4.2, the criticality bridge proposition II.4.7, the Part I monodromy wrapper theorem II.5.1, and the heterogeneous inequality theorem II.6.1. The reverse inclusion is independent of the Part I input: it comes from the sparse realization theorem II.7.3. Candidate nesting theorem II.8.4 is used only after both bounds are known, to pass from new-shell extrema to all orders. Thus neither realization nor nesting is used to establish criticality or monodromy.

The Part I assertion is existential: it selects one of the two adapted complex orientations and supplies monodromy data in that orientation. No uniqueness of invariant polygons, contact topology, or stochastic realizers is required.

Background scope. No prior Karpelevič boundary reduction or parametrization is used. We use elementary finite-dimensional convexity and compactness for the invariant-polygon criterion and radial structure. The intrinsic critical-polygon geometry, including hereditary saturation, contact reduction, return towers, and projective no-skipping, is proved in Part I, culminating in theorem 1.4. Part II uses only the complex monodromy and Jensen-sheet conclusions through theorem II.5.1; every other problem-specific implication used in Part II is proved below.

II.2 Farey data, candidate arcs, and direct extraction

II.2.1 Farey cells

Let F_n denote the Farey sequence of order n on $[0,1]$. We use

$$F_n^+ = F_n \cap \left[0, \frac{1}{2}\right]$$

for the upper half-plane.

Lemma II.2.1 (Farey adjacency criterion)**.** *Two reduced fractions $a/b < c/d$ belonging to F_n are consecutive in F_n if and only if*

$$bc - ad = 1, \qquad b + d > n. \tag{II.2.1}$$

Proof. Assume first that $bc - ad = 1$. If a reduced fraction h/k lies strictly between a/b and c/d, define

$$m = ck - dh, \qquad \ell = bh - ak.$$

Cross multiplication shows that m and ℓ are positive integers. The determinant-one identity gives the exact decomposition

$$(k, h) = m(b, a) + \ell(d, c).$$

Looking at the first coordinate yields $k = mb + \ell d \geq b + d$. Hence, if $b + d > n$, no fraction of denominator at most n can lie between the two endpoints, so they are consecutive in F_n.

Conversely, suppose the two fractions are consecutive in F_n and put $D = bc - ad$. The determinant is a positive integer. If $D > 1$, the half-open parallelogram spanned by the primitive integer vectors $u = (b, a)$ and $v = (d, c)$ contains a nonzero integer point

$$w = \alpha u + \beta v, \qquad 0 < \alpha, \beta < 1.$$

Indeed, the half-open fundamental parallelogram contains exactly $D = |\det(u, v)|$ representatives of the quotient group $\mathbb{Z}^2/(\mathbb{Z}u + \mathbb{Z}v)$. When $D > 1$, one of these representatives is different from the origin. It cannot lie on either open coordinate edge of the parallelogram: such a point would be a nonintegral proper multiple of the primitive vector u or v. The strict inequalities follow because a primitive vector has no integer point in the relative interior of its first unit segment. Both w and $u + v - w$ are positive combinations of u, v, so their slopes lie strictly between a/b and c/d. Their first coordinates are positive integers k and $b + d - k$; one is at most $(b + d)/2 \leq n$ because $b, d \leq n$. After reducing that intermediate slope, its denominator can only decrease. This produces an element of F_n strictly between the alleged consecutive fractions, a contradiction. Therefore $D = 1$. Finally, if $b + d \leq n$, the reduced mediant $(a + c)/(b + d)$ has denominator at most n and lies strictly between the endpoints. Thus consecutiveness also forces $b + d > n$. □

Let $f < g$ be consecutive elements of F_n^+. Label the two endpoints, independently of their left-right order, as

$$\frac{p}{q}, \qquad \frac{r}{s}, \qquad q \le s, \tag{II.2.2}$$

so that $|rq - ps| = 1$. Put

$$d = \left\lfloor \frac{n}{q} \right\rfloor, \qquad e = s - dq. \tag{II.2.3}$$

The integer d is the number of complete q-returns that fit into the vertex budget n.

Definition II.2.2 (Ito polynomial family)**.** For $0 \le \alpha \le 1$, put $\beta = 1 - \alpha$. The Ito equation associated with the cell $[f, g]$ is

$$\lambda^s(\lambda^q - \beta)^d = \alpha^d \lambda^{dq}. \tag{II.2.4}$$

For $\lambda \neq 0$, equivalently, after cancelling the common power of λ and clearing a possible negative exponent,

$$\begin{cases} \lambda^e(\lambda^q - \beta)^d = \alpha^d, & e \ge 0, \\ (\lambda^q - \beta)^d = \alpha^d \lambda^{-e}, & e < 0. \end{cases} \tag{II.2.5}$$

We define the carrier by its radial graph, rather than by an unproved choice of an α-parametrized algebraic root branch. The next proposition constructs exactly one algebraic point on every open ray.

II.2.2 The scalar radius equation

Let $x \in (f, g)$ and write $\lambda = \rho e^{2\pi i x}$. Define

$$A = 2\pi\,|qx - p|, \qquad B = \frac{2\pi}{d}\,|sx - r|. \tag{II.2.6}$$

To verify the range explicitly, put

$$t = \frac{|x - p/q|}{|r/s - p/q|} \in (0, 1).$$

The determinant-one identity gives

$$|qx - p| = \frac{t}{s}, \qquad |sx - r| = \frac{1-t}{q},$$

regardless of which labeled endpoint is on the left. Hence

$$\frac{A+B}{2\pi} = \frac{t}{s} + \frac{1-t}{dq}.$$

For an order-$n \ge 3$ Farey cell labeled by $q \le s$, one has $s \ge 3$ and $dq \ge 2$. If $dq = 2$, the coefficient $1/(dq) = 1/2$ is multiplied by the strict weight $1 - t < 1$, while the other coefficient is smaller than $1/2$. Thus the open-cell convex combination is strictly smaller than $1/2$, and

$$A > 0, \qquad B > 0, \qquad A + B < \pi. \tag{II.2.7}$$

Proposition II.2.3 (One scalar equation per ray)**.** *There is a unique* $\rho^{(n)}_{f,g}(x) \in (0, 1)$ *satisfying*

$$\rho^{s/d} \sin A + \rho^q \sin B = \sin(A + B). \tag{II.2.8}$$

For this radius,

$$\alpha = \frac{\rho^{s/d} \sin A}{\sin(A+B)}, \qquad \beta = \frac{\rho^q \sin B}{\sin(A+B)} \tag{II.2.9}$$

are nonnegative and sum to one, and $\lambda = \rho e^{2\pi i x}$ *satisfies* (II.2.4)*. The rooted identity* (II.2.10)*, with the sheet specified in the proof, varies continuously with* x *throughout the open cell.*

Proof. The left side of (II.2.8) is strictly increasing in ρ. At $\rho = 0$ it is zero, while at $\rho = 1$ it exceeds the right side because

$$\sin A + \sin B - \sin(A+B) = 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{A+B}{2} > 0.$$

This proves existence and uniqueness. Equation (II.2.9) and (II.2.8) give $\alpha + \beta = 1$.

Set

$$z = \overline{\lambda}^{-1} = \rho^{-1} e^{2\pi i x}.$$

Choose the dth-root sheet anchored at the endpoint r/s by defining

$$\omega = e^{-2\pi i r/d}, \qquad \omega z^{s/d} := \rho^{-s/d} \exp\left(\frac{2\pi i(sx - r)}{d}\right).$$

The two quantities $qx - p$ and $sx - r$ have opposite signs. Hence, with a common choice of sign,

$$z^q = \rho^{-q} e^{\pm iA}, \qquad \omega z^{s/d} = \rho^{-s/d} e^{\mp iB}.$$

Moreover (II.2.9) gives

$$\beta \rho^{-q} = \frac{\sin B}{\sin(A+B)}, \qquad \alpha \rho^{-s/d} = \frac{\sin A}{\sin(A+B)}.$$

The cancellation in the vector identity is completely explicit. The imaginary part of the proposed right side is

$$\frac{\sin B \sin A - \sin A \sin B}{\sin(A+B)} = 0,$$

and its real part is

$$\frac{\sin B \cos A + \sin A \cos B}{\sin(A+B)} = 1.$$

Thus resolving the two vectors into real and imaginary parts gives

$$1 = \beta z^q + \alpha \omega z^{s/d}. \tag{II.2.10}$$

Raising to the dth power after moving the first term gives

$$(1 - \beta \overline{\lambda}^{-q})^d = \alpha^d \overline{\lambda}^{-s},$$

and therefore

$$(\overline{\lambda}^q - \beta)^d = \alpha^d \overline{\lambda}^{dq-s}.$$

Conjugation is exactly (II.2.4). All quantities in (II.2.10) depend continuously on x on the open cell. □

Proposition II.2.4 (Endpoint limits, including the order-three exception)**.** *On every open Farey cell, $x \mapsto \rho^{(n)}_{f,g}(x)$ is continuous. If $n \geq 4$, it extends continuously to the closed cell with value 1 at both endpoints, and the corresponding points tend to the two endpoint roots of unity. The same conclusion holds for $n = 3$ except on the terminal cell $[1/3, 1/2]$. On that cell,*

$$\lim_{x \downarrow 1/3} \rho^{(3)}_{1/3,1/2}(x) = 1, \qquad \lim_{x \uparrow 1/2} \rho^{(3)}_{1/3,1/2}(x) = \frac{1}{2}, \tag{II.2.11}$$

so the nonreal radial graph tends to $-1/2$, not to the endpoint root -1.

Proof. Open-cell continuity follows from strict monotonicity in ρ in (II.2.8), equivalently from the implicit-function theorem. Let x_k tend to an endpoint and pass to a subsequence for which $\rho(x_k) \to r \in [0,1]$.

At the endpoint p/q, where $A \to 0$, determinant one gives

$$B_0 = \frac{2\pi}{dq}.$$

Thus $0 < B_0 \le \pi$. Equality can occur only when $dq = 2$; under $n \ge 3$, $q \le s$, and the upper-half-plane Farey restrictions, this is exactly the order-three terminal cell, with $q = 2$, $d = 1$, endpoint $p/q = 1/2$, and opposite endpoint $r/s = 1/3$. Outside that single case, $B_0 \in (0,\pi)$ and the limiting equation

$$r^q \sin B_0 = \sin B_0$$

forces $r = 1$.

At the other endpoint r/s, where $B \to 0$, one has

$$A_0 = \frac{2\pi}{s} \in (0,\pi),$$

because $s \ge 3$. Hence $r^{s/d} \sin A_0 = \sin A_0$ and again $r = 1$.

It remains to evaluate the exceptional limit. Put $x = 1/2 - \varepsilon$ in the cell $[1/3, 1/2]$. Then

$$A = 4\pi\varepsilon, \qquad B = \pi - 6\pi\varepsilon, \qquad A + B = \pi - 2\pi\varepsilon,$$

and (II.2.8) is

$$\rho^3 \sin(4\pi\varepsilon) + \rho^2 \sin(6\pi\varepsilon) = \sin(2\pi\varepsilon).$$

After division by $2\pi\varepsilon$ and passage to the limit, every subsequential limit satisfies

$$2r^3 + 3r^2 = 1, \qquad (2r-1)(r+1)^2 = 0.$$

The unique solution in $[0,1]$ is $r = 1/2$. Since every subsequence has the same limit, all asserted endpoint limits follow. In the exceptional asymptotics, (II.2.9) also gives $\alpha \to (1/2)^3 \cdot 4/2 = 1/4$, so the nonreal arc meets the added real branch at the same algebraic parameter. At every nonexceptional endpoint, (II.2.10) tends to the corresponding root of unity. $\square$

Definition II.2.5 (Farey–Ito carrier)**.** Let

$$\Gamma^{(n),\circ}_{f,g} = \left\{ \rho^{(n)}_{f,g}(x) e^{2\pi i x} : f < x < g \right\}. \tag{II.2.12}$$

For every cell other than the order-three terminal cell, define

$$\Gamma^{(n)}_{f,g} = \overline{\Gamma^{(n),\circ}_{f,g}}.$$

For the exceptional cell, define

$$\Gamma^{(3)}_{1/3,1/2} = \overline{\Gamma^{(3),\circ}_{1/3,1/2}} \cup [-1, -1/2]. \tag{II.2.13}$$

Every point of the open radial graph satisfies the Ito equation by proposition II.2.3. On the added real segment, writing $\alpha = -\lambda(\lambda+1) \in [0, 1/4]$ gives

$$\lambda^3 - (1-\alpha)\lambda - \alpha = (\lambda - 1)(\lambda^2 + \lambda + \alpha) = 0,$$

so the exceptional segment belongs to the same algebraic family.

Algorithm II.2.6 (Boundary extraction)**.** For a fixed order $n \ge 3$:

1. Generate F_n^+ in increasing order.
2. For each consecutive pair $f < g$, label the endpoint with smaller denominator p/q and the other r/s, compute $d = \lfloor n/q \rfloor$, and record (II.2.5).
3. For a desired ray $x \in (f, g)$, compute A, B from (II.2.6) and solve the strictly monotone equation (II.2.8) by bisection or Newton iteration.
4. Recover α, β from (II.2.9). At this stage the corresponding point is the unique *scalar candidate* on that ray. Appendix II.7 constructs a stochastic matrix realizing it, and appendices II.8 and II.9 prove the nesting and outer-boundary statements needed to identify it as the boundary point and fill the radial segment below it.

II.3 Main theorem

Theorem II.3.1 (Karpelevič–Ito, standalone form)**.** *Let $n \geq 3$. For every consecutive pair $f < g$ in F_n^+, let $\Gamma_{f,g}^{(n)}$ be the carrier of definition II.2.5. Then the upper boundary of Θ_n is the ordered chain of these carriers, from 1 to -1. The lower boundary is its complex conjugate, and*

$$\Theta_n = \{t\zeta : 0 \leq t \leq 1,\ \zeta \in \partial\Theta_n\}\,. \tag{II.3.1}$$

The only points of Θ_n on the unit circle are roots of unity of order at most n.

For $n = 3$, the terminal carrier contains the boundary segment $[-1, -1/2]$ before leaving the real axis. For $n = 1$, one has $\Theta_1 = \{1\}$; for $n = 2$, one has $\Theta_2 = [-1, 1]$.

The proof occupies appendix II.4–appendix II.9. The principal upper-bound input is theorem 1.4 from Part I, restated for the new-shell setting in appendix II.5. Everything after it is one-dimensional convexity, explicit realization, and Farey nesting.

II.4 Invariant polygons and elementary structure

Proposition II.4.1 (Compactness, conjugation, and disk bound)**.** *For every n, the set Θ_n is compact, invariant under complex conjugation, and contained in the closed unit disk.*

Proof. Let $\mathcal{S}_n$ be the compact set of row-stochastic matrices. First note that every eigenvalue lies in the closed unit disk: if $Av = \lambda v$, choose an index with $|v_i| = \|v\|_\infty$; then

$$|\lambda|\|v\|_\infty = |(Av)_i| \leq \|v\|_\infty.$$

Consequently the eigenpair set

$$\mathcal{E}_n = \{(A, \lambda) \in \mathcal{S}_n \times \overline{\mathbb{D}} : \det(\lambda I - A) = 0\}$$

is closed in a compact space and is therefore compact. Its projection to the λ-coordinate is exactly Θ_n, so Θ_n is compact. Real characteristic polynomials give conjugation symmetry. $\square$

Theorem II.4.2 (Invariant-polygon criterion)**.** *Let $n \geq 2$. For $\lambda \in \mathbb{C}$, the following are equivalent.*

(i) $\lambda \in \Theta_n$.

(ii) *There is a non-singleton convex polygon $P \subset \mathbb{C}$ with at most n vertices such that*

$$\lambda P \subseteq P. \tag{II.4.1}$$

Proof. If $Av = \lambda v$, put $P = \operatorname{conv}\{v_1, \ldots, v_n\}$. Every coordinate satisfies

$$\lambda v_i = \sum_j a_{ij} v_j \in P,$$

so (II.4.1) holds. If the hull is a singleton, all coordinates of the nonzero eigenvector are equal to the same nonzero number, so $\lambda = 1$; in that case any fixed segment may be used.

Conversely, let $x_1, \ldots, x_m$ be the vertices of an invariant polygon, $m \le n$. Choose barycentric coordinates

$$\lambda x_i = \sum_{j=1}^{m} a_{ij} x_j, \qquad a_{ij} \ge 0, \qquad \sum_j a_{ij} = 1.$$

Then $A = (a_{ij})$ is row-stochastic and $Ax = \lambda x$. If $m < n$, append absorbing states and extend the eigenvector by zero coordinates. □

Corollary II.4.3 (Radial filling)**.** *Let $n \ge 2$. If $\lambda \in \Theta_n$, then $t\lambda \in \Theta_n$ for every $0 \le t \le 1$.*

Proof. If $|\lambda| < 1$, take an invariant polygon P. Iteration gives $\lambda^k P \subseteq P$ and $\lambda^k x \to 0$, hence $0 \in P$; convexity then gives $t\lambda P \subseteq P$.

Suppose $|\lambda| = 1$. If $\lambda = 1$, the segment $[0, 1]$ satisfies $t[0, 1] \subseteq [0, 1]$. Otherwise take an invariant polygon with the least number of vertices. If it has nonzero area, area equality forces $\lambda P = P$, so multiplication by λ permutes its vertices and λ has finite order $k \le n$. The regular orbit polygon

$$Q = \operatorname{conv}\{1, \lambda, \ldots, \lambda^{k-1}\}$$

contains 0, satisfies $\lambda Q = Q$, and therefore satisfies $t\lambda Q = tQ \subseteq Q$. If the minimal polygon is a segment, then $\lambda = -1$ and the same conclusion follows from $Q = [-1, 1]$. The invariant-polygon criterion completes the proof. □

Proposition II.4.4 (Unit-circle points)**.** *Let $n \ge 2$. If $\lambda \in \Theta_n$ and $|\lambda| = 1$, then λ is a root of unity of order at most n. Conversely every such root belongs to Θ_n.*

Proof. Take an invariant polygon with the least number of vertices. If it has nonzero area, then $\lambda P \subseteq P$ and equality of areas force $\lambda P = P$. Multiplication by λ permutes the vertices, so a nonzero vertex has an orbit of length at most n and $\lambda^k = 1$. If the polygon is a segment, $\lambda = \pm 1$. Conversely, a cyclic permutation matrix realizes every root of unity of order at most n, after padding. □

Lemma II.4.5 (Interior origin for a nonreal contraction)**.** *Let P be a non-singleton compact convex polygon and let $\lambda = \rho e^{i\theta}$ with $0 < \rho < 1$ and $\theta \not\equiv 0, \pi \pmod{2\pi}$. If $\lambda P \subseteq P$, then P has nonempty interior and $0 \in \operatorname{int} P$.*

Proof. Iteration gives $0 \in P$. If P were a segment, its affine line would pass through 0 and would have to contain the nondegenerate segment λP; hence multiplication by $e^{i\theta}$ would preserve a real line, forcing $\theta \equiv 0$ or π, a contradiction. Thus P has nonempty interior.

Suppose $0 \in \partial P$. Choose a nonzero real linear functional ℓ such that $\ell \ge 0$ on P and choose $z \in P$ with $\ell(z) > 0$. Invariance would give

$$\ell(e^{ik\theta} z) \ge 0 \qquad (k \ge 0).$$

If $e^{i\theta}$ has finite order m, then $m \ge 3$ and $\sum_{k=0}^{m-1} e^{ik\theta} z = 0$; summing the displayed nonnegative quantities forces them all to vanish, contrary to $\ell(z) > 0$. If $e^{i\theta}$ has infinite order, its powers are dense in the unit circle, so some $e^{ik\theta} z$ lies in the open half-plane where $\ell < 0$, again a contradiction. Hence 0 is interior. □

For $0 < \theta < \pi$, define the radial function

$$R_n(\theta) = \max\left\{\rho \ge 0 : \rho e^{i\theta} \in \Theta_n\right\}. \tag{II.4.2}$$

The maximum exists by compactness and radial filling. On an open Farey cell it is strictly less than one by proposition II.4.4.

Definition II.4.6 (Polygonal complexity and radial criticality)**.** For a real-linear map $A : \mathbb{R}^2 \to \mathbb{R}^2$, let

$$\nu_{\mathrm{poly}}(A) = \min\Big\{|\mathrm{Ext}\, P| : P \text{ is a nondegenerate compact convex polygon and } AP \subseteq P\Big\},$$

with value ∞ if no such polygon exists. An elliptic contraction T is *N-critical* when

$$\nu_{\mathrm{poly}}(T) = N, \qquad \nu_{\mathrm{poly}}(tT) > N \quad (t > 1).$$

Here elliptic means $(\mathrm{tr}\, T)^2 < 4 \det T$, and contraction means that the spectral radius lies in $(0, 1)$.

For $N \geq 4$ and $0 < \theta < \pi$, a *new-shell radial extremum* is a point satisfying

$$\lambda = R_N(\theta)e^{i\theta} \in \Theta_N \setminus \Theta_{N-1}, \qquad 0 < |\lambda| < 1. \tag{II.4.3}$$

Proposition II.4.7 (New-shell extrema are polygonally critical)**.** *If* (II.4.3) *holds and $T_\lambda : \mathbb{C} \to \mathbb{C}$ is multiplication by λ, regarded as a real-linear map, then T_λ is an N-critical elliptic contraction.*

Proof. In the real basis $(1, i)$,

$$[T_\lambda] = \begin{pmatrix} \mathrm{Re}\,\lambda & -\mathrm{Im}\,\lambda \\ \mathrm{Im}\,\lambda & \mathrm{Re}\,\lambda \end{pmatrix}.$$

Since $0 < \theta < \pi$, the multiplier is nonreal, and therefore

$$(\mathrm{tr}\, T_\lambda)^2 - 4 \det T_\lambda = -4(\mathrm{Im}\,\lambda)^2 < 0.$$

Its spectral radius is $|\lambda| \in (0, 1)$, so it is an elliptic contraction.

By theorem II.4.2, membership $\lambda \in \Theta_N$ gives a non-singleton invariant polygon with at most N vertices. By lemma II.4.5, every such polygon is nondegenerate, so $\nu_{\mathrm{poly}}(T_\lambda) \leq N$. If this complexity were at most $N-1$, the same invariant-polygon criterion would give $\lambda \in \Theta_{N-1}$, contrary to (II.4.3). Consequently

$$\nu_{\mathrm{poly}}(T_\lambda) = N.$$

Finally, radial maximality in the definition of $R_N(\theta)$ gives $t\lambda \notin \Theta_N$ for every $t > 1$. If $\nu_{\mathrm{poly}}(tT_\lambda) \leq N$, a nondegenerate polygon witnessing that inequality would put $t\lambda$ in Θ_N by theorem II.4.2, a contradiction. Hence $\nu_{\mathrm{poly}}(tT_\lambda) > N$ for all $t > 1$, which is precisely N-criticality. □

II.5 Critical-polygon input from Part I

For a nonzero complex number z that is not positive real, let

$$\arg_+(z) \in (0, 2\pi)$$

denote its positive lifted argument. The following is the complete interface with Part I used below.

Theorem II.5.1 (Part I critical-polygon monodromy)**.** *Let $N \geq 4$ and suppose* (II.4.3) *holds. Retain $\lambda = |\lambda|e^{i\theta}$ and $x = \theta/(2\pi)$ for the original upper-half-plane multiplier. There are a selected multiplier*

$$\mu \in \{\lambda, \overline{\lambda}\}, \qquad \vartheta = \arg_+(\mu), \qquad y = \frac{\vartheta}{2\pi},$$

and consecutive reduced fractions in F_N*,*

$$\frac{p}{q} < y < \frac{r}{s}, \qquad rq - ps = 1, \qquad q \le s.$$

Thus $y = x$ *when* $\mu = \lambda$ *and* $y = 1 - x$ *when* $\mu = \overline{\lambda}$*. Put*

$$d = \lfloor N/q \rfloor, \qquad e = s - dq.$$

There are parameters

$$0 \le \beta_j < 1, \qquad \alpha_j = 1 - \beta_j > 0 \qquad (1 \le j \le d) \tag{II.5.1}$$

such that

$$\mu^s \prod_{j=1}^{d} (\mu^q - \beta_j) = \mu^{dq} \prod_{j=1}^{d} \alpha_j. \tag{II.5.2}$$

Equivalently, since $\mu \ne 0$*,*

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta_j) = \prod_{j=1}^{d} \alpha_j. \tag{II.5.3}$$

Any zero factor introduced only to complete a return strip is algebraic padding and is not claimed to be an additional strict contact.

Set

$$A = q\vartheta - 2\pi p, \qquad M = \arg_+(\mu^q - 1).$$

For the unique arguments $u_j \in (0, \pi)$ *determined by*

$$e^{iu_j} = \frac{\mu^q - \beta_j}{|\mu^q - \beta_j|},$$

one has

$$0 < A < M < \pi, \qquad u_j \in [A, M) \quad (1 \le j \le d), \tag{II.5.4}$$

and the exact lifted phase identity

$$e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(r - dp). \tag{II.5.5}$$

Proof. By proposition II.4.7, T_λ is an N-critical elliptic contraction. Apply the complex monodromy and Farey-carrier theorem of Part I, theorem 1.4, including its Jensen-sheet conclusion. The determinant identity for the consecutive Farey endpoints follows from lemma II.2.1. The two adapted complex orientations give precisely the choice between λ and $\overline{\lambda}$. □

Lemma II.5.2 (Reflection dictionary for the selected orientation)**.** *In theorem II.5.1, suppose* $\mu = \overline{\lambda}$*, so that* $y = 1 - x$*. Then*

$$\frac{s - r}{s} < x < \frac{q - p}{q}$$

is the reflected order-N *Farey cell. With the denominator-based labeling used in* (II.2.6)*,*

$$\begin{aligned} 2\pi\,|qx - (q - p)| &= 2\pi(qy - p) = A, \\ \frac{2\pi}{d}\,|sx - (s - r)| &= \frac{2\pi}{d}(r - sy) = \frac{2\pi r - s\vartheta}{d}. \end{aligned}$$

Thus reflection preserves q, s, d, e*, the modulus* $|\lambda| = |\mu|$*, and the absolute scalar equation* (II.2.8)*. Conjugating a constant-profile carrier for* μ *gives the Ito carrier* (II.2.4) *for* λ *in this reflected cell.*

Proof. The map $t \mapsto 1-t$ reverses the endpoint order and preserves denominators, giving the displayed cell. Since $x = 1-y$,

$$qx - (q-p) = p - qy < 0, \qquad sx - (s-r) = r - sy > 0,$$

which proves the two identities. All remaining assertions follow directly from the definitions and complex conjugation. $\square$

II.6 Log-sine equalization and the sharp radial inequality

The Part I monodromy input supplies two scalar constraints. Taking moduli of the product turns the factors into a sum of one potential $F(u_j)$, while the lifted phase identity fixes their arithmetic mean at $A+B$. On the common sheet proved above, $F'' > 0$, so Jensen's inequality is strict unless all factor arguments, and hence all contact parameters, agree. At a new-shell radial maximum the sparse realization supplies the matching equality point; the upper inequality is thus sharp rather than merely qualitative.

We work in the oriented cell

$$\frac{p}{q} < x < \frac{r}{s}, \qquad rq - ps = 1, \qquad q \le s. \tag{II.6.1}$$

Put $\theta = 2\pi x$ and

$$A = q\theta - 2\pi p, \qquad B = \frac{2\pi r - s\theta}{d}. \tag{II.6.2}$$

These are the non-absolute versions of (II.2.6). The same determinant-one computation as in (II.2.7) gives

$$A > 0, \qquad B > 0, \qquad A + B < \pi. \tag{II.6.3}$$

II.6.1 The factor potential

Fix $\lambda = \rho e^{i\theta}$ with $0 < \rho < 1$ and choose

$$M = \operatorname{Arg}(\lambda^q - 1)$$

so that $A < M < \pi$. Since $\lambda^q = \rho^q e^{iA}$ lies in the open upper half-plane, the horizontal segment $\lambda^q - [0,1)$ never meets the origin. Its argument is a strictly increasing continuous function of β with range $[A, M)$. For $0 \le \beta < 1$, let

$$u = \operatorname{Arg}(\lambda^q - \beta) \in [A, M)$$

be this uniquely determined branch. The triangle with vertices β, 1, and λ^q gives

$$F(u) := \log \frac{|\lambda^q - \beta|}{1-\beta} = \log \sin M - \log \sin(M-u), \tag{II.6.4}$$

and hence

$$F''(u) = \csc^2(M-u) > 0. \tag{II.6.5}$$

At $\beta = 0$, the same triangle gives

$$\rho^q = \frac{\sin M}{\sin(M-A)}. \tag{II.6.6}$$

Theorem II.6.1 (Heterogeneous sharp inequality)**.** *Suppose* λ *and parameters* (II.5.1) *satisfy* (II.5.3) *in the cell* (II.6.1)*. For each factor let* $u_j \in [A, M)$ *be the branch just defined, and suppose these arguments satisfy* (II.5.5)*. Then*

$$\rho^{s/d} \sin A + \rho^q \sin B \leq \sin(A+B). \tag{II.6.7}$$

Equality holds if and only if

$$\beta_1 = \cdots = \beta_d. \tag{II.6.8}$$

Proof. The phase identity gives

$$\sum_{j=1}^{d} u_j = 2\pi(r - dp) - (s - dq)\theta = d(A+B).$$

Taking moduli in (II.5.3) gives

$$\sum_{j=1}^{d} F(u_j) = (dq - s)\log\rho.$$

Because every u_j belongs to the interval $[A, M)$, their average $A+B$ belongs to the same interval and F is strictly convex on the entire convex hull of the arguments. Strict Jensen convexity gives

$$dF(A+B) \leq (dq - s)\log\rho. \tag{II.6.9}$$

Equality holds exactly when all u_j coincide. The argument map $\beta \mapsto \operatorname{Arg}(\lambda^q - \beta)$ is strictly increasing, so this is equivalent to all β_j coinciding.

Exponentiating (II.6.9) and using (II.6.6) yields

$$\frac{\sin(M - A - B)}{\sin M} \geq \rho^{s/d - q}.$$

Using (II.6.6) and the identity

$$\sin(M-A)\sin(A+B) - \sin M \sin B = \sin A \sin(M - A - B)$$

gives

$$\frac{\sin(M-A-B)}{\sin M} = \frac{\rho^{-q}\sin(A+B) - \sin B}{\sin A}. \tag{II.6.10}$$

Multiplication by $\rho^q \sin A$ gives (II.6.7). ☐

Corollary II.6.2 (Outermost equality profile)**.** *For a new-shell radial extremum in an open Farey cell, the parameters in the monodromy output selected by theorem II.5.1 satisfy*

$$\beta_1 = \cdots = \beta_d = \beta, \qquad \alpha = 1 - \beta,$$

and the original point lies on (II.2.4)*. Its modulus is the unique solution of* (II.2.8)*.*

Proof. Let μ, ϑ, y, and the Farey data be supplied by theorem II.5.1, and put

$$B_\mu = \frac{2\pi r - s\vartheta}{d}.$$

The same theorem supplies exactly the product, phase, and argument-sheet hypotheses of theorem II.6.1. Applying that theorem to μ gives

$$\rho^{s/d} \sin A + \rho^q \sin B_\mu \leq \sin(A + B_\mu), \qquad \rho = |\mu|.$$

If $\mu = \lambda$, the oriented quantities are the absolute quantities of the original cell. If $\mu = \overline{\lambda}$, lemma II.5.2 identifies them with those absolute quantities. Hence in both cases, with $\rho = |\lambda|$,

$$\rho^{s/d} \sin A + \rho^q \sin B \leq \sin(A + B)$$

for the original ray.

Let ρ_* be the unique equality radius supplied by proposition II.2.3. By theorem II.7.3, it is attained in order N, so $\rho_* \leq \rho = R_N(\theta)$. Strict monotonicity of the scalar left side and the preceding inequality give the reverse inequality. Therefore $\rho = \rho_*$ and equality holds in theorem II.6.1. Its equality statement gives a constant profile for μ. For $\mu = \lambda$ this is (II.2.4) directly; for $\mu = \overline{\lambda}$ it becomes that carrier for the original cell by lemma II.5.2. $\square$

II.7 Explicit stochastic realization and attainment

We now prove the inner inclusion directly, without importing an Ito realization theorem.

Throughout this section we use the tail–row convention: an arrow $u \to v$ of weight $w(u, v)$ is encoded by

$$A_{uv} = w(u, v), \tag{II.7.1}$$

with $A_{uv} = 0$ when the arrow is absent. Thus the sum of the weights of arrows leaving u is the row sum of A at u.

Lemma II.7.1 (Cycle-cover coefficient rule)**.** *Let A be the weighted adjacency matrix of a finite directed graph. In $\det(tI - A)$, a collection of k pairwise vertex-disjoint directed cycles contributes*

$$(-1)^k t^{N-L}$$

times the product of their edge weights, where L is the total number of vertices on the cycles. Vertices not on a selected cycle contribute diagonal factors t.

Proof. With the convention (II.7.1), a directed cycle $u_1 \to u_2 \to \cdots \to u_\ell \to u_1$ is selected in the Leibniz expansion by the permutation cycle $u_i \mapsto u_{i+1}$ and contributes the edge product $A_{u_1 u_2} \cdots A_{u_\ell u_1}$. Its permutation sign is $(-1)^{\ell-1}$, while selecting the $-A$ entry at its ℓ vertices contributes $(-1)^\ell$, so the net sign is -1. This also covers $\ell = 1$: one selects $-A_{uu}$ from the diagonal factor $t - A_{uu}$. Disjoint selected cycles multiply, and at every remaining fixed vertex one selects t. Hence a collection of k cycles has the stated sign and power of t. $\square$

Lemma II.7.2 (Cycle collections in the sparse block graph)**.** *Consider the block graph used in theorem II.7.3: each block is a deterministic directed path to its terminal vertex, each terminal has one local return edge to the start of the same block and one cross edge to the next block, and at most one cross edge is subdivided by deterministic intermediate vertices. Then every directed simple cycle is either one local q-cycle inside a single block or the unique global cycle using all d cross edges. Moreover every local cycle shares its terminal vertex with the global cycle. Consequently the vertex-disjoint directed cycle collections are exactly either*

(i) *an arbitrary subset of the d local cycles, or*

(ii) *the singleton global cycle.*

Proof. A directed cycle that uses no cross edge is trapped in one block. The only way to leave the terminal and remain in that block is the local return, after which the deterministic path forces the whole local q-cycle.

Now let a simple cycle use a cross edge into some block. Once it enters that block, all following edges are forced along the deterministic suffix until the terminal is reached. If the cycle then takes the local return, the deterministic path reaches the previously visited entry vertex before the original cycle has closed; hence a simple cycle using one cross edge cannot use any local return. It must therefore take the cross edge at every terminal it reaches. Since the cross edges advance cyclically through the d blocks, this produces exactly the unique global cross cycle. The subdivision vertices, when present, lie on that global route and introduce no choice.

The global cycle passes through the terminal vertex of every block, and the local cycle in that block also passes through that terminal. Thus no local cycle is vertex-disjoint from the global cycle, while distinct local cycles live in disjoint blocks. This proves the stated classification of cycle collections. □

Theorem II.7.3 (Sparse carrier realization)**.** *Let $p/q, r/s$ be a Farey pair of order n, labeled by $q \le s$, and let $d = \lfloor n/q \rfloor$. For every $\alpha \in [0,1]$, $\beta = 1 - \alpha$, there is a row-stochastic matrix of order*

$$N_0 = \max\{dq, s\} \le n \tag{II.7.2}$$

whose spectrum contains every root of the reduced carrier (II.2.5)*, and hence every nonzero root of* (II.2.4)*.*

Proof. Take d blocks

$$B_j = \{v_{j,0}, \dots, v_{j,q-1}\}.$$

Inside each block put deterministic edges of weight 1, $v_{j,k} \to v_{j,k+1}$ for $k < q-1$. At the terminal vertex $v_{j,q-1}$ put a local return $v_{j,q-1} \to v_{j,0}$ of weight β and a cross-block edge of weight α entering block $j+1$, with block indices read cyclically modulo d.

Suppose first $s \le dq$. Farey consecutiveness gives $q + s > n \ge dq$, hence $s > (d-1)q$. Put

$$\ell_1 = s - (d-1)q, \qquad \ell_2 = \cdots = \ell_d = q.$$

The cross edge entering block j is, by convention, the edge whose tail is the terminal of block $j-1$. Route it to $v_{j,q-\ell_j}$. The local cycles have length q and weight β. The unique global cross cycle uses all d cross edges, has weight α^d, and visits exactly ℓ_j vertices in block j, hence has length s. By lemma II.7.2, the only cycle-cover collections are arbitrary local-cycle subsets or the singleton global cycle. Therefore the local subsets contribute

$$\sum_{J \subseteq \{1,\dots,d\}} (-1)^{|J|} \beta^{|J|} t^{dq - q|J|} = (t^q - \beta)^d,$$

and the global cycle contributes $-\alpha^d t^{dq-s}$. By lemma II.7.1,

$$\det(tI - A) = (t^q - \beta)^d - \alpha^d t^{dq-s}. \tag{II.7.3}$$

If $s > dq$, route every cross edge into the first vertex of the next block. Set $K = s - dq > 0$. On one such edge $u \to v$, introduce exactly the K new vertices $w_1, \dots, w_K$ and replace that edge by the path

$$u \xrightarrow{\alpha} w_1 \xrightarrow{1} \cdots \xrightarrow{1} w_K \xrightarrow{1} v.$$

The global cycle has length s and weight α^d. The local cycles remain the d disjoint q-cycles of weight β, and every local-cycle collection leaves all intermediate vertices unused. Using lemmas II.7.1 and II.7.2, the local subsets contribute

$$t^{s-dq} \sum_{J \subseteq \{1,\dots,d\}} (-1)^{|J|} \beta^{|J|} t^{dq - q|J|} = t^{s-dq}(t^q - \beta)^d,$$

while the singleton global cycle contributes $-\alpha^d$. Therefore

$$\det(tI - A) = t^{s-dq}(t^q - \beta)^d - \alpha^d. \tag{II.7.4}$$

By the tail–row convention, every nonterminal block vertex and every subdivision vertex has one outgoing edge of weight 1, while every block terminal has outgoing weight $\alpha + \beta = 1$. Hence A is row-stochastic. Equations (II.7.3) and (II.7.4) are exactly the two reduced forms in (II.2.5). Finally, pad by absorbing states to reach order n. □

Corollary II.7.4 (Attainment of the scalar boundary). *For every open Farey ray x, the point $\rho^{(n)}_{f,g}(x)e^{2\pi ix}$ belongs to Θ_n.*

Proof. Use (II.2.9) in theorem II.7.3. □

II.8 Farey refinement and nesting in the order

The Part I monodromy input applies only to new-shell points. To close the induction, we prove directly that the candidate radial regions are nested as n increases.

II.8.1 A log-line lemma

Let L be a line not through the origin. On any interval of angles for which L has a positive radial intersection, write the logarithm of that radius as $\ell(\phi)$. For suitable constants c, ϕ_0,

$$\ell(\phi) = c - \log\cos(\phi - \phi_0), \qquad \ell''(\phi) = \sec^2(\phi - \phi_0) > 0. \tag{II.8.1}$$

We shall use the following signed form of the chord calculation. Identify complex numbers with vectors in $\mathbb{R}^2$. For

$$X = Pe^{iA}, \qquad Y = Qe^{-iC}, \qquad P, Q > 0, \qquad A, C > 0, \qquad A + C < \pi,$$

define

$$\Delta_{X,Y}(r) := \det(Y - X, r - X) = r(P\sin A + Q\sin C) - PQ\sin(A + C), \qquad r > 0. \tag{II.8.2}$$

Thus the positive-real intercept of the line through X and Y is

$$I(P, Q) = \frac{PQ\sin(A + C)}{P\sin A + Q\sin C}, \tag{II.8.3}$$

and

$$I(P, Q) > 1 \iff \Delta_{X,Y}(1) < 0. \tag{II.8.4}$$

The endpoint monotonicities also have fixed signs:

$$\frac{\partial I}{\partial P} = \frac{Q^2\sin(A + C)\sin C}{(P\sin A + Q\sin C)^2} > 0, \qquad \frac{\partial I}{\partial Q} = \frac{P^2\sin(A + C)\sin A}{(P\sin A + Q\sin C)^2} > 0.$$

All chord comparisons below will be reduced to (II.8.2); no diagrammatic side convention is used.

Lemma II.8.1 (Mediant chord expansion in both denominator orientations). *Let $n \geq 4$, let $a/b < c/d$ be consecutive in F_{n-1}, and suppose the mediant*

$$\frac{a + c}{b + d}$$

is newly inserted in F_n, so $n = b + d$. Fix an interior ray of one of the two new subcells and let $R > 1$ be the reciprocal of the old order-$(n - 1)$ candidate radius on that ray. At this same R, the positive-real intercept of the order-n rooted chord is strictly larger than 1, the intercept of the old rooted chord. At the common Farey endpoints the two candidate outer radii are both 1 by definition. This holds whether $b < d$ or $d < b$.

Proof. We first treat $b < d$. Put

$$m = \left\lfloor \frac{n-1}{b} \right\rfloor, \qquad U = R^b e^{iA}, \qquad V = R^{d/m} e^{-iB},$$

where

$$A = 2\pi(bx - a), \qquad B = \frac{2\pi}{m}(c - dx).$$

At the old candidate radius, (II.2.10) gives

$$1 = \beta U + \alpha V, \qquad \alpha, \beta \in (0,1).$$

Consequently $1 \in \operatorname{relint}[U, V]$. Let L be the line through U, V, and let ℓ be its log-radial function on the angular interval $[-B, A]$. Then

$$\ell(0) = 0, \qquad \ell(A) = b \log R, \qquad \ell(-B) = \frac{d}{m} \log R,$$

and (II.8.1) says that ℓ' is strictly increasing.

Consider first the left subcell

$$\frac{a}{b} < x < \frac{a+c}{b+d},$$

equivalently $B > A/m$. Suppose initially that $b > 1$. Since $\gcd(b, d) = 1$, one has $b \nmid d$, and hence

$$\left\lfloor \frac{n}{b} \right\rfloor = \left\lfloor \frac{n-1}{b} \right\rfloor = m.$$

The new rooted chord has endpoints

$$U, \qquad W = VU^{1/m}, \qquad \arg W = -B + \frac{A}{m}.$$

Put $\phi = -B + A/m \in (-B, 0)$. Strict monotonicity of ℓ' gives

$$\begin{aligned} \ell(\phi) - \ell(-B) &= \int_0^{A/m} \ell'(-B + t)\, dt \\ &< \int_0^{A/m} \ell'(mt)\, dt = \frac{1}{m}\ell(A). \end{aligned} \tag{II.8.5}$$

Therefore

$$\log |W| = \ell(-B) + \frac{1}{m}\ell(A) > \ell(\phi).$$

Let

$$V_* = \exp(\ell(\phi)) e^{i\phi} \in [V, 1], \qquad W = \gamma V_*, \qquad \gamma > 1.$$

Since $U, V_*, 1$ are collinear,

$$\det(V_* - U, 1 - U) = 0.$$

The signed determinant of the new chord at 1 is therefore

$$\begin{aligned} \Delta_{U,W}(1) &= \det(W - U, 1 - U) \\ &= (\gamma - 1) \det(U, 1) \\ &= -(\gamma - 1) R^b \sin A < 0. \end{aligned} \tag{II.8.6}$$

By (II.8.4), its positive-real intercept is strictly larger than 1.

The right subcell satisfies

$$\frac{a+c}{b+d} < x < \frac{c}{d}, \qquad \text{equivalently} \qquad A > mB.$$

This calculation is valid also when $b=1$. Since $b<d$, the new multiplicity at the endpoint of denominator d is one. With the positive-angle endpoint written first, the new rooted endpoints are

$$X = UV^m, \qquad Y = V^m;$$

indeed their arguments are $A - mB > 0$ and $-mB < 0$. The new line is $V^m L$, because $1, U \in L$. Moreover

$$\ell(mB) + m\ell(-B) = m\int_0^B (\ell'(mt) - \ell'(-t))\,dt > 0. \tag{II.8.7}$$

Put

$$Z = \exp(\ell(mB))e^{imB} \in [1, U], \qquad \eta = V^{-m}.$$

Equation (II.8.7) says that $\eta = \gamma Z$ for some $0 < \gamma < 1$. Multiplication by V^m multiplies real determinants by the positive number $|V|^{2m}$, so

$$\begin{aligned}\Delta_{X,Y}(1) &= \det(V^m(1-U), 1 - V^m U)\\ &= |V|^{2m}\det(1-U, \eta - U)\\ &= |V|^{2m}(\gamma - 1)\det(1-U, U)\\ &= |V|^{2m}(\gamma-1)R^b \sin A < 0.\end{aligned} \tag{II.8.8}$$

Here the third equality uses the collinearity of $1, U, Z$. Again (II.8.4) gives an intercept strictly larger than 1.

It remains to handle the left subcell when $b=1$. Farey consecutiveness then gives the old cell

$$\frac{0}{1} < \frac{1}{d},$$

and the new left cell has right endpoint $1/(d+1)$. Set

$$\phi_{\text{old}} = \frac{2\pi}{d}, \qquad \phi_{\text{new}} = \frac{2\pi}{d+1}.$$

For $0 < \theta < \phi_{\text{new}}$, the radial intersection of the chord from 1 to $e^{i\phi}$ is

$$r_\phi(\theta) = \frac{\sin\phi}{\sin\theta + \sin(\phi-\theta)} = \frac{\cos(\phi/2)}{\cos(\theta - \phi/2)}.$$

Its logarithmic derivative has the explicit sign

$$\begin{aligned}\frac{\partial}{\partial\phi}\log r_\phi(\theta) &= -\frac{1}{2}\left(\tan\frac{\phi}{2} + \tan\left(\theta - \frac{\phi}{2}\right)\right)\\ &= -\frac{\sin\theta}{2\cos(\phi/2)\cos(\theta-\phi/2)} < 0.\end{aligned} \tag{II.8.9}$$

Hence

$$r_{\phi_{\text{new}}}(\theta) > r_{\phi_{\text{old}}}(\theta).$$

At $R = r_{\phi_{\text{old}}}(\theta)^{-1}$, the intercept of the new reciprocal rooted chord is, by (II.8.3),

$$I_{\text{new}} = \frac{R\sin\phi_{\text{new}}}{\sin\theta + \sin(\phi_{\text{new}} - \theta)} = \frac{r_{\phi_{\text{new}}}(\theta)}{r_{\phi_{\text{old}}}(\theta)} > 1.$$

Finally suppose $d < b$. Complex conjugation sends $[a/b, c/d]$ to the oppositely oriented interval $[-c/d, -a/b]$. Translating both endpoint fractions by the same integer does not change their roots, reciprocal radii, scalar equations, or positive-real chord intercepts. The reflected interval has smaller denominator d at its left endpoint, so the case already proved applies and conjugation gives the two required inequalities in the original interval. Equality $b=d$ would force $b=d=1$ from $bc - ad = 1$, and then $b+d=2$, contrary to $n \geq 4$.

At every inherited Farey endpoint the two candidate radii are both 1 by definition. $\square$

Lemma II.8.2 (Multiplicity padding)**.** *Suppose a Farey cell is unchanged from order $n-1$ to order n, but*

$$\left\lfloor \frac{n}{q} \right\rfloor = \left\lfloor \frac{n-1}{q} \right\rfloor + 1 = M.$$

Let $\lambda = \rho e^{2\pi i x}$ be the order-$(n-1)$ equal-profile candidate on an open ray of this cell. Orient the cell—conjugating λ and reflecting its Farey endpoints when necessary—to obtain $\mu = \rho e^{i\vartheta}$ and $y = \vartheta/(2\pi)$ such that

$$\frac{p}{q} < y < \frac{r}{s}, \qquad rq - ps = 1, \qquad q \le s. \tag{II.8.10}$$

Then the order-$(n-1)$ equal-profile candidate is a heterogeneous order-n product obtained by adjoining one factor with $\beta_M = 0$, $\alpha_M = 1$. Its factors lie on the Jensen sheet and satisfy the order-n phase identity. Consequently its radius does not exceed the order-n equal-profile radius.

Proof. Put $m = M - 1$ and define

$$A = q\vartheta - 2\pi p, \qquad B_m = \frac{2\pi r - s\vartheta}{m}. \tag{II.8.11}$$

If the smaller-denominator endpoint is already on the left, take $\mu = \lambda$ and $y = \vartheta/(2\pi)$. Otherwise replace λ by $\overline{\lambda}$ and the cell by its reflected interval; denominators, moduli, and the scalar candidate equation are unchanged. Thus (II.8.10) holds, and the determinant-one calculation used in (II.6.3) gives

$$A > 0, \qquad B_m > 0, \qquad A + B_m < \pi.$$

The order-$(n-1)$ candidate has the equal profile $\beta_1 = \cdots = \beta_m = \beta$ and $\alpha_1 = \cdots = \alpha_m = \alpha = 1 - \beta$, where the scalar formulas at multiplicity m are

$$\beta = \frac{\rho^q \sin B_m}{\sin(A + B_m)}, \qquad \alpha = \frac{\rho^{s/m} \sin A}{\sin(A + B_m)}. \tag{II.8.12}$$

Since $\mu^q = \rho^q e^{iA}$, direct resolution into real and imaginary parts gives

$$\mu^q - \beta = \frac{\rho^q \sin A}{\sin(A + B_m)} e^{i(A + B_m)}. \tag{II.8.13}$$

Hence every old factor has its unique Jensen-sheet argument

$$u_1 = \cdots = u_m = A + B_m.$$

In particular, the old phase relation is not an additional assumption:

$$\begin{aligned} (s - mq)\vartheta + \sum_{j=1}^{m} u_j &= (s - mq)\vartheta + m(A + B_m) \\ &= 2\pi(r - mp). \end{aligned} \tag{II.8.14}$$

The old equal-profile carrier equation is

$$\mu^{s - mq} (\mu^q - \beta)^m = \alpha^m. \tag{II.8.15}$$

Now set

$$\beta_M = 0, \qquad \alpha_M = 1, \qquad u_M = A.$$

Because $M = m+1$, multiplying (II.8.15) by $\mu^{-q}(\mu^q - 0) = 1$ gives the order-n heterogeneous product

$$\mu^{s-Mq}(\mu^q - \beta)^m(\mu^q - 0) = \alpha^m. \tag{II.8.16}$$

Likewise, (II.8.14) and $A = q\vartheta - 2\pi p$ give

$$\begin{aligned}(s - Mq)\vartheta + \sum_{j=1}^{M} u_j &= 2\pi(r - mp) - q\vartheta + A \\ &= 2\pi(r - Mp).\end{aligned} \tag{II.8.17}$$

Finally, $0 < \beta < 1$ on an open ray. The argument map $t \mapsto \operatorname{Arg}(\mu^q - t)$ is strictly increasing on $[0, 1)$, so with $M_* = \operatorname{Arg}(\mu^q - 1)$ one has

$$u_1 = \cdots = u_m = A + B_m \in (A, M_*), \qquad u_M = A.$$

Thus all factors in (II.8.16) lie in the interval $[A, M_*)$ required by theorem II.6.1, and (II.8.17) is exactly its phase hypothesis at multiplicity M. Put

$$B_M = \frac{2\pi r - s\vartheta}{M} = \frac{m}{M}B_m.$$

Applying theorem II.6.1 to (II.8.16)–(II.8.17) gives the explicit order-n scalar sign

$$\rho^{s/M} \sin A + \rho^q \sin B_M - \sin(A + B_M) < 0. \tag{II.8.18}$$

The inequality is strict: the M factor parameters consist of m copies of $\beta \in (0, 1)$ and one copy of 0, so they are not all equal, and the equality condition in theorem II.6.1 cannot hold. The left side of (II.8.18) is strictly increasing as a function of $\rho > 0$ and vanishes at the order-n equal-profile radius. Thus the old radius is strictly smaller than the new one on every open ray. Conjugating back preserves the modulus. □

For $n \geq 3$, define the candidate outer radius K_n as follows. For $x = \theta/(2\pi)$ in an open Farey cell and $0 < \theta < \pi$, use the unique scalar radius of proposition II.2.3; at a Farey endpoint in $[0, \pi)$ set $K_n = 1$; and set

$$K_n(\pi) = 1. \tag{II.8.19}$$

For $n \geq 4$ this is the continuous closed-cell scalar graph. For $n = 3$ it has the necessary jump $\lim_{\theta\uparrow\pi} K_3(\theta) = 1/2$ but $K_3(\pi) = 1$, representing the outer endpoint -1 of the terminal radial segment.

Lemma II.8.3 (Exhaustive order-refinement split)**.** *Let $n \geq 4$ and let $\theta/(2\pi) \in [0, 1/2]$. Comparing F_{n-1}^+ with F_n^+, exactly one of the following alternatives applies.*

(i) *The ray is an inherited Farey endpoint; both candidate outer radii are defined to be* 1.
(ii) *The ray is a newly inserted endpoint of denominator n; the order n candidate outer radius is* 1*, while the order $n-1$ ray is interior to one old cell.*
(iii) *The ray is interior to an old cell which is not split at this order. Then the endpoint pair is unchanged, and for the smaller denominator q the multiplicity changes from $\lfloor (n-1)/q \rfloor$ to $\lfloor n/q \rfloor$, hence either stays the same or increases by exactly one.*
(iv) *The ray is interior to one of the two cells obtained by inserting the mediant $(a+c)/(b+d)$ between consecutive old endpoints $a/b < c/d$; in this case $b + d = n$.*

There are no further cases.

Proof. The only fractions which can appear in F_n but not in F_{n-1} have reduced denominator exactly n. If such a fraction h/n is inserted between consecutive old endpoints $a/b < c/d$, the Farey adjacency criterion lemma II.2.1 gives $bc - ad = 1$ and $b + d \geq n$. Since h/n lies between them, the decomposition argument in the proof of lemma II.2.1 writes

$$(n, h) = m(b, a) + \ell(d, c)$$

with positive integers m, ℓ. Thus $n = mb + \ell d \geq b + d$. Hence $b + d = n$, $m = \ell = 1$, and $h/n = (a + c)/(b + d)$. This proves that every new interior split is exactly the mediant case and that no other old cell is split.

If the ray itself is a new fraction, it is case (ii). If it is an old endpoint, it is case (i). Otherwise it is interior to an old cell. When that old cell is split, case (iv) applies; when it is not split, the endpoint pair is unchanged. Increasing the order from $n - 1$ to n can change $\lfloor n/q \rfloor$ by at most one, so the unchanged-cell alternatives in case (iii) are exhaustive. The upper-half restriction merely discards the reflected lower-half copies and does not introduce another case. □

Theorem II.8.4 (Candidate nesting)**.** *With this definition,*

$$K_{n-1}(\theta) \leq K_n(\theta) \tag{II.8.20}$$

for every $0 \leq \theta \leq \pi$ *and every* $n \geq 4$*; the lower-half-plane version follows by conjugation.*

Proof. Fix first an angle θ which is interior to a cell of both F_{n-1}^+ and F_n^+. Label the endpoints of its order-n cell by $p/q, r/s$ with $q \leq s$, put

$$D = \left\lfloor \frac{n}{q} \right\rfloor,$$

and let A, B be the positive quantities in (II.2.6). Define the order-n scalar defect

$$\mathcal{F}_{n,\theta}(t) := t^{s/D} \sin A + t^q \sin B - \sin(A + B), \qquad t > 0. \tag{II.8.21}$$

Its derivative has the fixed sign

$$\mathcal{F}'_{n,\theta}(t) = \frac{s}{D} t^{s/D-1} \sin A + q t^{q-1} \sin B > 0. \tag{II.8.22}$$

By proposition II.2.3, $K_n(\theta)$ is the unique zero of $\mathcal{F}_{n,\theta}$.

Set

$$\rho_- = K_{n-1}(\theta), \qquad R = \rho_-^{-1}.$$

By lemma II.8.3, there are three interior-ray possibilities.

If the Farey cell and its multiplicity are unchanged, then its scalar equation is unchanged and

$$\mathcal{F}_{n,\theta}(\rho_-) = 0.$$

If the cell is unchanged but its multiplicity increases, then (II.8.18) gives

$$\mathcal{F}_{n,\theta}(\rho_-) < 0.$$

It remains to consider a cell created by inserting a new denominator-n mediant. The two reciprocal rooted endpoints of the new scalar chord at ρ_- are, after choosing their common orientation,

$$R^q e^{iA}, \qquad R^{s/D} e^{-iB}.$$

Its positive-real intercept is

$$I_R = \frac{R^{q+s/D}\sin(A+B)}{R^q \sin A + R^{s/D}\sin B}.$$

Direct subtraction, without a side convention, gives

$$I_R - 1 = \frac{R^{q+s/D}(-\mathcal{F}_{n,\theta}(\rho_-))}{R^q \sin A + R^{s/D}\sin B}. \tag{II.8.23}$$

By lemma II.8.1, $I_R > 1$; every other factor in (II.8.23) is positive. Hence

$$\mathcal{F}_{n,\theta}(\rho_-) < 0.$$

Thus in all three cases

$$\mathcal{F}_{n,\theta}(\rho_-) \le 0.$$

Together with (II.8.22) and $\mathcal{F}_{n,\theta}(K_n(\theta)) = 0$, this proves

$$K_{n-1}(\theta) = \rho_- \le K_n(\theta)$$

on every ray interior to both Farey decompositions. The inequality is strict whenever the cell is split or its multiplicity increases.

Now let $\theta/(2\pi)$ be a newly inserted fraction. It is interior to one order-$(n-1)$ cell, so

$$K_{n-1}(\theta) < 1$$

by proposition II.2.3, whereas $K_n(\theta) = 1$ by the endpoint definition. At every inherited Farey endpoint below π, both values are 1. At $\theta = 0$ the same is true, and at $\theta = \pi$ both sides are 1 by (II.8.19). The latter separate clause is essential for $n = 4$, because $\lim_{\theta\uparrow\pi} K_3(\theta) = 1/2$ although $K_3(\pi) = 1$. This proves (II.8.20) everywhere. Conjugation gives the lower-half-plane statement. □

II.9 Completion of the proof

Lemma II.9.1 (Compact radial sets with continuous radius)**.** *Let $S \subset \mathbb{C}$ be compact and radially filled, and suppose that*

$$r_S(\phi) := \max\{r \ge 0 : re^{i\phi} \in S\}$$

is positive and continuous on the circle. Then

$$S = \{t\, r_S(\phi)e^{i\phi} : 0 \le t \le 1,\ -\pi < \phi \le \pi\}, \qquad \partial S = \{r_S(\phi)e^{i\phi} : -\pi < \phi \le \pi\}. \tag{II.9.1}$$

Proof. Compactness makes each maximum attainable, and radial filling then says that the intersection of S with the ray of angle ϕ is exactly $\{re^{i\phi} : 0 \le r \le r_S(\phi)\}$. Taking the union over ϕ gives the first formula. A point $r_S(\phi)e^{i\phi}$ cannot be interior, since a sufficiently small outward radial displacement would contradict maximality.

Conversely, let $z = re^{i\phi}$ with $0 < r < r_S(\phi)$. Choose $\eta > 0$ with $r + 3\eta < r_S(\phi)$. By continuity, $r_S(\psi) > r + 2\eta$ for all ψ in some neighborhood of ϕ. A sufficiently small Euclidean ball about z consists of points w with $\arg w$ in that neighborhood and $|w| < r + \eta$; the ray description then puts the whole ball in S. Thus every nonzero point strictly below the radial graph is interior. Finally, positivity and compactness of the circle give $c := \min_\phi r_S(\phi) > 0$, so the disk $|z| < c$ lies in S and 0 is interior. No further boundary points remain. □

II.9.1 The base orders

Proposition II.9.2 (Orders one, two, and three)**.** *One has*

$$\Theta_1 = \{1\}, \qquad \Theta_2 = [-1, 1],$$

and, with $\omega = e^{2\pi i/3}$,

$$\Theta_3 = \operatorname{conv}\{1, \omega, \overline{\omega}\} \cup [-1, -1/2]. \tag{II.9.2}$$

Proof. For order one the only matrix is [1]. For order two, every stochastic matrix has the form

$$\begin{pmatrix} a & 1-a \\ b & 1-b \end{pmatrix}, \qquad 0 \le a, b \le 1,$$

and its second eigenvalue is $a - b$; hence the first two assertions follow.

Let a 3×3 stochastic matrix have nonreal eigenvalues $\lambda = x + iy$ and $\overline{\lambda}$. Its spectrum is $1, \lambda, \overline{\lambda}$. Since the diagonal is nonnegative,

$$1 + 2x = \operatorname{tr} A \ge 0,$$

so $x \ge -1/2$. Also

$$(\operatorname{tr} A)^2 \le 3 \sum_i a_{ii}^2 \le 3 \operatorname{tr}(A^2),$$

because every term $a_{ij}a_{ji}$ in $\operatorname{tr}(A^2)$ is nonnegative. Substituting

$$\operatorname{tr} A = 1 + 2x, \qquad \operatorname{tr}(A^2) = 1 + 2(x^2 - y^2)$$

gives

$$3y^2 \le (1 - x)^2.$$

Thus the nonreal region lies in the triangle in (II.9.2); every real stochastic eigenvalue lies in $[-1, 1]$.

The chord $[1, \omega]$ is realized by $(1 - \alpha)I + \alpha C_3$, $0 \le \alpha \le 1$. The terminal sparse realization of theorem II.7.3 has polynomial

$$\lambda^3 - (1 - \alpha)\lambda - \alpha = (\lambda - 1)(\lambda^2 + \lambda + \alpha). \tag{II.9.3}$$

For $0 \le \alpha \le 1/4$, the root $(-1 - \sqrt{1 - 4\alpha})/2$ runs from -1 to $-1/2$. For $1/4 < \alpha \le 1$, the upper root is

$$\lambda = -\frac{1}{2} + \frac{i}{2}\sqrt{4\alpha - 1},$$

and runs from $-1/2$ to ω. Conjugation supplies the lower edges, and radial filling fills the triangle and the real segment. This proves (II.9.2).

It remains to identify the two nonreal boundary pieces with the canonical radius K_3 used by the induction. On the Farey cell $[0, 1/3]$, the denominator labeling of (II.2.2) is

$$(q, s) = (1, 3), \qquad d = \lfloor 3/1 \rfloor = 3.$$

For $0 < \theta = 2\pi x < 2\pi/3$, (II.2.6) gives

$$A = \theta, \qquad B = \frac{2\pi}{3} - \theta, \qquad A + B = \frac{2\pi}{3}.$$

Thus (II.2.8) is

$$\rho \sin \theta + \rho \sin\left(\frac{2\pi}{3} - \theta\right) = \sin \frac{2\pi}{3}. \tag{II.9.4}$$

With the weights from (II.2.9), this equation is exactly $\alpha+\beta = 1$, and direct real and imaginary parts give

$$\rho e^{i\theta} = \beta + \alpha\omega.$$

Indeed, the imaginary part is $\alpha \sin(2\pi/3) = \rho \sin\theta$, while the real part is

$$\frac{\rho \sin(2\pi/3 - \theta) + \rho \sin\theta \cos(2\pi/3)}{\sin(2\pi/3)} = \rho\cos\theta.$$

For $\lambda \neq 0$, (II.2.4) reduces here to $(\lambda-\beta)^3 = \alpha^3$, and its upper branch is precisely $\lambda = \beta+\alpha\omega$. Moreover

$$\frac{d}{d\alpha} \arg(1 + \alpha(\omega - 1)) = \frac{\Im\omega}{|1 + \alpha(\omega - 1)|^2} > 0.$$

Hence the chord meets every ray with angle in $(0, 2\pi/3)$ exactly once; the uniqueness in proposition II.2.3 identifies that modulus with $K_3(\theta)$.

On the terminal cell $[1/3, 1/2]$, denominator order gives

$$(q, s) = (2, 3), \qquad d = \lfloor 3/2\rfloor = 1.$$

For $2\pi/3 < \theta = 2\pi x < \pi$ one has

$$A = 2(\pi - \theta), \qquad B = 3\theta - 2\pi, \qquad A + B = \theta,$$

and the scalar equation becomes

$$\rho^3 \sin(2(\pi - \theta)) + \rho^2 \sin(3\theta - 2\pi) = \sin\theta. \tag{II.9.5}$$

The nonzero carrier equation is now $\lambda(\lambda^2-\beta) = \alpha$, equivalently the polynomial (II.9.3). From $\lambda^3 = \beta\lambda + \alpha$, with $\lambda = \rho e^{i\theta}$, imaginary and real parts give

$$\beta = \frac{\rho^2 \sin(3\theta - 2\pi)}{\sin\theta}, \qquad \alpha = \frac{\rho^3 \sin(2(\pi - \theta))}{\sin\theta}.$$

Consequently $\alpha + \beta = 1$ is exactly (II.9.5). Along the upper root $\lambda = -1/2 + iy$,

$$\frac{d}{dy} \arg\left(-\frac{1}{2} + iy\right) = \frac{-1/2}{1/4 + y^2} < 0.$$

As y increases from 0 to $\sqrt{3}/2$, the argument therefore decreases strictly from π to $2\pi/3$. This boundary piece meets every ray in $(2\pi/3, \pi)$ exactly once, and proposition II.2.3 again identifies its modulus with $K_3(\theta)$. The real range $0 \le \alpha \le 1/4$ supplies the additional segment $[-1, -1/2]$ on the endpoint ray; it is not an open-cell branch.

The trace bounds proved that no order-three point lies beyond these pieces, while the displayed matrices attain them. At the Farey endpoint rays $0, 2\pi/3, \pi$, the outer points are respectively $1, \omega, -1$, and the endpoint convention gives $K_3 = 1$. We have therefore proved the base identity required at order four:

$$R_3(\theta) = K_3(\theta) \qquad (0 \le \theta \le \pi), \tag{II.9.6}$$

where at 0 and π the notation R_3 denotes the attained outer radial modulus. The jump from the terminal nonreal arc to -1 is exactly the exceptional order-three behavior recorded in proposition II.2.4. □

II.9.2 Induction on the order

Proof of theorem II.3.1. The orders $1, 2, 3$ are proposition II.9.2. Assume now that $n \geq 4$ and that the theorem holds at order $n - 1$.

First note the natural nesting

$$\Theta_{n-1} \subseteq \Theta_n : \tag{II.9.7}$$

if A is an $(n-1) \times (n-1)$ row-stochastic matrix, then $A \oplus [1]$ is an $n \times n$ row-stochastic matrix with every eigenvalue of A. Consequently

$$R_{n-1}(\theta) \leq R_n(\theta) \qquad (0 < \theta < \pi).$$

Fix an angle θ lying in an open cell of F_n^+ and put

$$\lambda_n = R_n(\theta) e^{i\theta}.$$

By corollary II.7.4,

$$K_n(\theta) e^{i\theta} \in \Theta_n, \qquad \text{hence} \qquad 0 < K_n(\theta) \leq R_n(\theta).$$

Since $\theta/(2\pi)$ is not a Farey endpoint of order n, $e^{i\theta}$ is not a root of unity of order at most n. Thus proposition II.4.4 gives

$$0 < R_n(\theta) < 1.$$

Suppose first that $\lambda_n \notin \Theta_{n-1}$. Then λ_n satisfies precisely the new-shell hypotheses (II.4.3). By corollary II.6.2, its modulus is the unique solution of the scalar radius equation on this ray. By definition that solution is $K_n(\theta)$, and hence

$$R_n(\theta) = K_n(\theta).$$

Suppose instead that $\lambda_n \in \Theta_{n-1}$. Its membership and the definition of R_{n-1} give

$$R_n(\theta) \leq R_{n-1}(\theta),$$

whereas (II.9.7) gives the reverse inequality. Thus

$$R_n(\theta) = R_{n-1}(\theta).$$

Because an open order-n ray cannot be an inherited Farey endpoint, the induction hypothesis applies to this same ray and gives

$$R_{n-1}(\theta) = K_{n-1}(\theta).$$

Now theorem II.8.4 and attainment give the fully explicit chain

$$R_n(\theta) = K_{n-1}(\theta) \leq K_n(\theta) \leq R_n(\theta).$$

Hence equality holds throughout, and again

$$R_n(\theta) = K_n(\theta).$$

We have proved this identity on every open cell. If $\theta/(2\pi) = p/q \in F_n^+$ is a Farey endpoint, then $q \leq n$ and the cyclic permutation matrix of order q, padded to order n, realizes $e^{i\theta}$. The disk bound in proposition II.4.1 therefore shows that the outer radial modulus on that endpoint ray is $1 = K_n(\theta)$. Extend R_n to $\theta = 0, \pi$ by these outer radial moduli. Then

$$R_n(\theta) = K_n(\theta) \qquad (0 \leq \theta \leq \pi). \tag{II.9.8}$$

For $n \geq 4$, proposition II.2.4 now joins the open-cell graphs continuously at their Farey endpoints. Thus

$$\zeta_n(\theta) = K_n(\theta)e^{i\theta}, \qquad 0 \leq \theta \leq \pi,$$

traces exactly the ordered upper chain of the carriers $\Gamma^{(n)}_{f,g}$, from 1 to -1. Every point of this chain belongs to Θ_n by corollary II.7.4 and the preceding endpoint argument. Conversely, the equality $R_n = K_n$ says that no point of Θ_n lies farther out on any upper-half-plane ray. Conjugation gives the corresponding lower chain.

For completeness, define the full-circle function

$$\widehat{K}_n(\phi) = \begin{cases} K_n(\phi), & 0 \leq \phi \leq \pi, \\ K_n(-\phi), & -\pi \leq \phi \leq 0. \end{cases}$$

Conjugation symmetry and (II.9.8) show that this is the radial maximum of Θ_n on every ray. It is positive by attainment and continuous on the circle by proposition II.2.4; at the identified angles $-\pi$ and π both values are 1. The set Θ_n is compact by proposition II.4.1 and radially filled by corollary II.4.3. Therefore lemma II.9.1, applied with $S = \Theta_n$ and $r_S = \widehat{K}_n$, gives

$$\Theta_n = \{t\widehat{K}_n(\phi)e^{i\phi} : 0 \leq t \leq 1, -\pi < \phi \leq \pi\}, \qquad \partial\Theta_n = \{\widehat{K}_n(\phi)e^{i\phi} : -\pi < \phi \leq \pi\}.$$

Thus the two carrier chains are exactly the topological boundary, not merely the outer points on individual rays, and the first formula is the claimed radial hull. The unit-circle assertion is proposition II.4.4. The discontinuous terminal graph and the segment $[-1, -1/2]$ occur only at order three and were handled directly in proposition II.9.2; the continuous-radius lemma is not invoked for that exceptional base case. □

II.10 The complete order-seven example

The upper Farey sequence of order seven is

$$0, \frac{1}{7}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{2}{7}, \frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \frac{1}{2}. \tag{II.10.1}$$

For each cell, the table lists the denominator pair (q, s) with $q \leq s$, the multiplicity $d = \lfloor 7/q \rfloor$, the exponent $e = s - dq$, and the reduced carrier polynomial. Throughout $\beta = 1 - \alpha$.

cell	(q, s)	d	e	reduced carrier
$0 \to 1/7$	$(1, 7)$	7	0	$(z-\beta)^7 - \alpha^7$
$1/7 \to 1/6$	$(6, 7)$	1	1	$z(z^6 - \beta) - \alpha$
$1/6 \to 1/5$	$(5, 6)$	1	1	$z(z^5 - \beta) - \alpha$
$1/5 \to 1/4$	$(4, 5)$	1	1	$z(z^4 - \beta) - \alpha$
$1/4 \to 2/7$	$(4, 7)$	1	3	$z^3(z^4 - \beta) - \alpha$
$2/7 \to 1/3$	$(3, 7)$	2	1	$z(z^3 - \beta)^2 - \alpha^2$
$1/3 \to 2/5$	$(3, 5)$	2	-1	$(z^3 - \beta)^2 - \alpha^2 z$
$2/5 \to 3/7$	$(5, 7)$	1	2	$z^2(z^5 - \beta) - \alpha$
$3/7 \to 1/2$	$(2, 7)$	3	1	$z(z^2 - \beta)^3 - \alpha^3$

II.10.1 A worked ray: $x = 3/8$

The ray $x = 3/8$ lies in the cell $1/3 < x < 2/5$. Here

$$q = 3, \qquad s = 5, \qquad d = 2, \qquad e = -1,$$

and

$$A = 2\pi\left(3\cdot\frac{3}{8}-1\right) = \frac{\pi}{4}, \qquad B = \pi\left(2-5\cdot\frac{3}{8}\right) = \frac{\pi}{8}.$$

The boundary radius is the unique root of

$$\rho^{5/2}\sin\frac{\pi}{4} + \rho^3\sin\frac{\pi}{8} = \sin\frac{3\pi}{8}, \tag{II.10.2}$$

namely

$$\begin{aligned} &\rho = 0.940100221928822853\ldots, \\ &\alpha = 0.655850787368397414\ldots, \quad \beta = 0.344149212631602586\ldots. \end{aligned} \tag{II.10.3}$$

Thus

$$\lambda = \rho e^{3\pi i/4} = -0.664751241920849 + 0.664751241920849\, i$$

and

$$(\lambda^3-\beta)^2 = \alpha^2\lambda. \tag{II.10.4}$$

A six-state active realization, padded by one absorbing state, is

$$A_7 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \beta & 0 & 0 & \alpha & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & \alpha & 0 & \beta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{II.10.5}$$

The two local 3-cycles have weight β, while the unique cross cycle has length five and weight α^2. Hence

$$\det(tI - A_7) = (t-1)\big((t^3-\beta)^2 - \alpha^2 t\big),$$

which verifies (II.10.4) directly.

II.10.2 The order-seven region

II.11 Concluding structural remarks

The proof separates three statements that are often conflated.

First, the boundary equation is not a generic determinant identity for stochastic matrices. A new-shell radial extremum first becomes an N-critical planar multiplier; only then does critical-polygon monodromy force the heterogeneous product and phase identity.

Second, the Part I monodromy theorem is existential. It selects one of the two adapted complex orientations and supplies the corresponding Farey data. The reflection dictionary returns those data to the original upper-half-plane point; no uniqueness of invariant polygons or stochastic realizers is used.

Third, the scalar Farey geometry is downstream of the Part I input. Once the product, phase identity, and common Jensen sheet are available, the remainder is strict one-dimensional convexity: all nonconstant cell profiles lie radially inside the equal profile, and Farey refinement moves the equal-profile envelope outward. The intrinsic contact and return theory that establishes this input is developed in Part I and culminates in theorem 1.4.

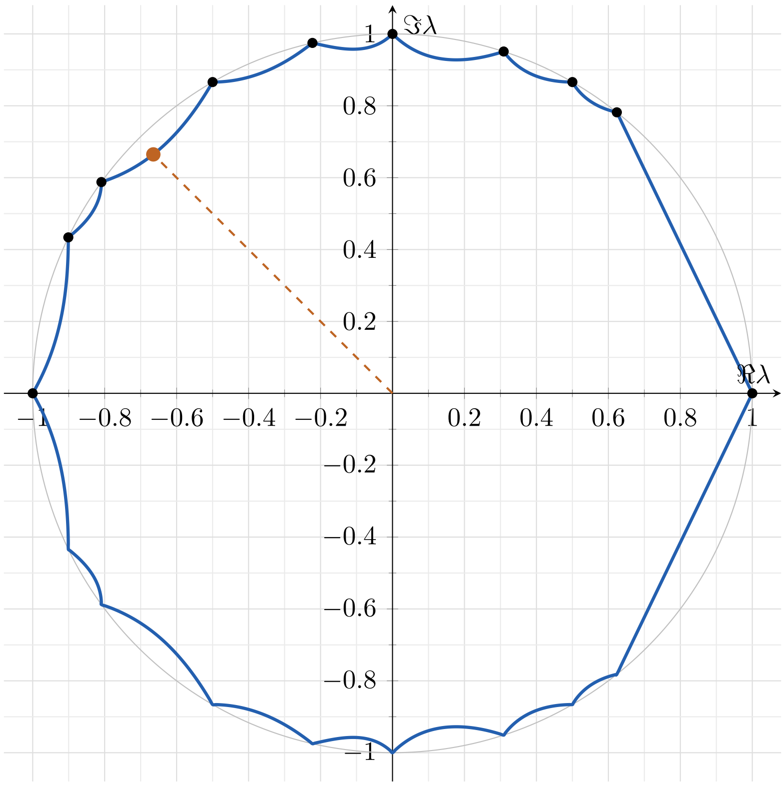

Figure 1: The boundary of Θ_7. The dashed ray is $x = 3/8$, and the marked point is the solution of (II.10.2).

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